

SHELBY

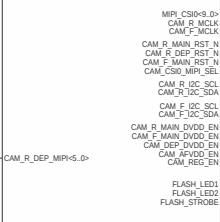
SYSTEM 2018

BASED ON 80-P7747-41 REV E

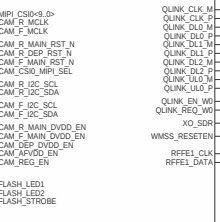
REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED

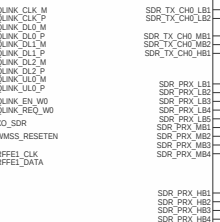
IMAGERS



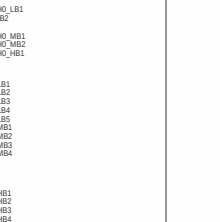
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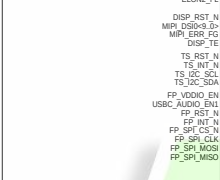
SDR660



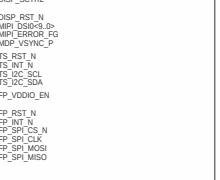
RFFE



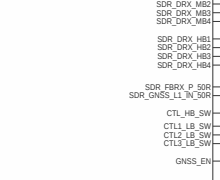
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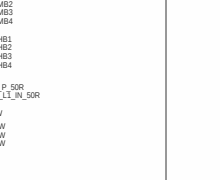
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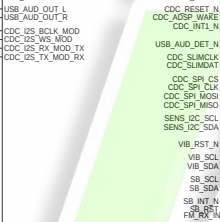
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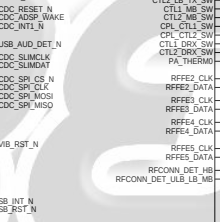
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AUDIO



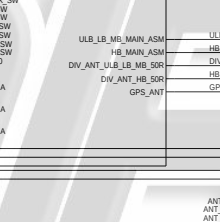
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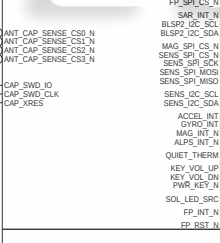
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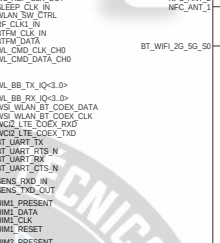
SENSOR



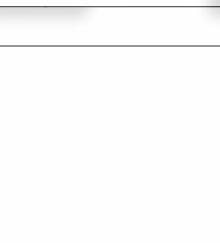
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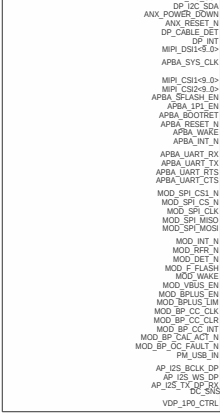
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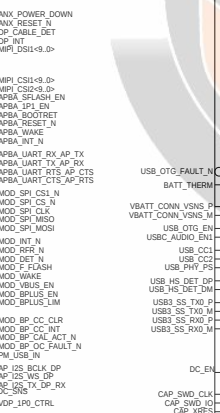
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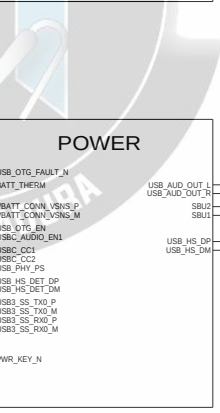
MODS



AP_PMIC



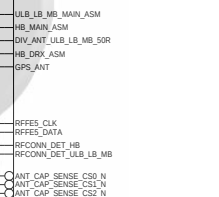
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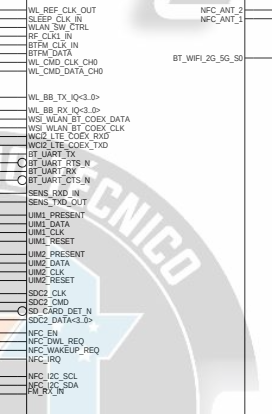
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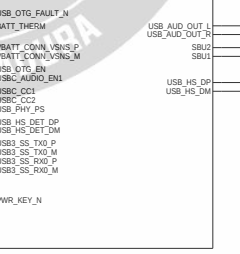
ANTENNAS



CONNECTIVITY



POWER



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Mon Apr 02 15:57:46 2018		8401672201_a_180402	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME 8401672201_a_180402.cpm	REV 03
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MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

SHELBY ME/SHIELDS

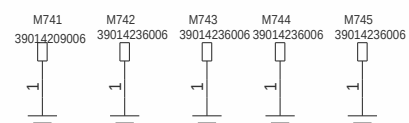
REFERENCE DESIGNATOR RANGES BY BLOCK

000 - 499: ANTENNA
500 - 699: POWER
700 - 799: AMPS
800 - 899: USER INPUT
1000-2999: SDM660 + PM660 + PM660A
3000-3199: WTR5975
4000-4499: CONNECTIVITY
4500-4999: AUDIO FE
5000-5499: DISPLAY
5500-5999: IMAGERS
6000-6999: XMCS
7000-8999: RFFE
9000-9999: PRODUCT-SPECIFIC (FILTERING, 100PF CAPS)

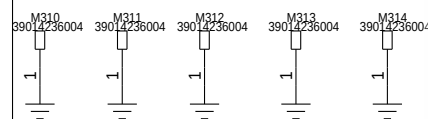
GROUND CONTACTS FROM PCB TO FRONT



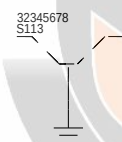
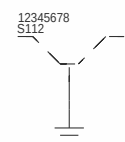
GROUND CONTACTS FOR MODS CONNECTOR



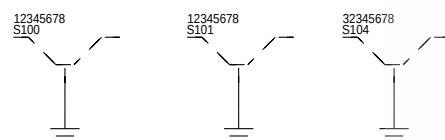
GROUND CONTACTS FROM PCB TO FRONT



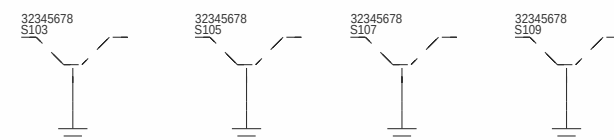
2.5*3.5MM SHAPED SCREW 2.1*3.5MM SHAPED SCREW



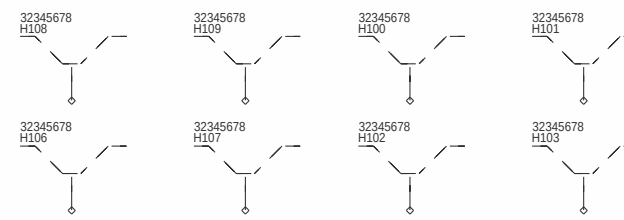
2.1*3.5MM SCREW



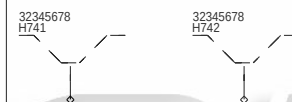
1.8*3.0MM SCREW



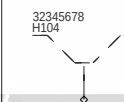
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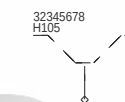
1.5MM NPTH



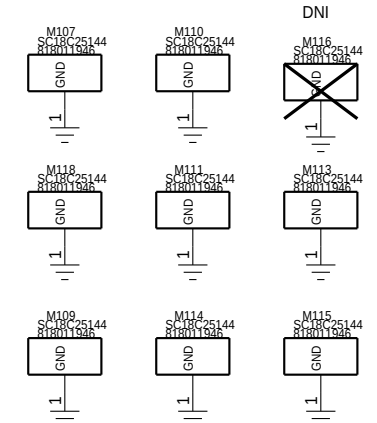
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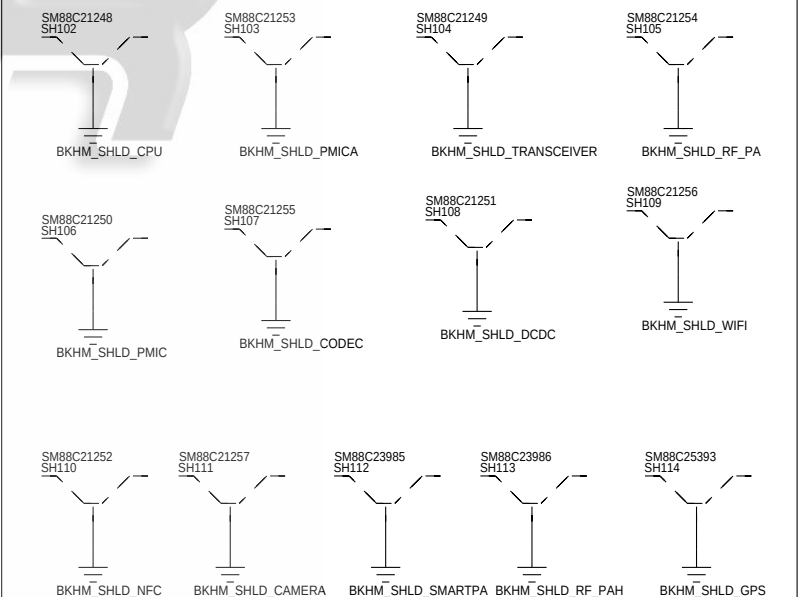
2.1MM NPTH



COAX GND CLIPS



SHIELDS



MOTOROLA MOBILIZACION RESTRICTED-MDB TRACKED TRACKED AND BLOCKED



ENGINEER: MOTOROLA

BLOCK TITLE:

LAST MODIFIED BY: MOTOROLA
Thu Mar 01 14:04:30 2018

8401672201_a_180402

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PROJECT NAME
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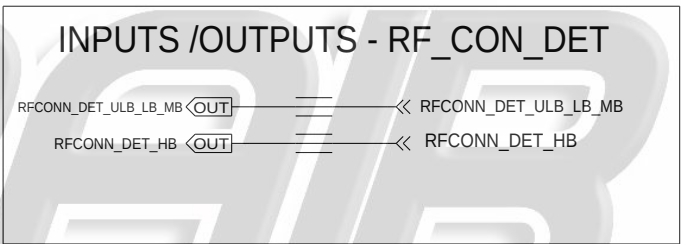
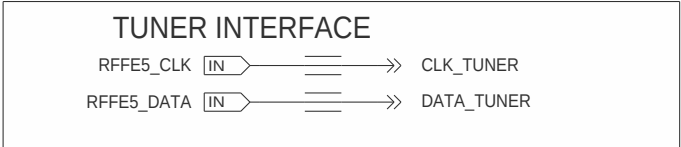
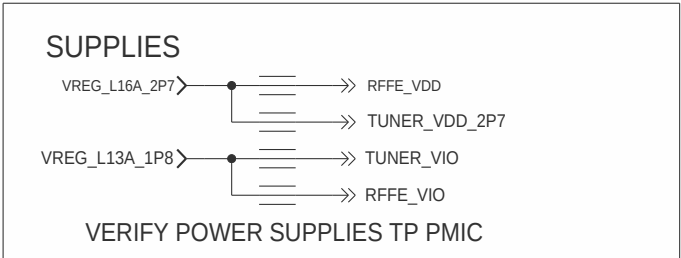
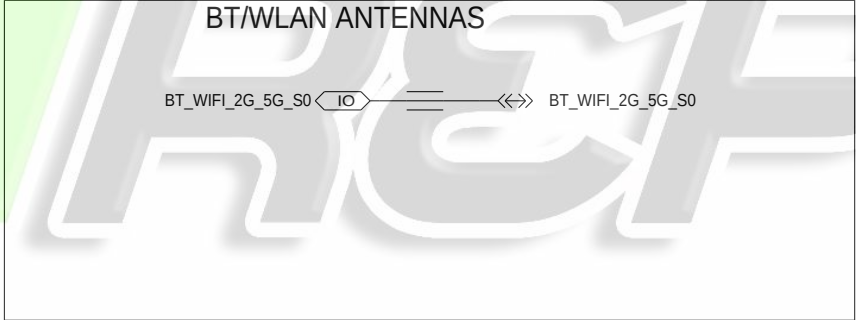
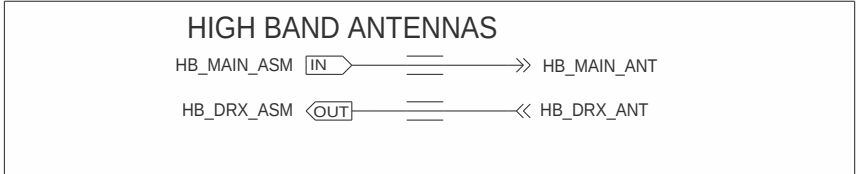
REV
03

CAD FILE NAME:

PAGE 2 OF 70

ANT: PORTS

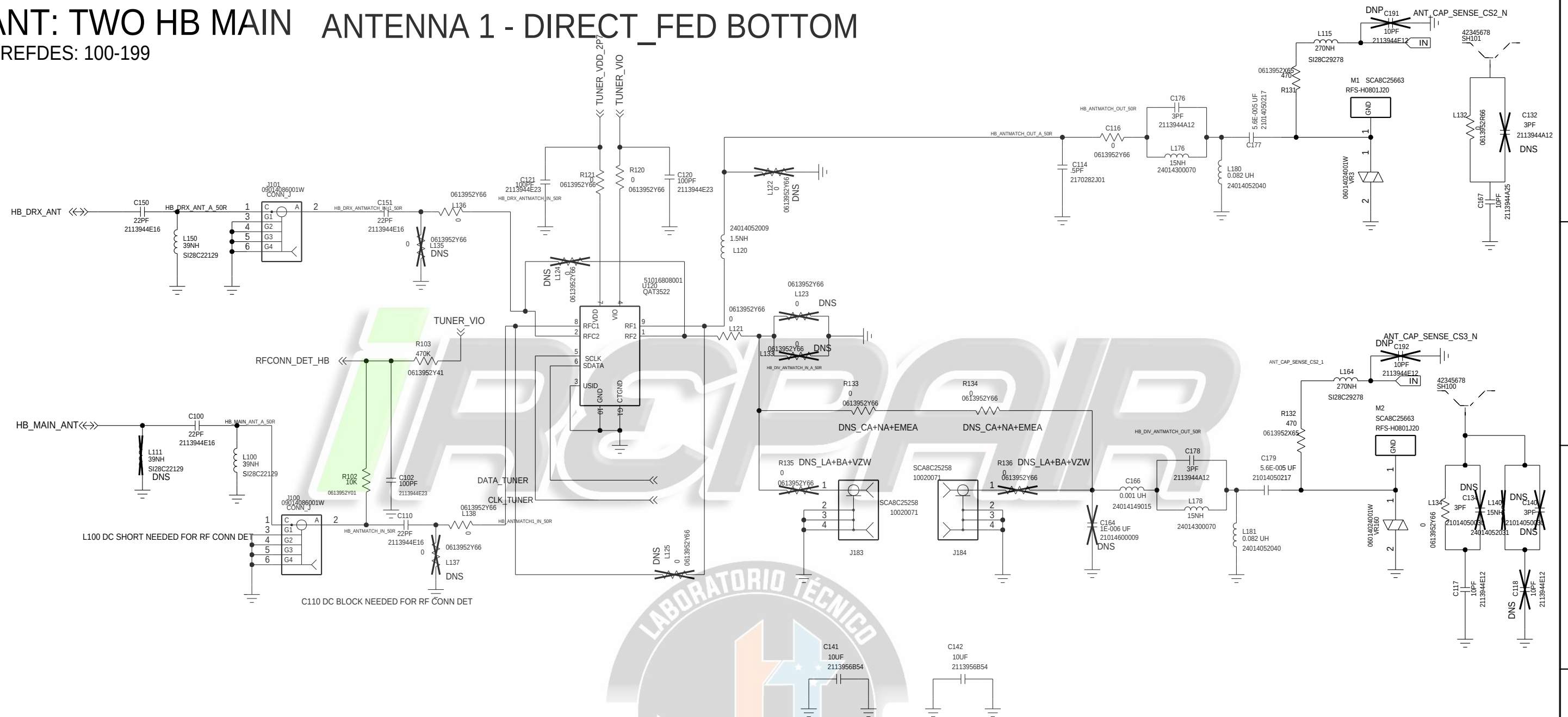
REFDES: 000 - 499



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ENGINEER: MOTOROLA		BLOCK TITLE: antennas		
LAST MODIFIED BY: MOTOROLA Thu Mar 08 11:15:16 2018				
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		B 8401672201_a_180402.cpm		03
		CAD FILE NAME:		PAGE 3 OF 70

ANT: TWO HB MAIN ANTENNA 1 - DIRECT FED BOTTOM

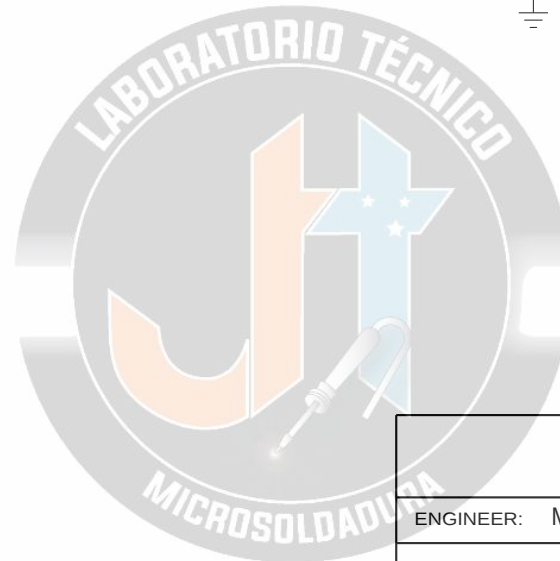
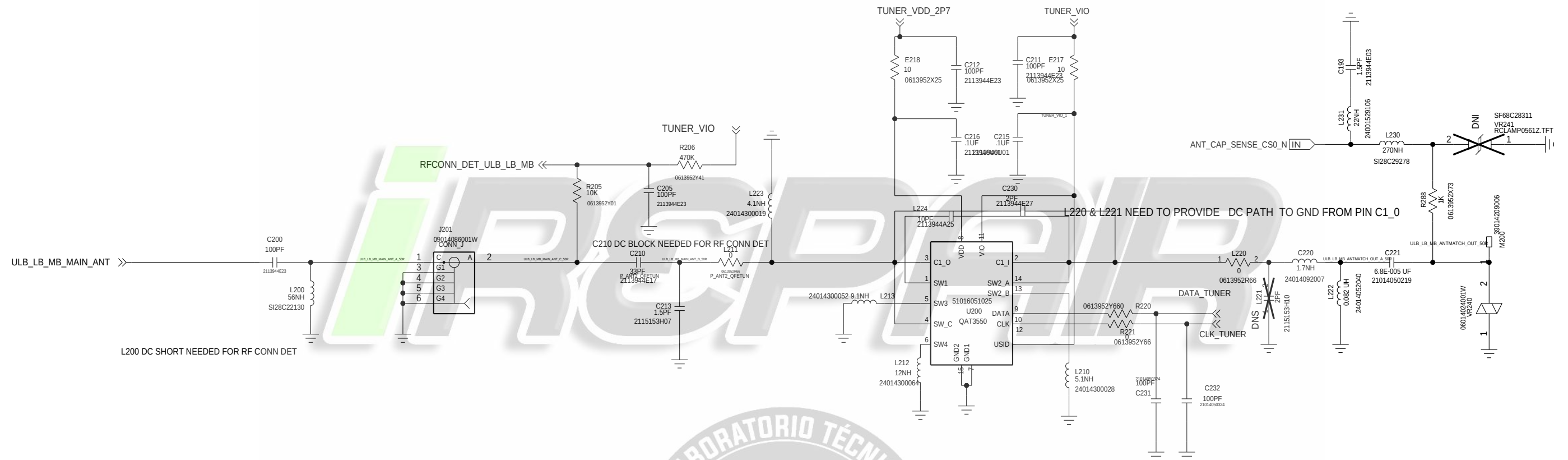
REFDES: 100-199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Fri Mar 09 11:36:47 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		CAD FILE NAME:	
		PAGE 4 OF 70	

REFDES: 200-299

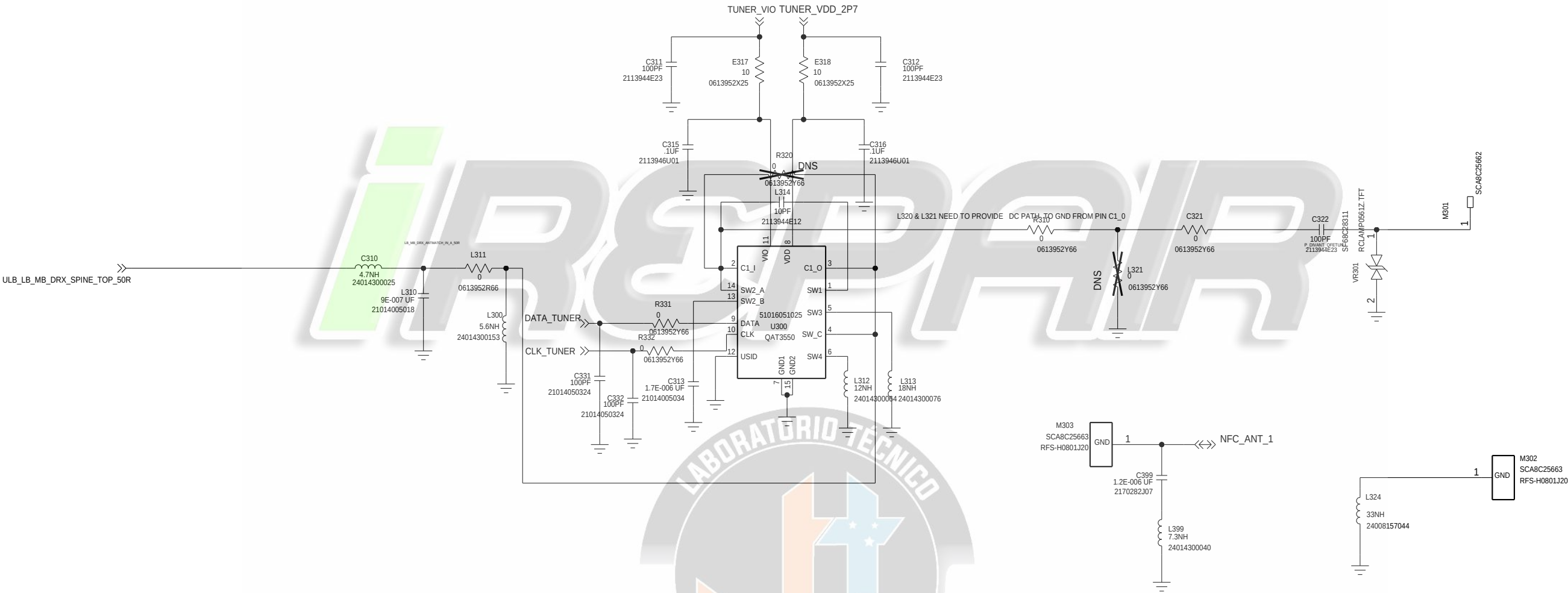
ANTENNA 0 - COUPLED FED LOOP BOTTOM



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: antennas		
LAST MODIFIED BY: MOTOROLA Thu Mar 08 11:15:17 2018			
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		CAD FILE NAME:	PAGE 5 OF 70

ANT: ULB+LB+MB DIV
REFDES: 300-399

ANTENNA 2 - DIRECT FED LOOP TOP



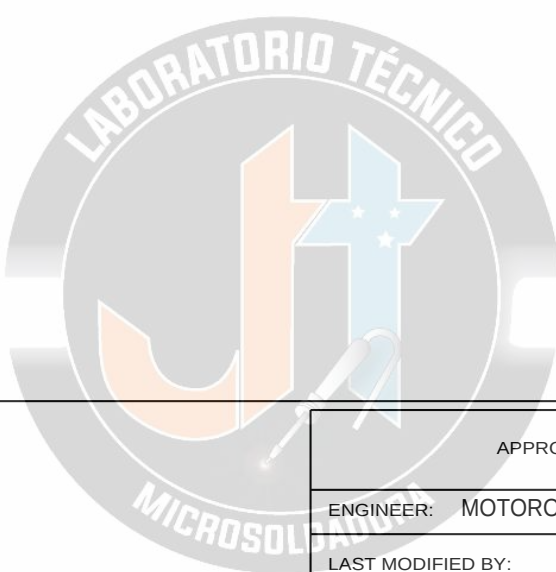
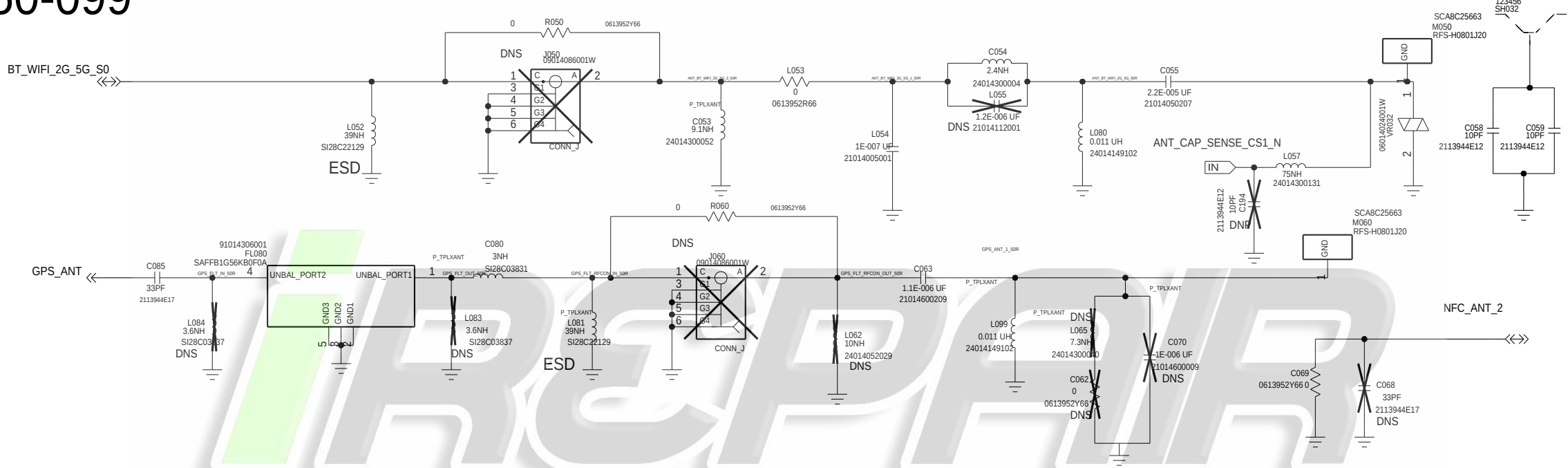
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 08 11:15:18 2018		antennas	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME 8401672201_a_180402.cpm
		REV 03	
		PAGE 6 OF 70	

ANT5 : GPS+WLAN2.4GHZ+WLAN5GHZ S1
ANT6 : WLAN 2.4GHZ +WLAN 5GHZ S0

REF:050-099

ANTENNA 5 - DIRECT FED LOOP TOP

ANTENNA 6 - DIRECT FED LOOP TOP



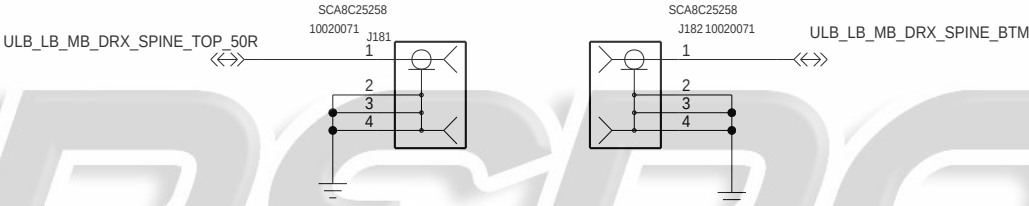
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		CAD FILE NAME: PAGE 7 OF 70	

ANT: SPINE COAX

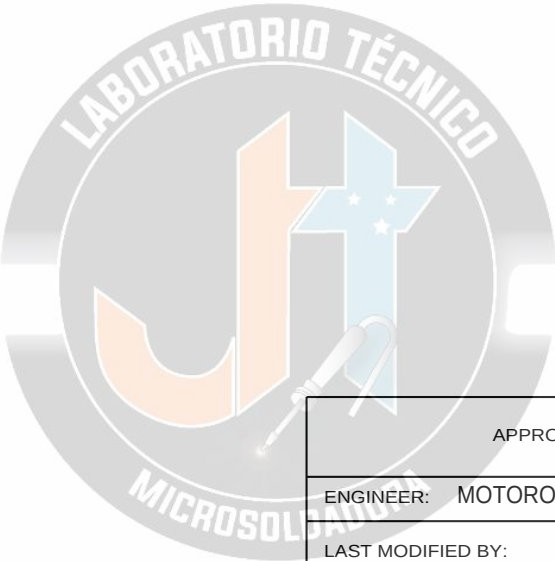
REFDES: 400-499

TOP OF SPINE

BOTTOM OF SPINE



REPAIR



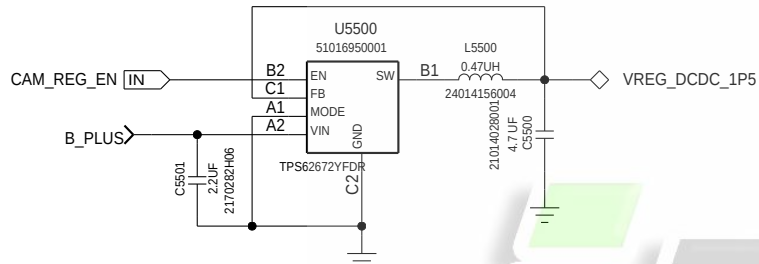
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Thu Mar 08 11:15:20 2018			
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		REV 03	
		PAGE 8 OF 70	

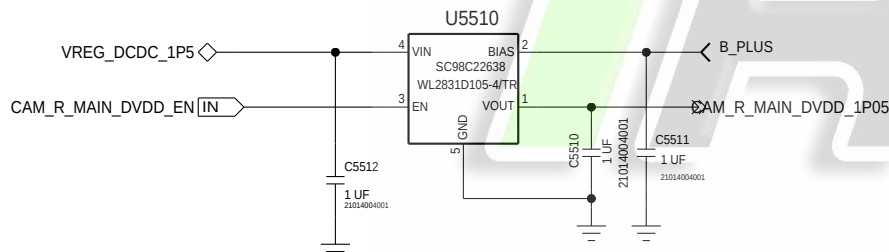
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REF:5500-5549

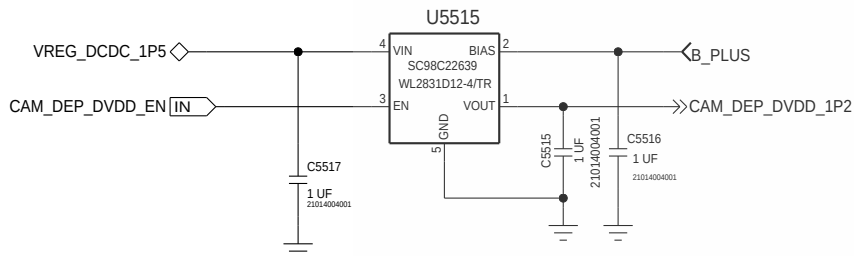
DCDC 1.5V FOR SUB-REGULATION



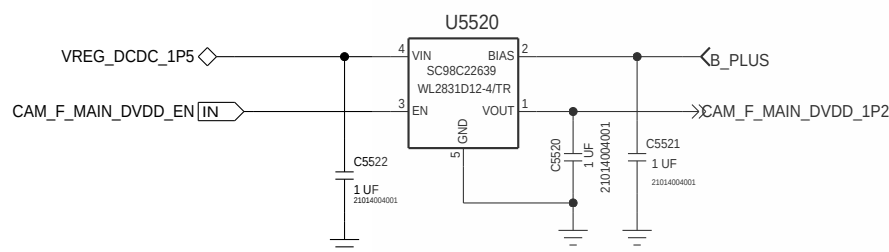
REAR MAIN IMAGER DVDD LDO 1.05V



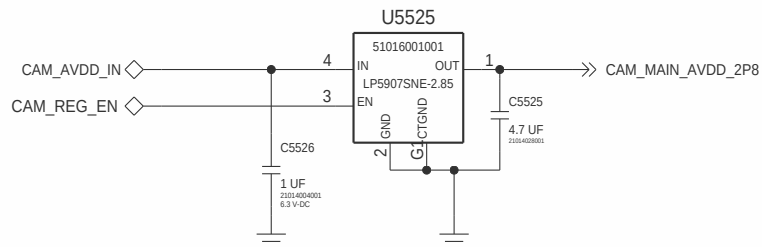
DEPTH IMAGER DVDD LDO 1.2V



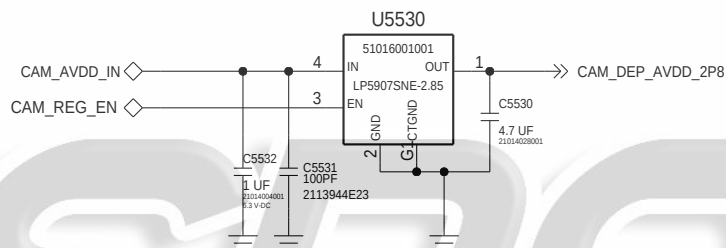
FRONT MAIN IMAGER DVDD LDO 1.2V



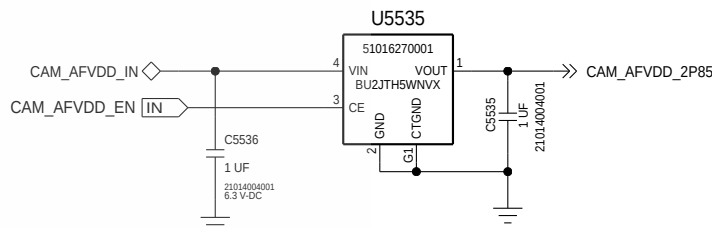
MAIN IMAGER AVDD LDO 2.85V



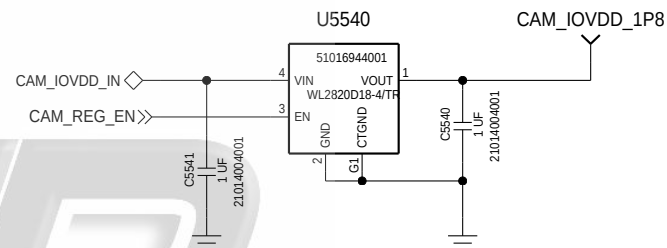
DEPTH IMAGER AVDD LDO 2.85V



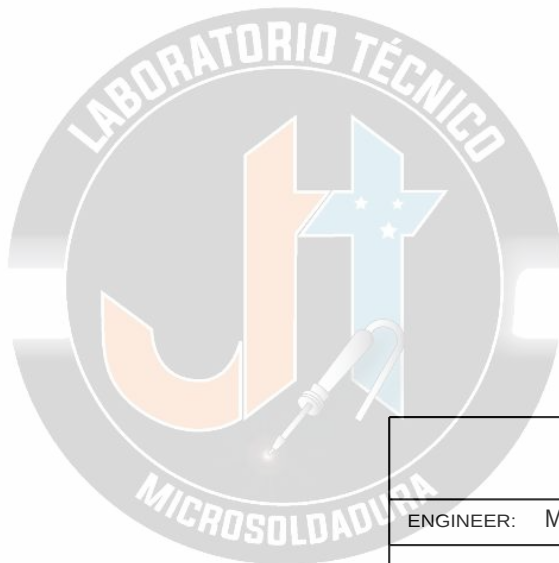
MAIN IMAGER AFVDD LDO 2.85V



IMAGER IOVDD LDO 1.8V



SUPPLIES



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Thu Mar 01 10:01:22 2018

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BLOCK TITLE:

imagers

PROJECT NAME
B 8401672201_a_180402.cpm

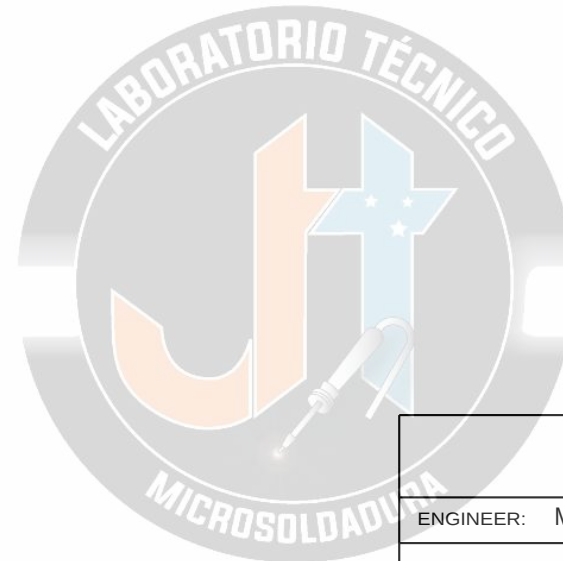
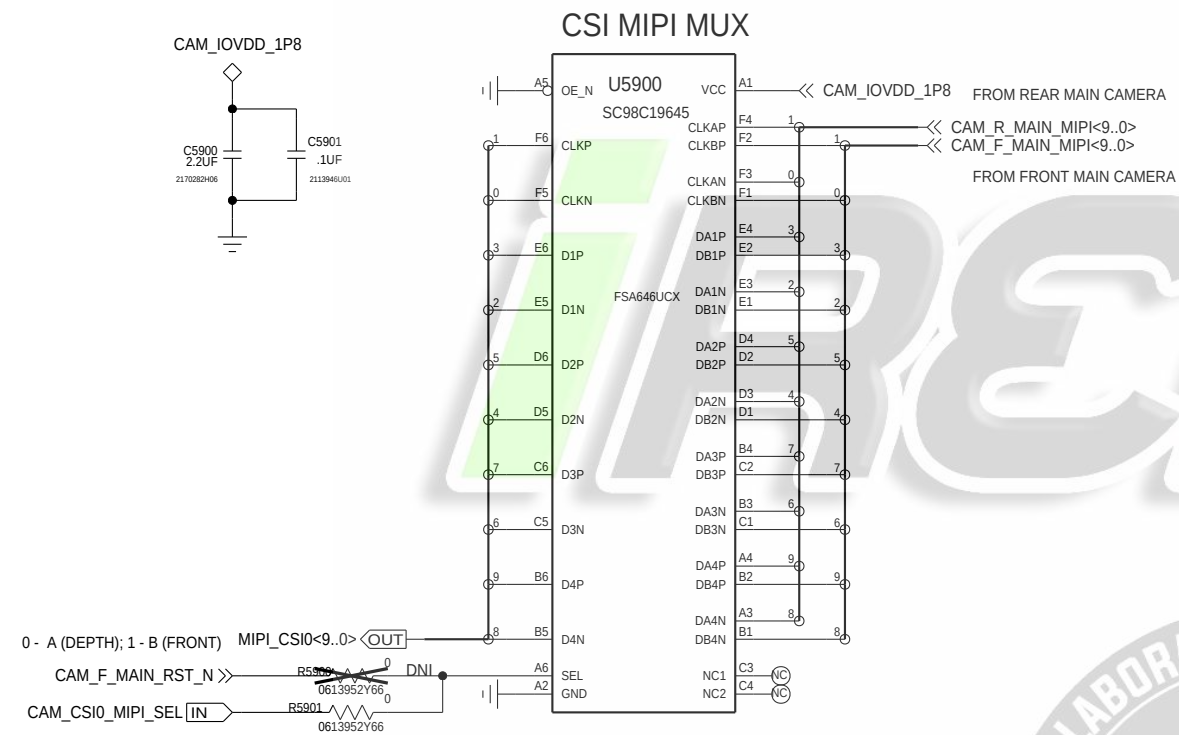
CAD FILE NAME:

REV
03

PAGE 9 OF 70

MIPI MUX
REF:5900-05919

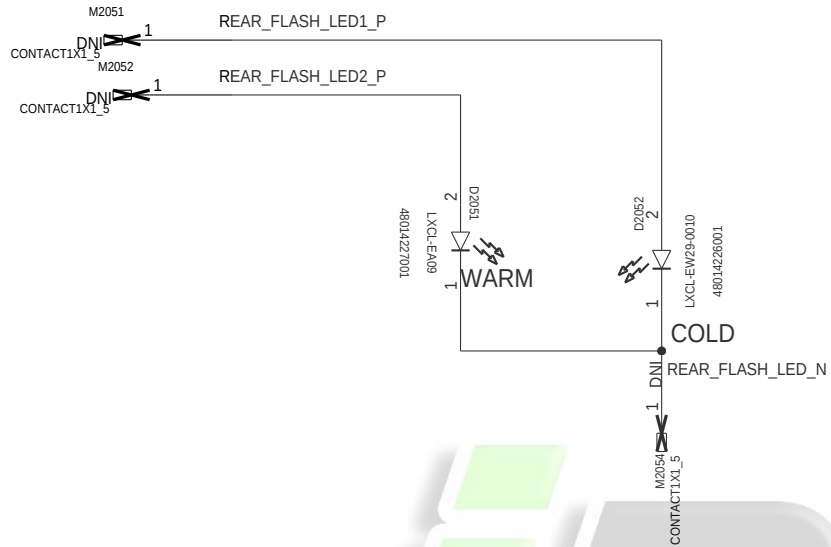
FRONT/REAR MAIN IMAGER MIPI SWITCH



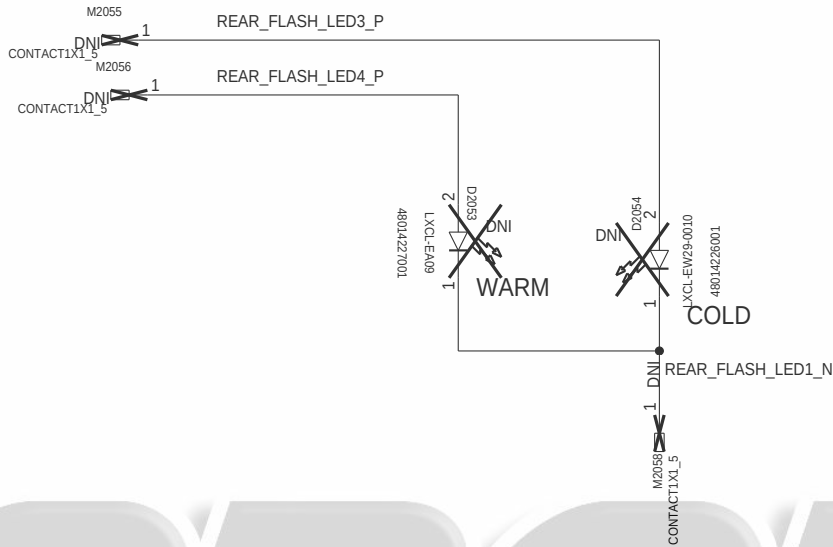
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ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:23 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		CAD FILE NAME:	
		PAGE 10 OF 70	

CAM: REAR FLASH

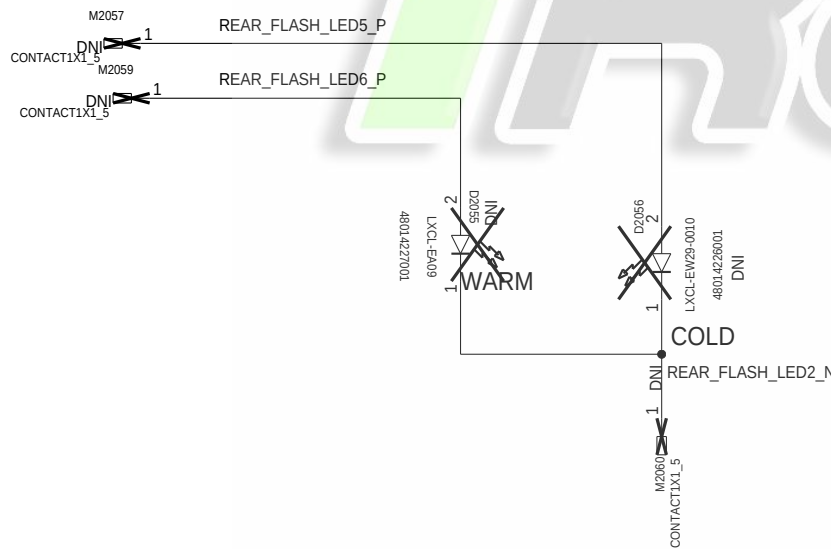
FLASH



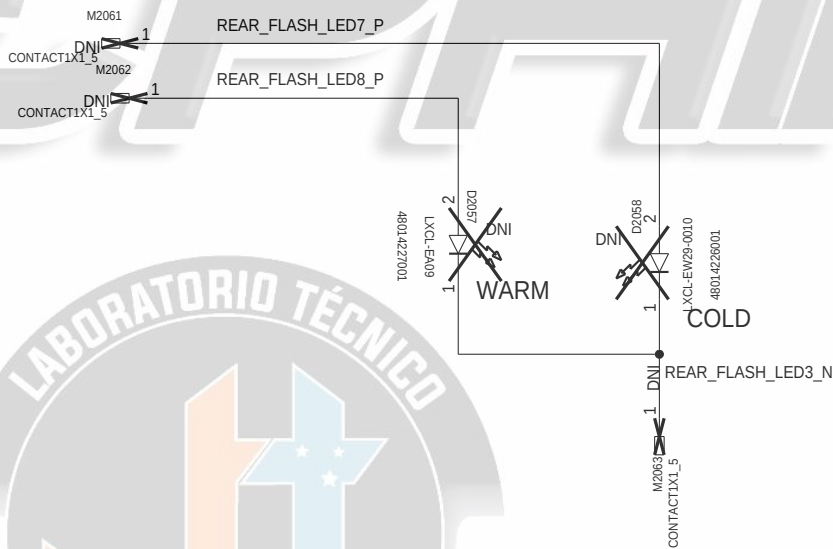
FLASH



FLASH



FLASH



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Fri Nov 24 14:18:51 2017

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 **MOTOROLA**
MOBILE DEVICES

BLOCK TITLE:

imagers

PROJECT NAME
B 8401672201_a_180402.cpm

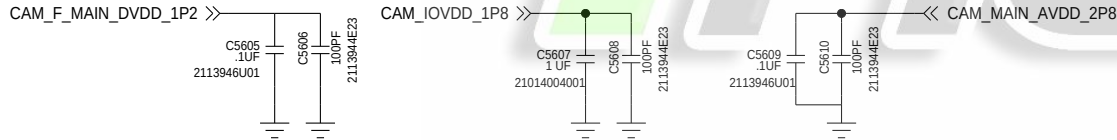
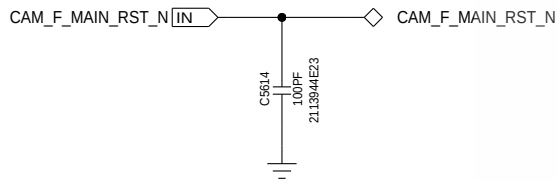
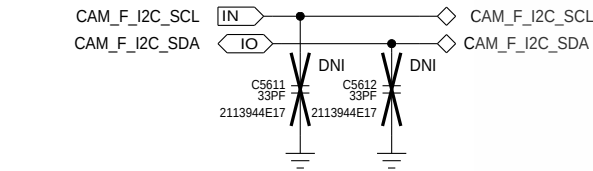
REV
03

CAD FILE NAME:

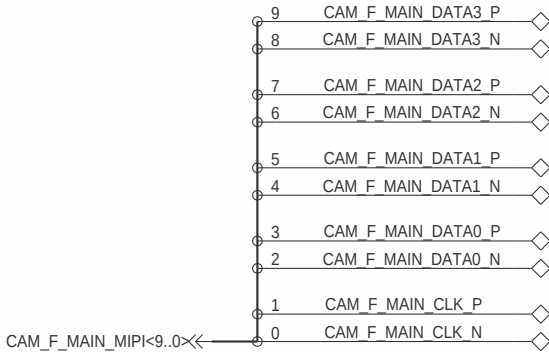
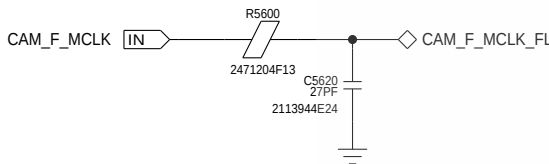
PAGE 11 OF 70

CAM: FRONT CAMERA

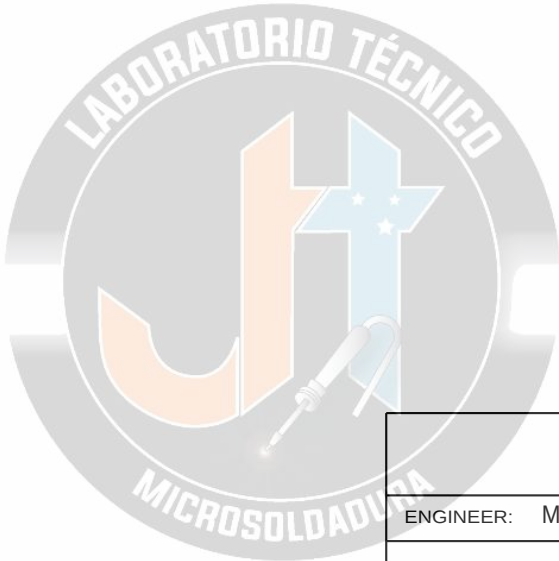
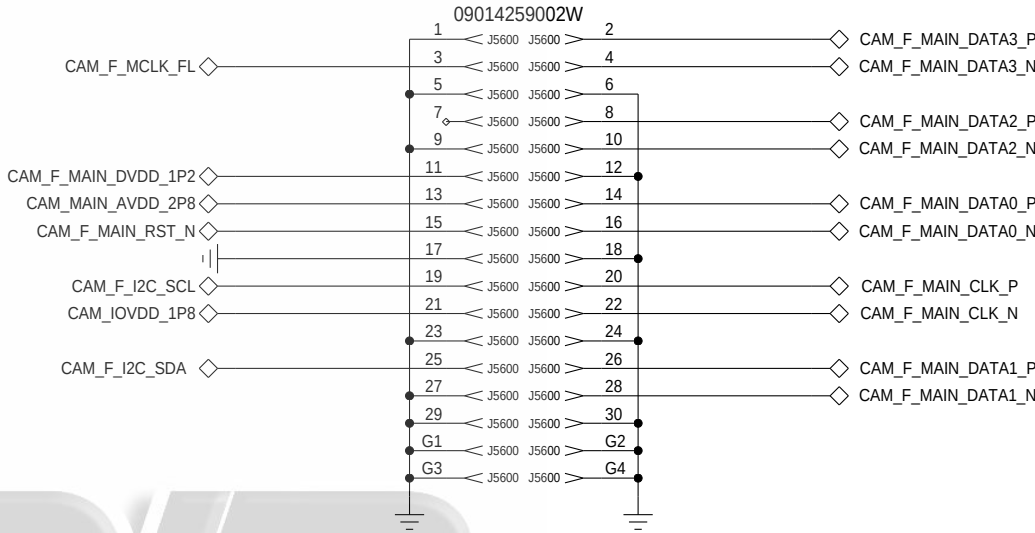
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EMI/ESD FILTER



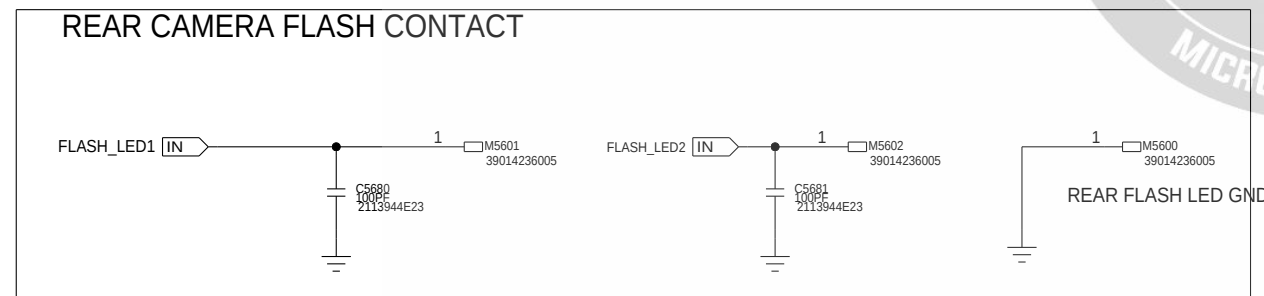
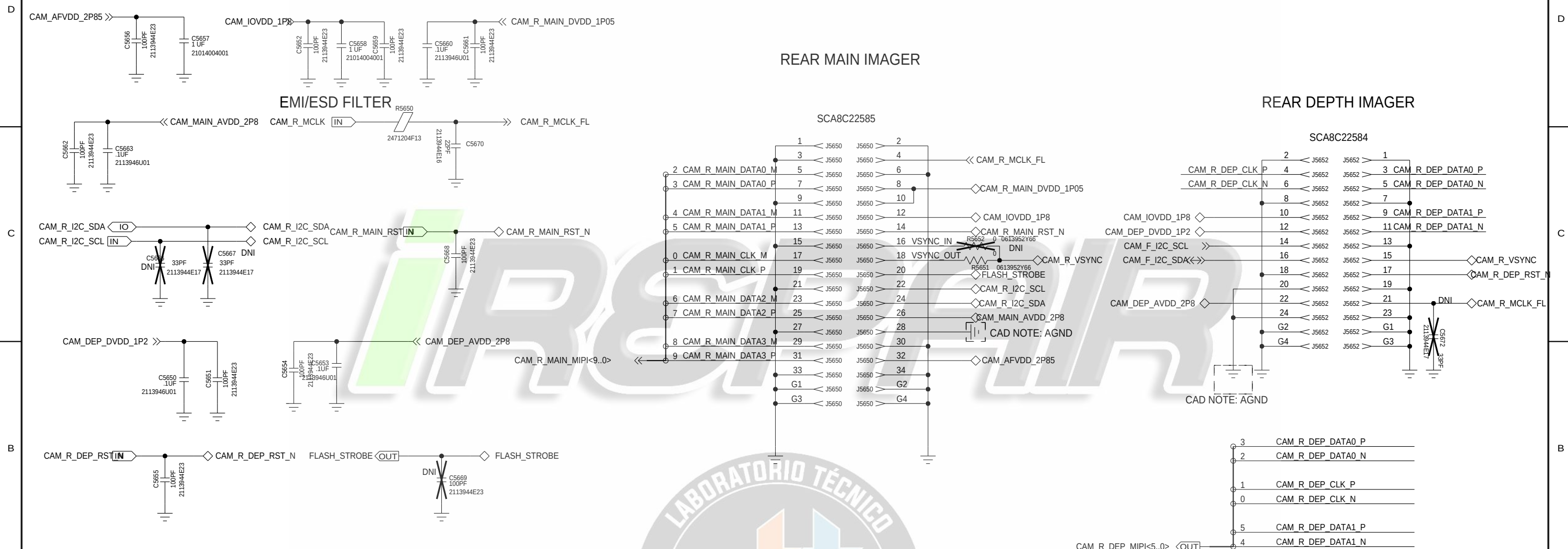
FRONT MAIN IMAGER



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:24 2018		PROJECT NAME 8401672201_a_180402.cpm	
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		PAGE 12 OF 70	

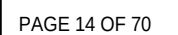
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REF 5650-5699

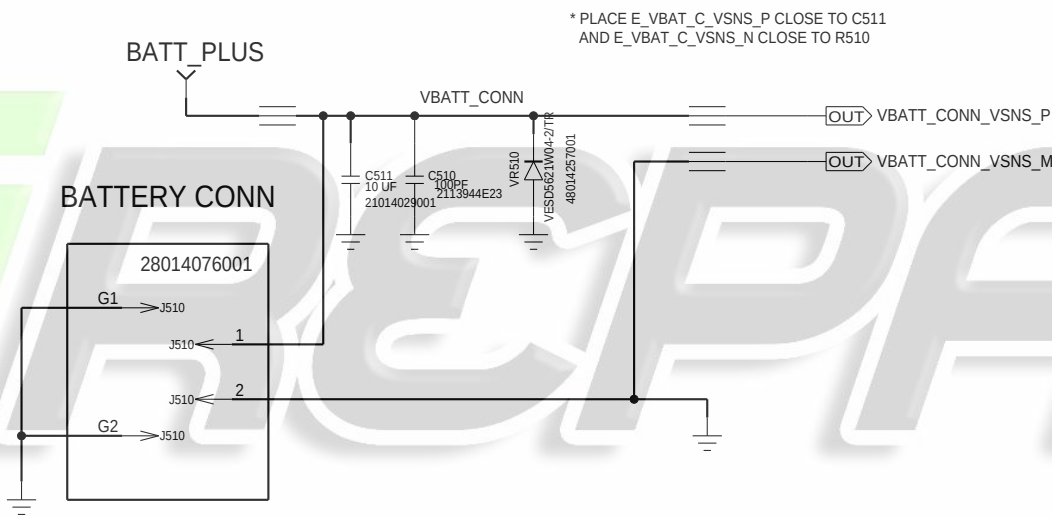


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:24 2018			
<u>MOTOROLA CONFIDENTIAL RESTRICTED</u> THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT.  COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME 8401672201_a_180402.cpm	REV 03
		CAD FILE NAME:	PAGE 13 OF 70

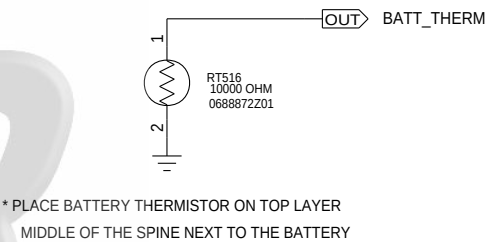
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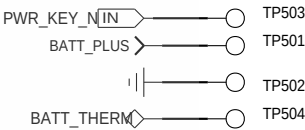
PWR: BATT
REF: 510-519, 650-659



BATT_THERM



FACTORY TEST POINT



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

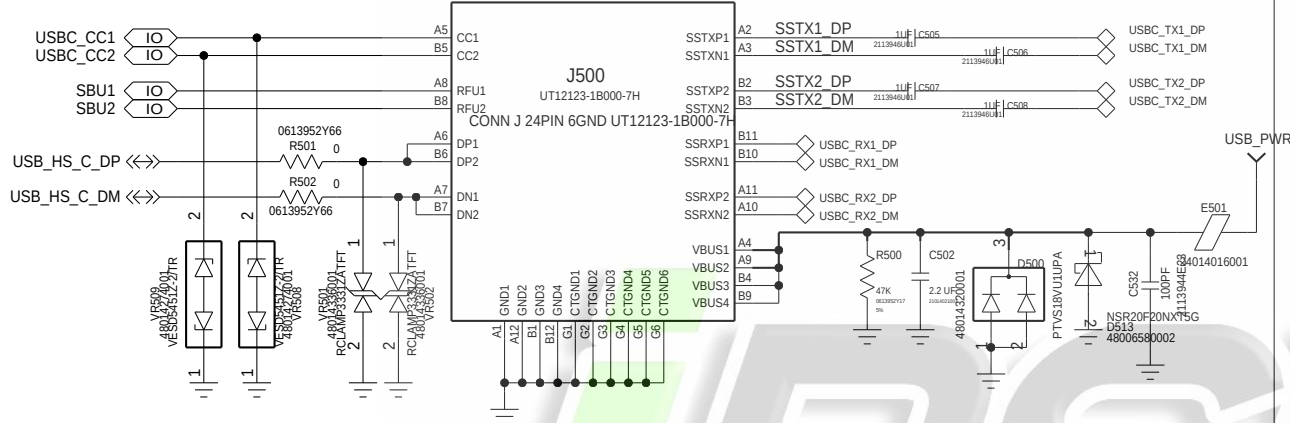
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:20 2018		power	
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		REV 03	
		PAGE 15 OF 70	

PWR: USB TYPE C

REF: 500-509,520-539,550-559

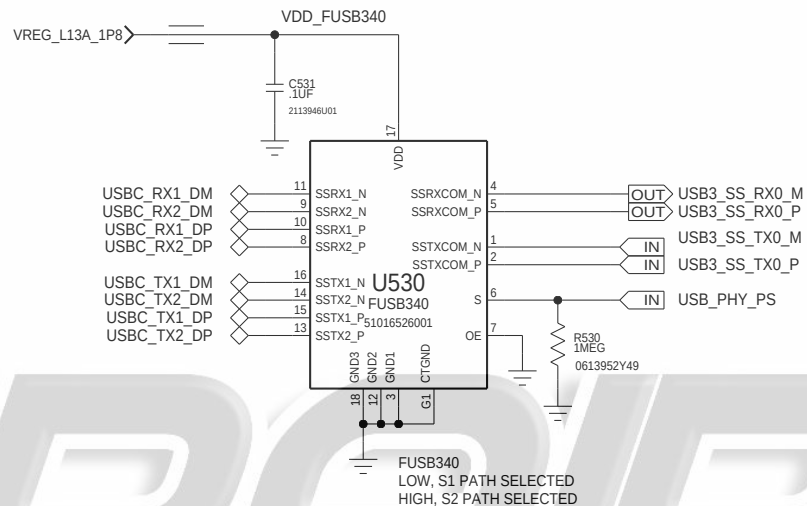
TYPE-C USB CONNECTOR

REF: 500-509



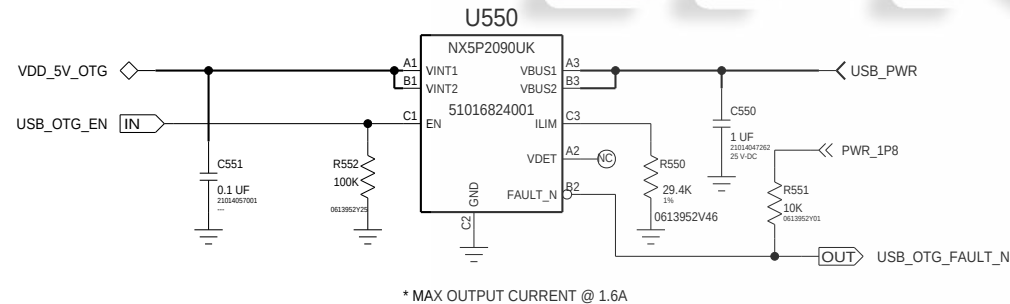
USB SS MUX

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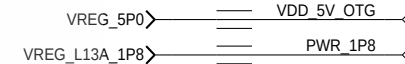
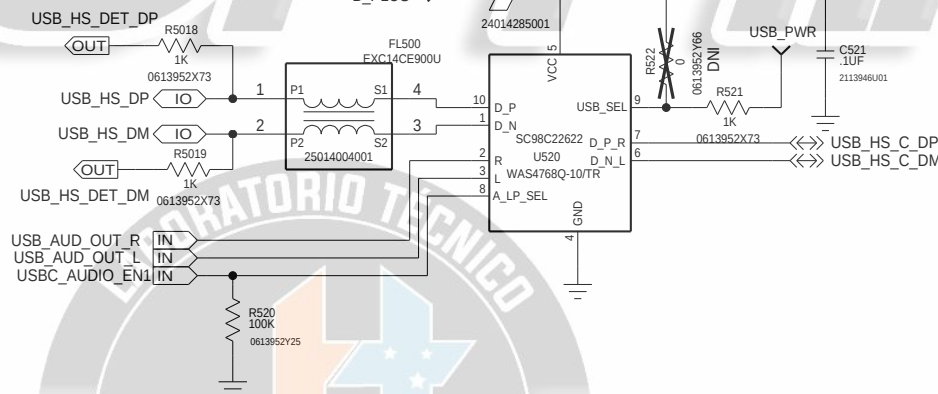
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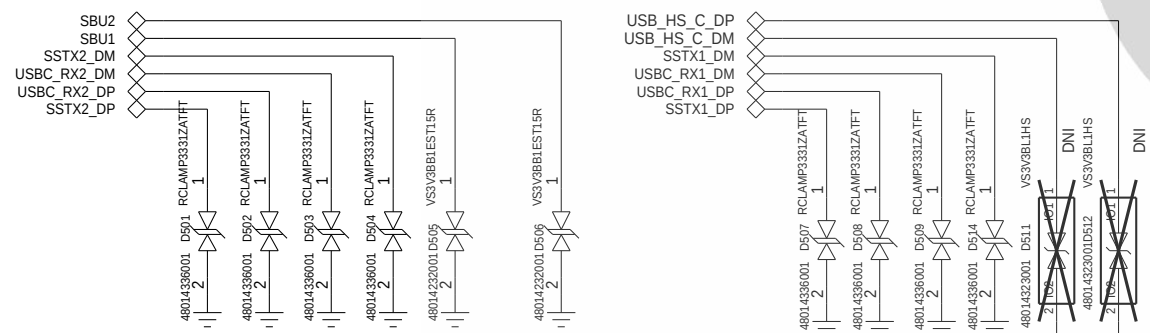
USB HS / HSJ AUDIO MUX

REF: 520-529



USB SS/HS/SBU ESD

REF: 500-509



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

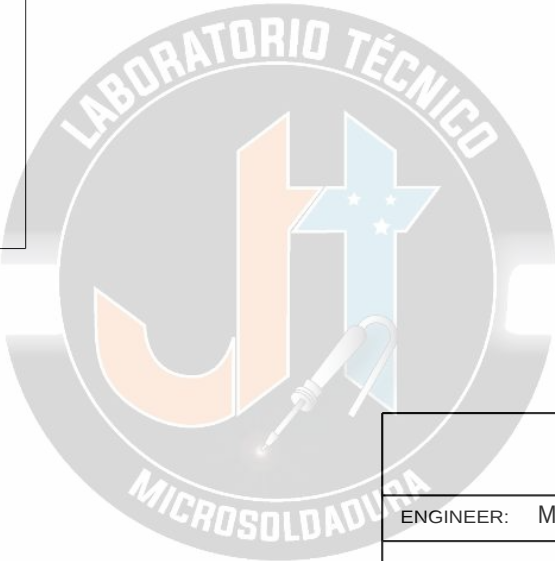
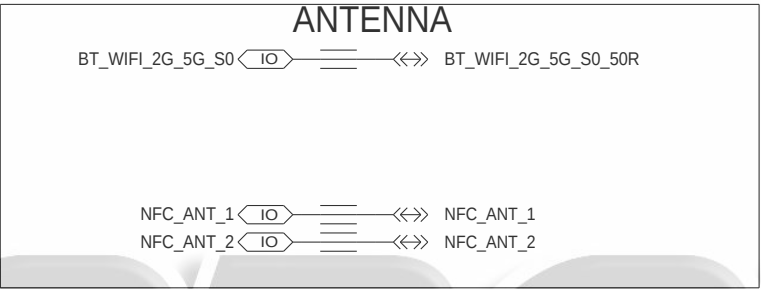
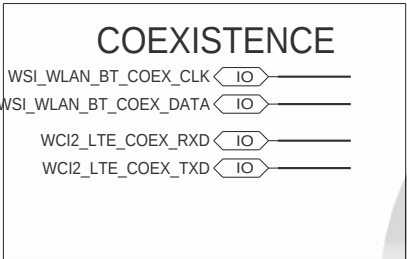
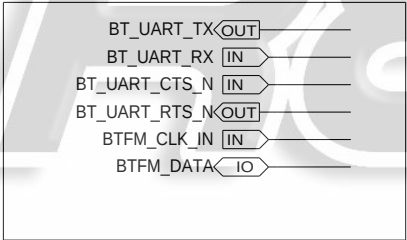
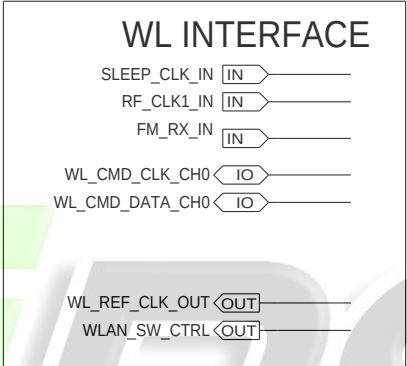
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ENGINEER: MOTOROLA		BLOCK TITLE: power	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:20 2018		PROJECT NAME B 8401672201_a_180402.cpm	
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		PAGE 16 OF 70	

8	7	6	5	4	3	2	1																																								
PWR: EXT CHARGER REF: 560-579 iR&PAIR LABORATORIO TÉCNICO MS Microsoft <tr><td>D</td><td colspan="6"></td><td>D</td></tr> <tr><td>C</td><td colspan="6"></td><td>C</td></tr> <tr><td>B</td><td colspan="6"></td><td>B</td></tr> <tr><td>A</td><td colspan="6"></td><td>A</td></tr> <tr><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr>								D							D	C							C	B							B	A							A	8	7	6	5	4	3	2	1
D							D																																								
C							C																																								
B							B																																								
A							A																																								
8	7	6	5	4	3	2	1																																								

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: power	
LAST MODIFIED BY: MOTOROLA Tue Aug 01 17:32:39 2017			
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		REV 03	
		CAD FILE NAME: PAGE 17 OF 70	

CONN: WCN3980 POWER / PORT
REF 4000-4099



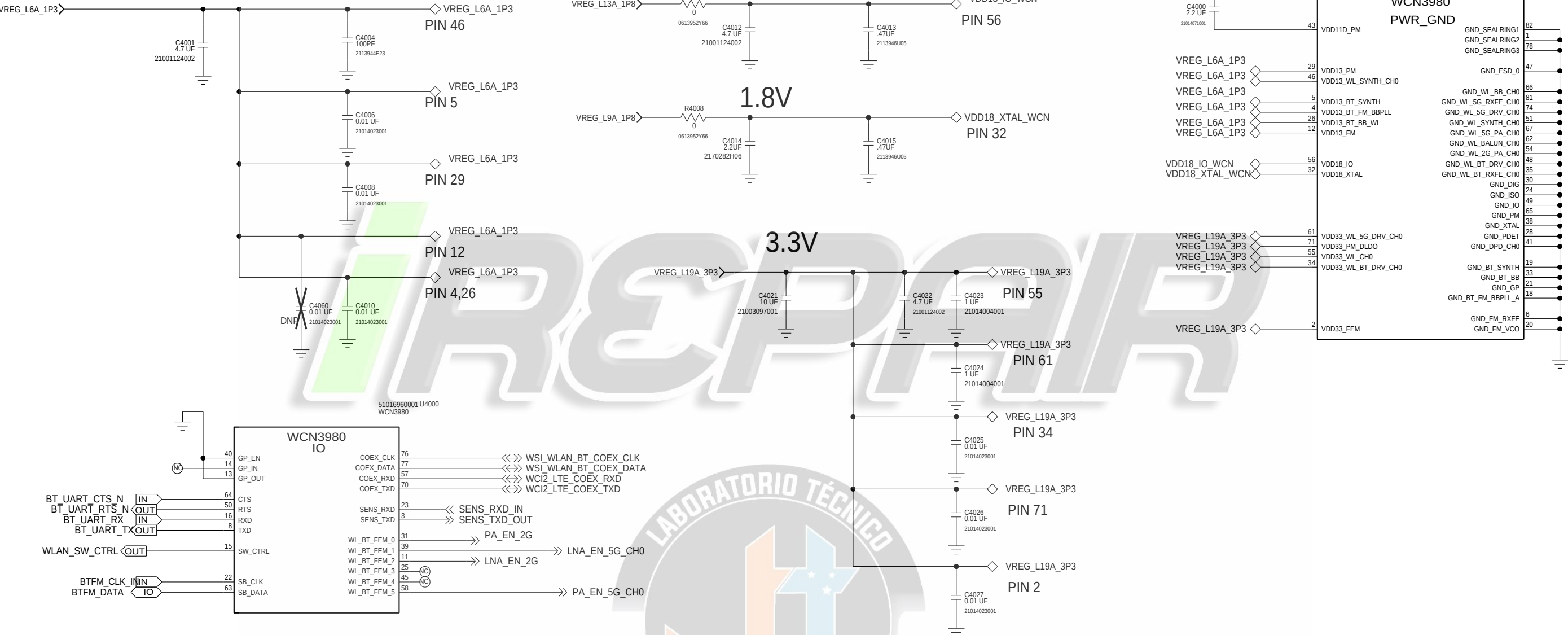
APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Fri Mar 16 13:55:30 2018			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME 8401672201_a_180402.cpm	
		CAD FILE NAME: PAGE 18 OF 70	

REF: 4000-4099

PAGE 19 OF 70

CONN: WCN3980 POWER/GND
REF 4000-4029

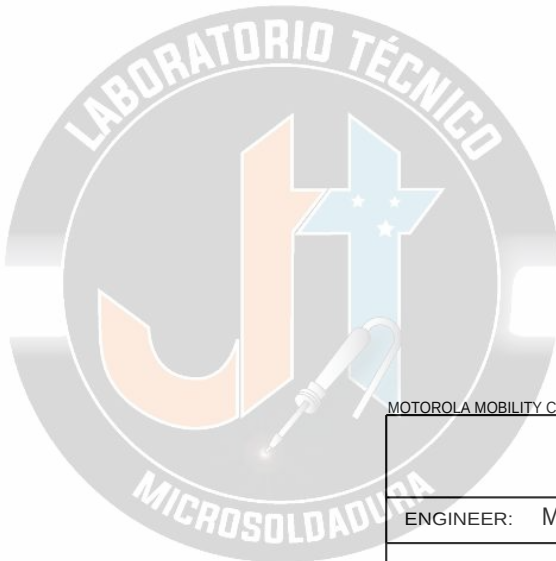
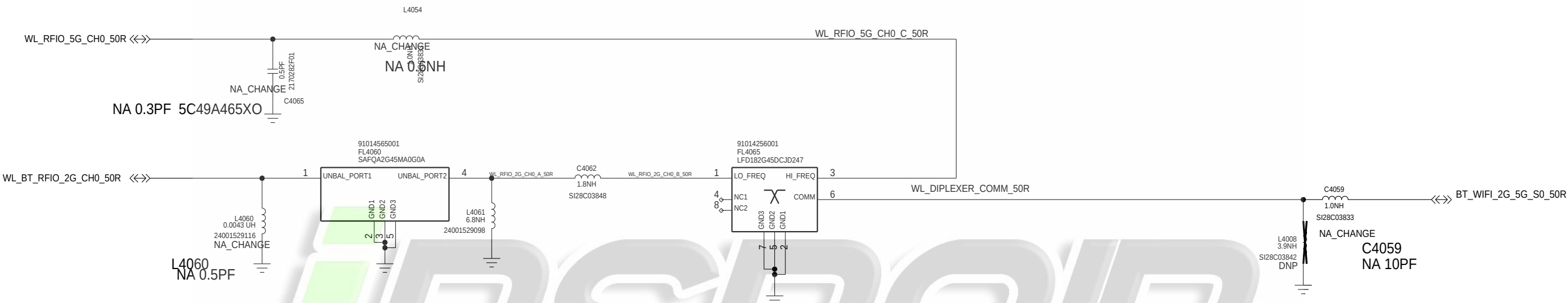
BT_FM_WLAN_1P3



BASED ON 80-WL020-41 REV.N

APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		connectivity	
LAST MODIFIED BY: MOTOROLA Fri Mar 16 13:55:38 2018		PROJECT NAME	
MOTOROLA CONFIDENTIAL RESTRICTED		REV	
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CONN: WCN3980 RADIO
REF 4040-4079

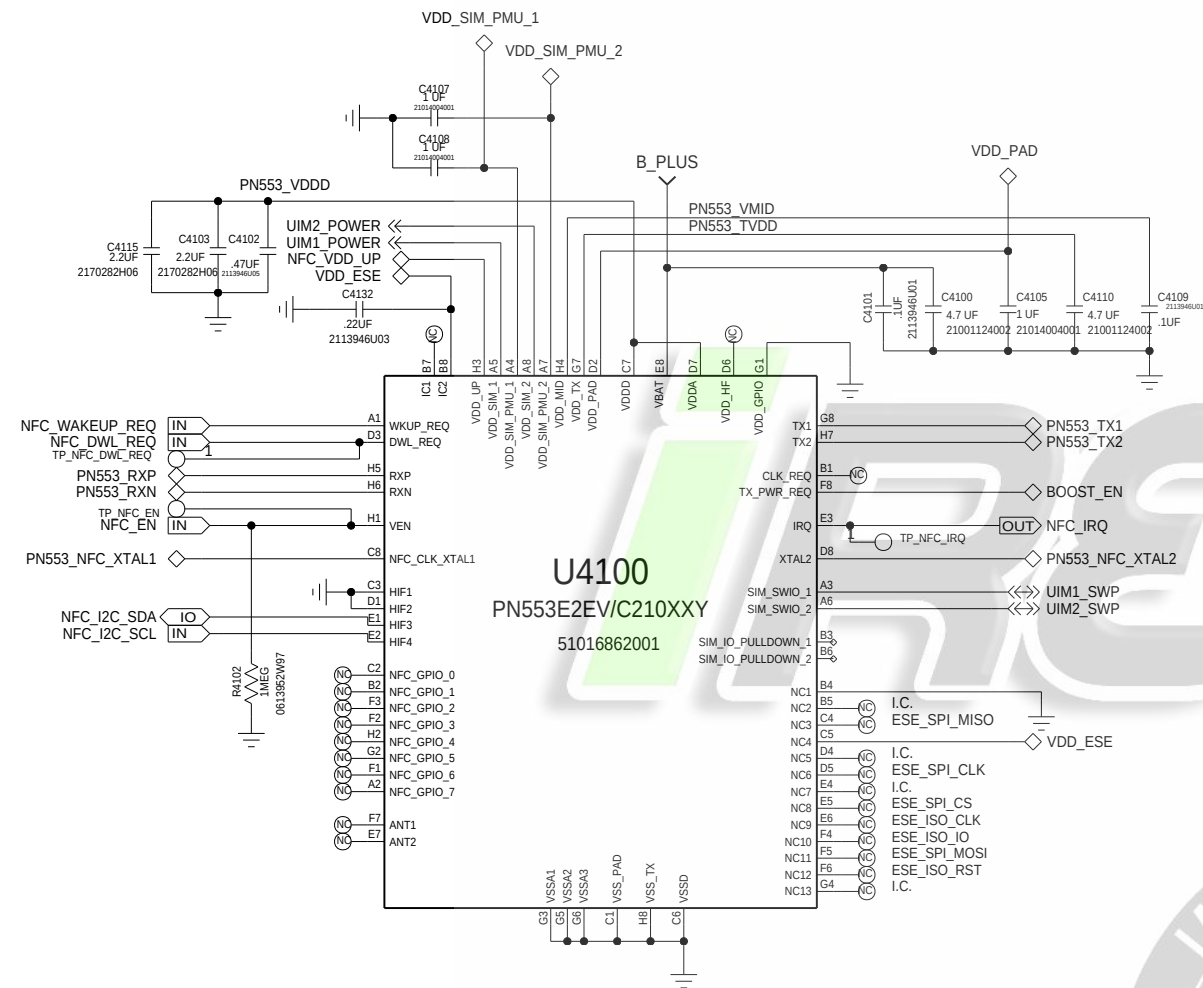


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APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: connectivity	
LAST MODIFIED BY: MOTOROLA Fri Mar 16 13:55:42 2018			
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		CAD FILE NAME: PAGE 21 OF 70	

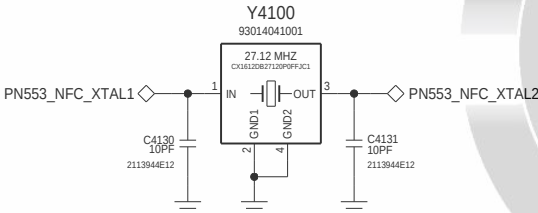
CONN: PN553 NFC

REF: 4100-4199

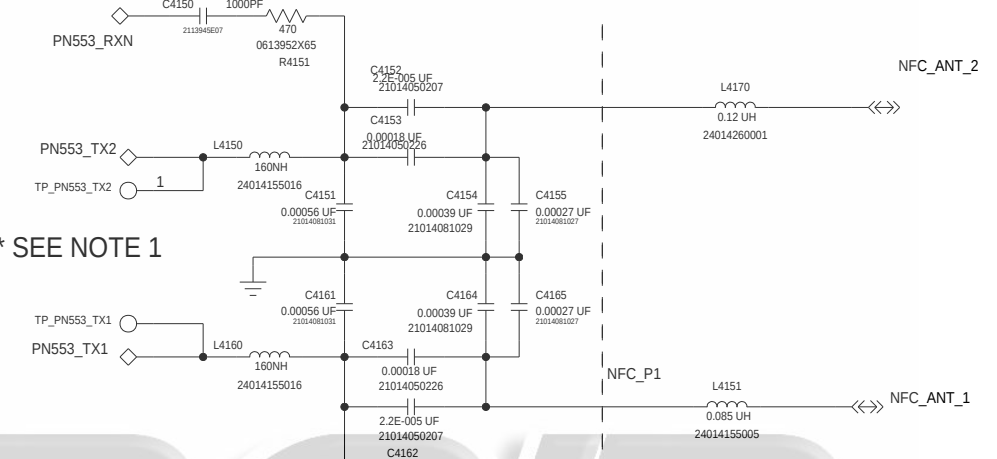


PN553 I2C ADDRESS MAP

HIF2	HIF1	BASE	WRITE	READ
LOW	LOW	0X28	0X50	0X51
LOW	HIGH	0X29	0X52	0X53
HIGH	LOW	0X2A	0X54	0X55
HIGH	HIGH	0X2B	0X56	0X57



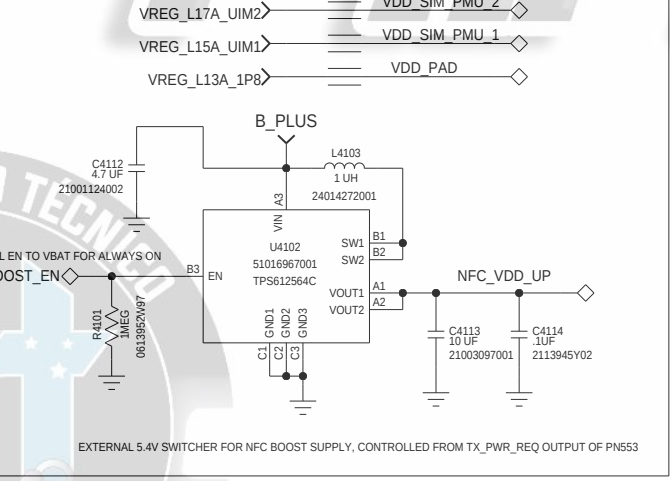
ANTENNA MATCHING



* SEE NOTE 1

NOTES:
1. L4150 AND L4160 PLACEMENT MUST MINIMIZE MUTUAL INDUCTIVE COUPLING, OR CIRCUIT'S WIRELESS FUNCTION WILL BE SEVERELY IMPAIRED.

POWER



APPROVALS

ENGINEER: MOTOROLA
LAST MODIFIED BY: MOTOROLA
Thu Mar 01 10:01:18 2018

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BLOCK TITLE:
connectivity
PROJECT NAME
8401672201_a_180402.cpm
REV
03

CAD FILE NAME:
PAGE 22 OF 70

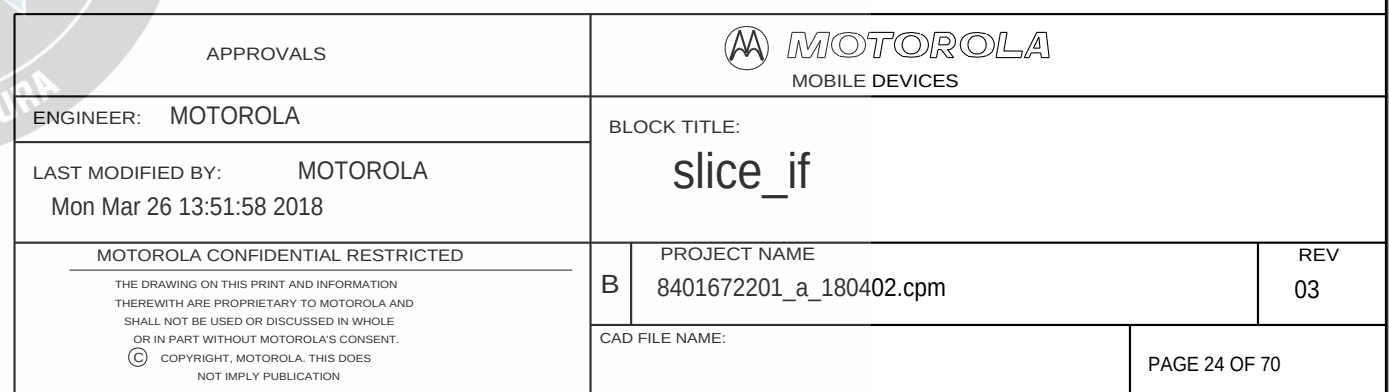
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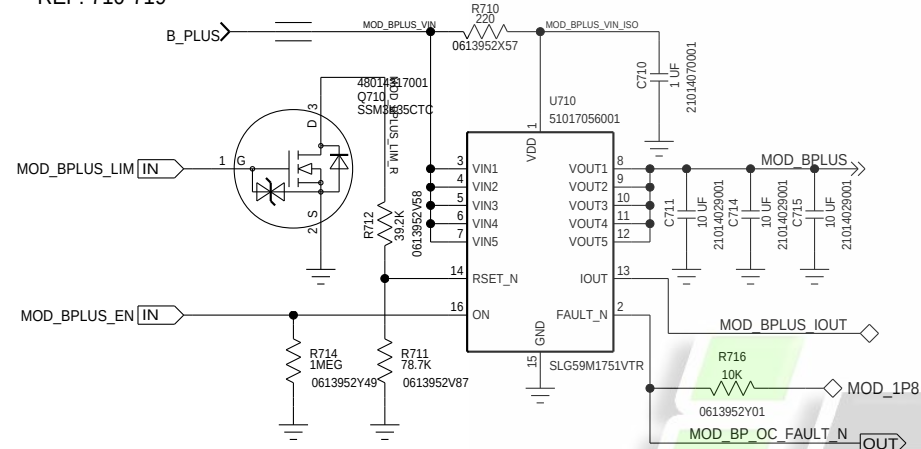
PAGE 23 OF 70

REF: 740-759

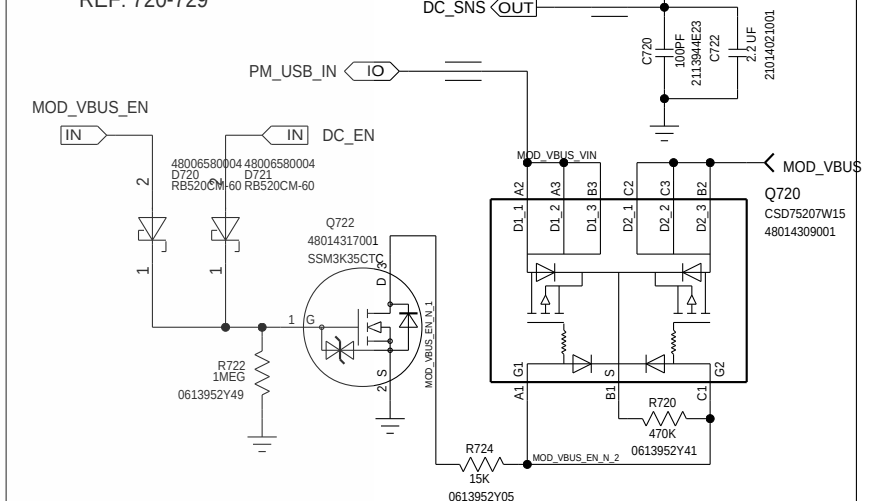


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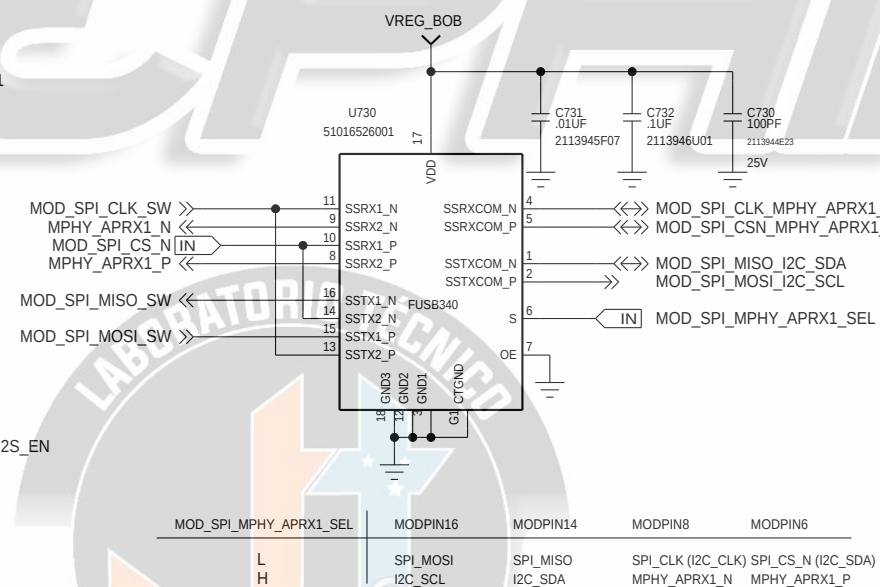
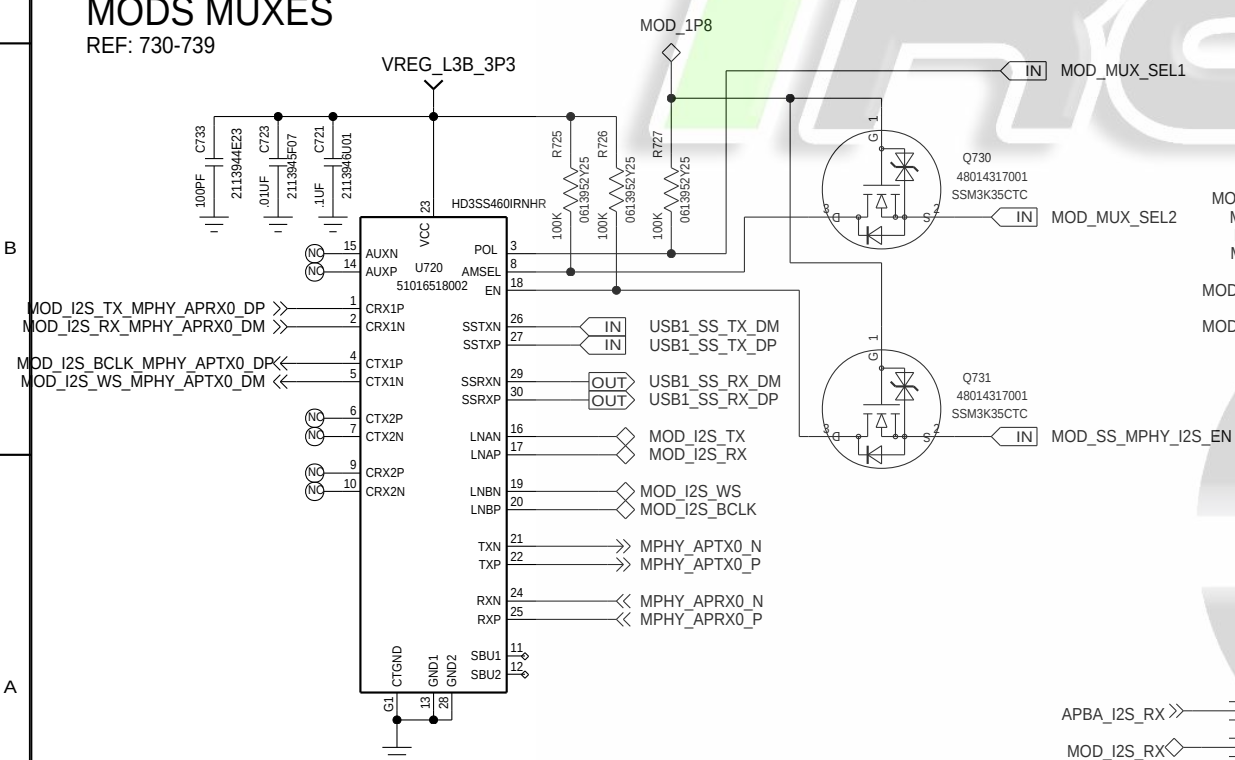
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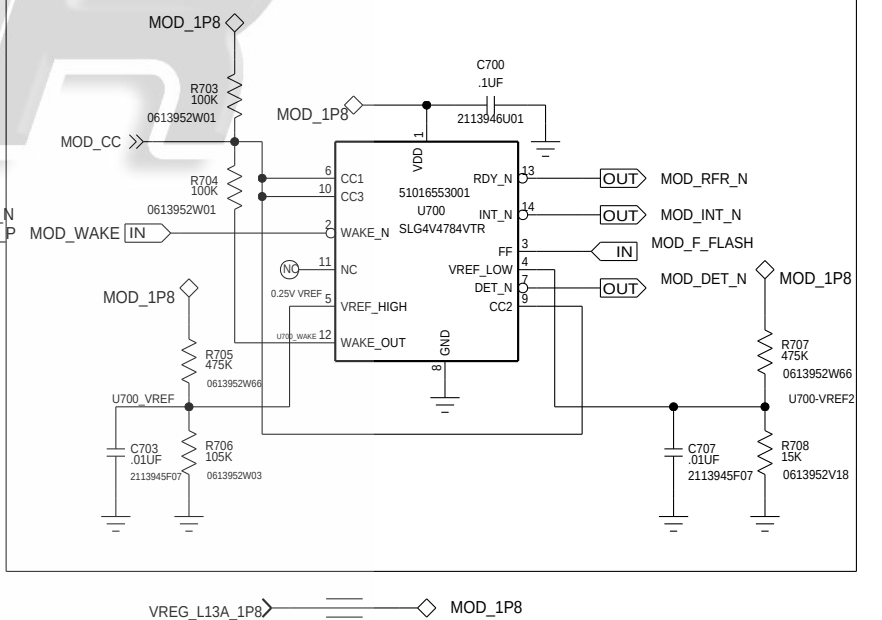
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REF: 730-739



REF: 700-709



MOD_SS_MPHY_I2S_EN	MOD_MPHY_SEL1	MOD_MPHY_SEL2	PATH
H	L	L	USB SS
H	L	H	MPHY
H	H	H	I2S
L	X	X	HIGH Z

Figure 10 illustrates the pin connections for the CDC_I2S module. The connections are as follows:

- APBA_I2S_RX and MOD_I2S_RX are connected to the IN pin of CDC_I2S_TX_MOD_RX.
- APBA_I2S_WS and MOD_I2S_WS are connected to the IN pin of CDC_I2S_WS_MOD.
- APBA_I2S_BCLK and MOD_I2S_BCLK are connected to the IN pin of CDC_I2S_BCLK_MOD.
- APBA_I2S_TX and MOD_I2S_TX are connected to the OUT pin of CDC_I2S_RX_MOD_TX.

ENGINEER: MOTOROLA

MOTOROLA CONFIDENTIAL RESTRICTED


MOTOROLA
 MOBILE DEVICES

BLOCK TITLE:

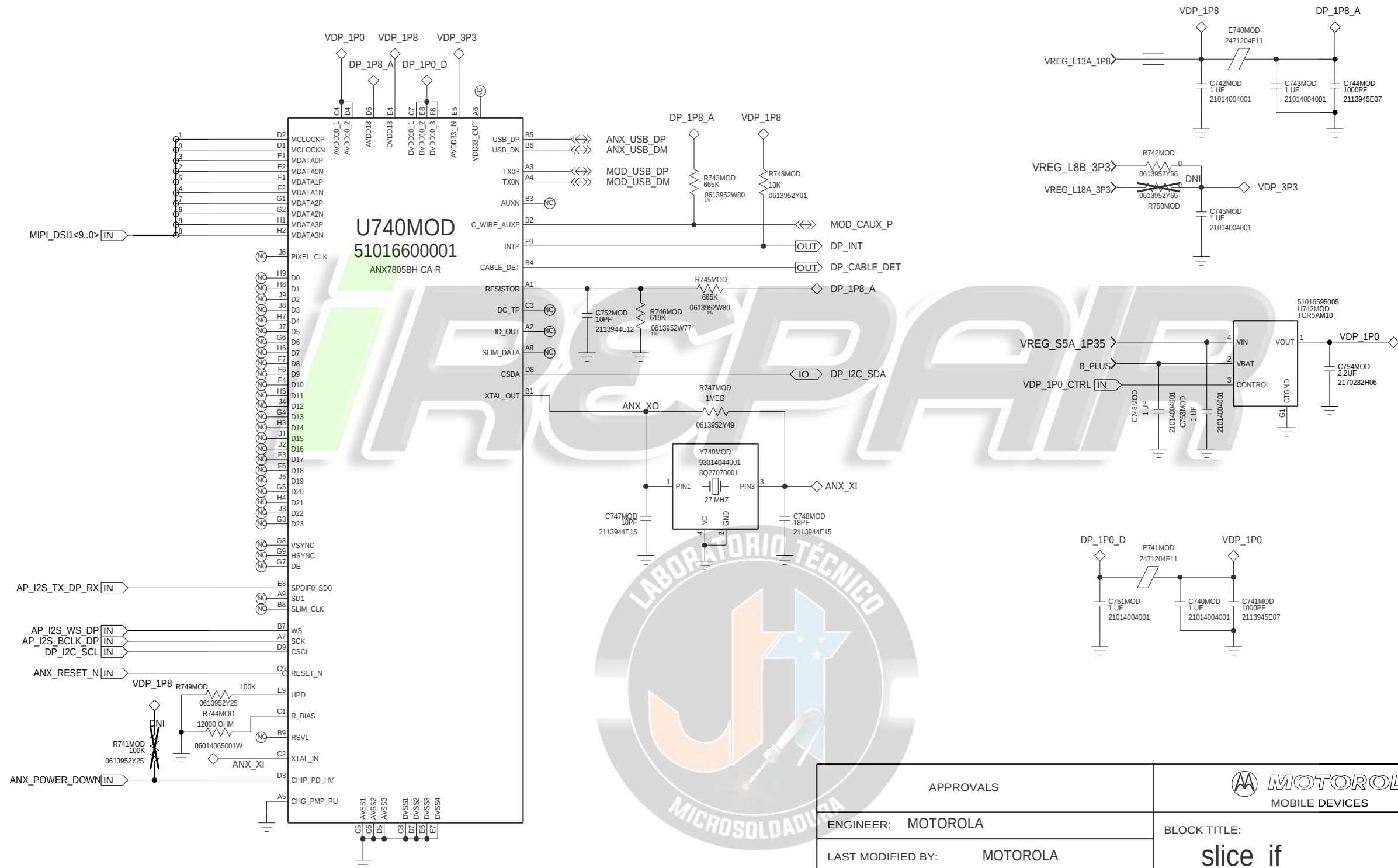
slice_if

REV
03

CAD FILE NAME:

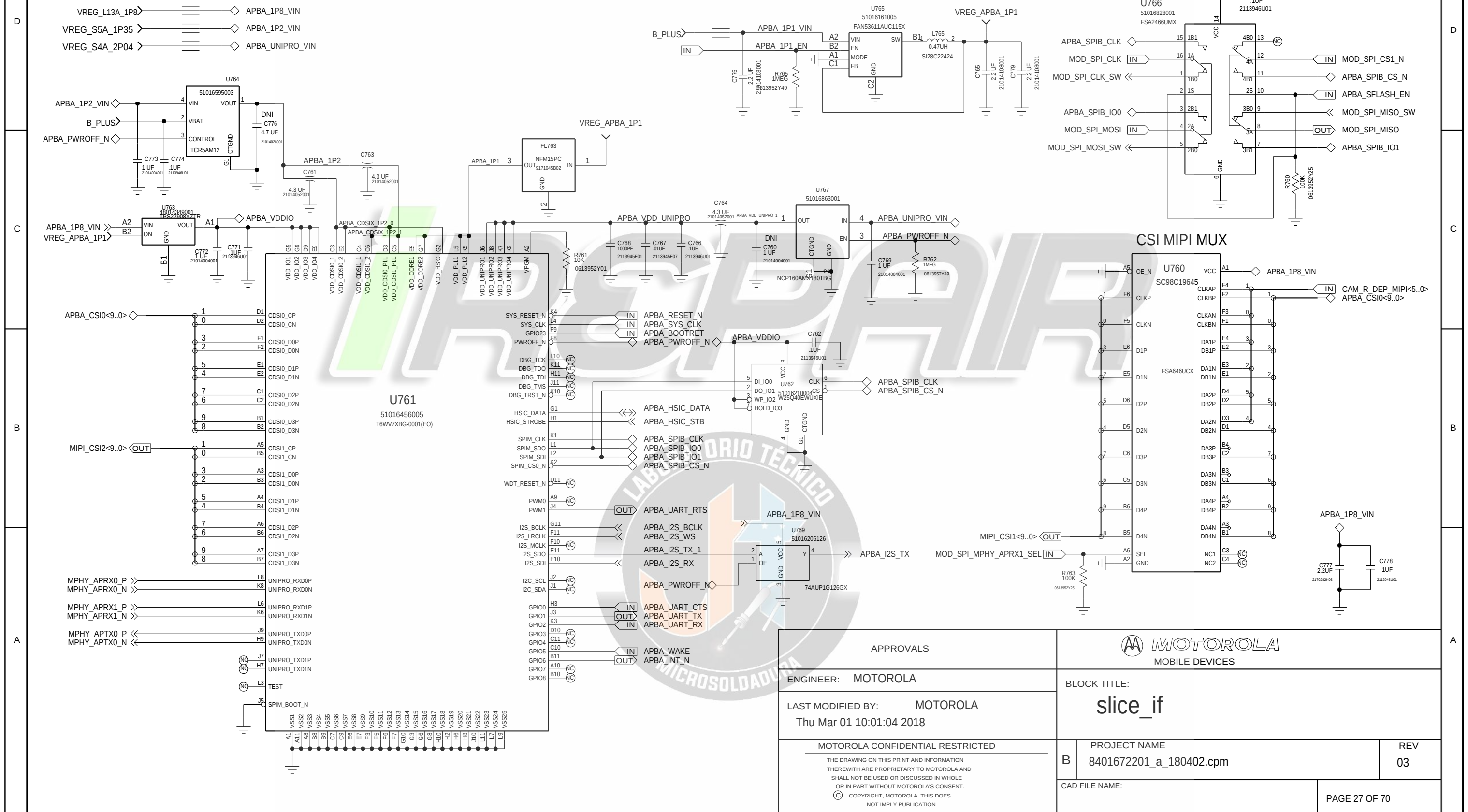
PAGE 25 OF 70

MODS: DRM INTERFACE



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: slice_if	
LAST MODIFIED BY: MOTOROLA Fri Mar 02 09:48:24 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 03	
		CAD FILE NAME:	
		PAGE 26 OF 70	

REF: 760-779



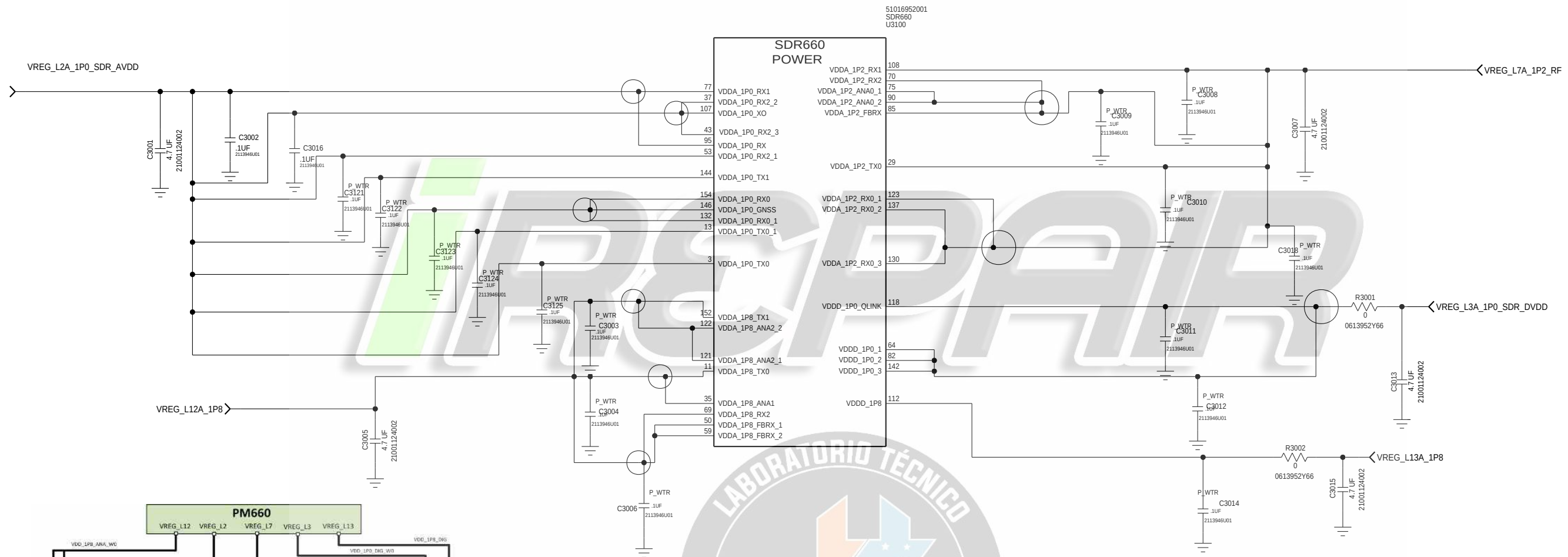
REFDES: 1250-1279



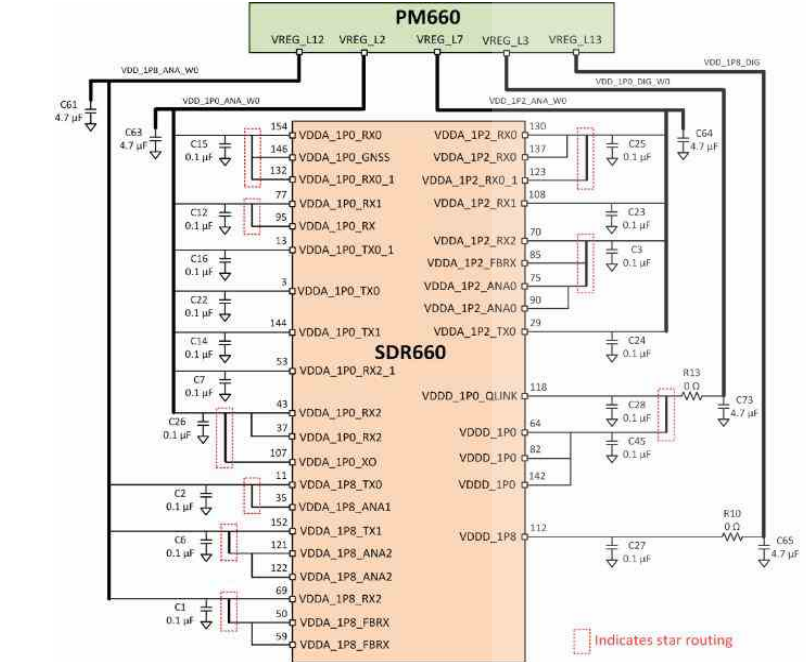
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ENGINEER: MOTOROLA		BLOCK TITLE: slice_if	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:04 2018			
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CAD FILE NAME:			PAGE 28 OF 70

SDR660: POWER

REF 3000-3099

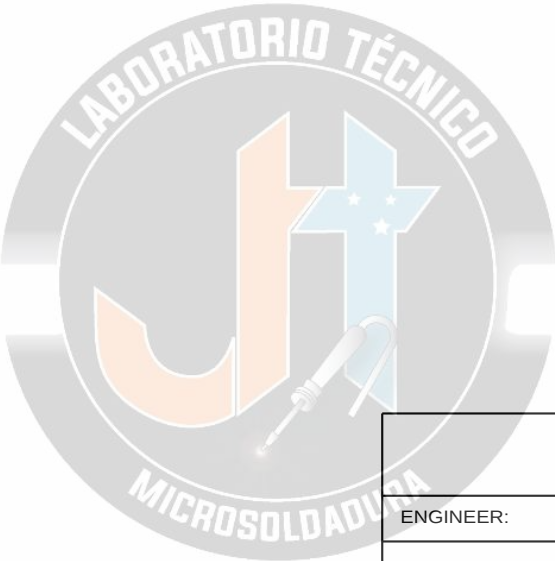
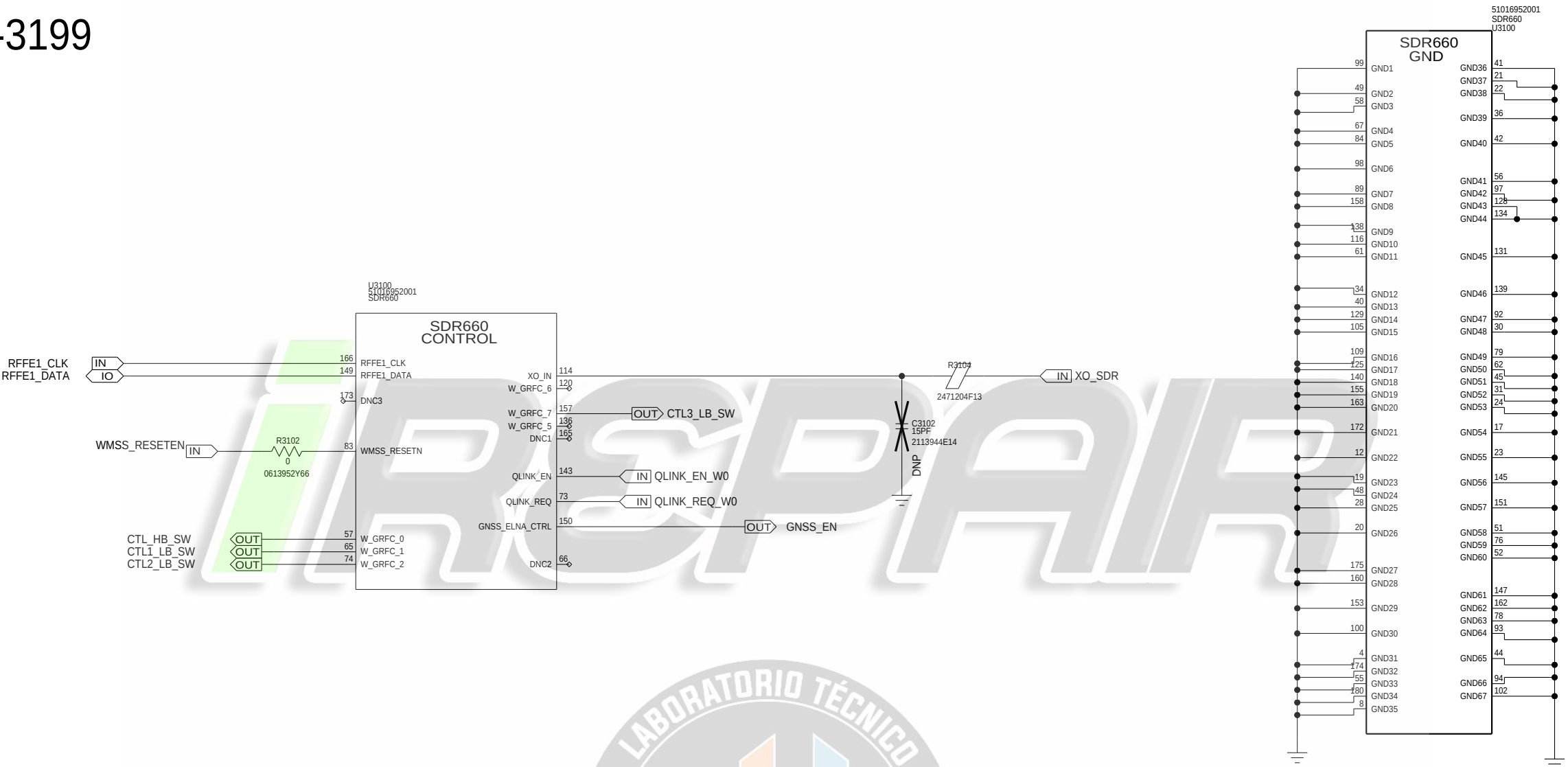


CIRCLE PRESENT SDR660 POWER SUPPLY STAR ROUTING



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: blk_sdr660	
LAST MODIFIED BY: Thu Mar 01 10:01:00 2018		PROJECT NAME B 800167201_a_180402.cpm	
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CAD FILE NAME:		PAGE 29 OF 70	

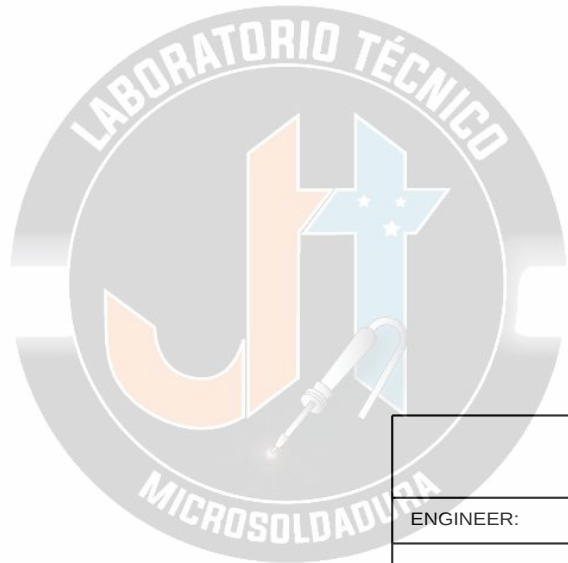
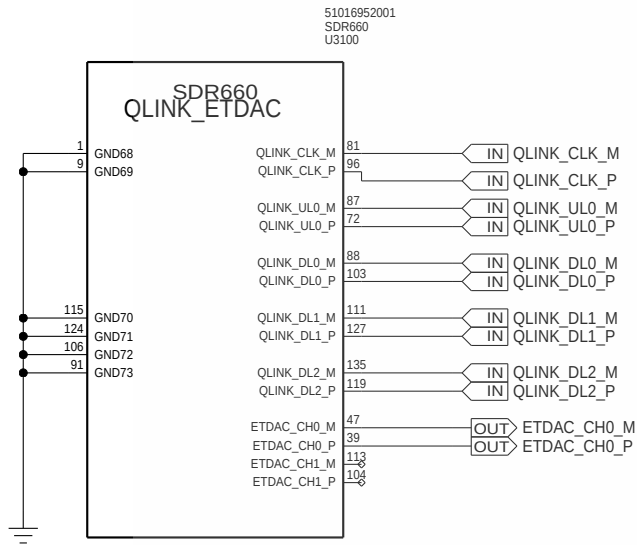
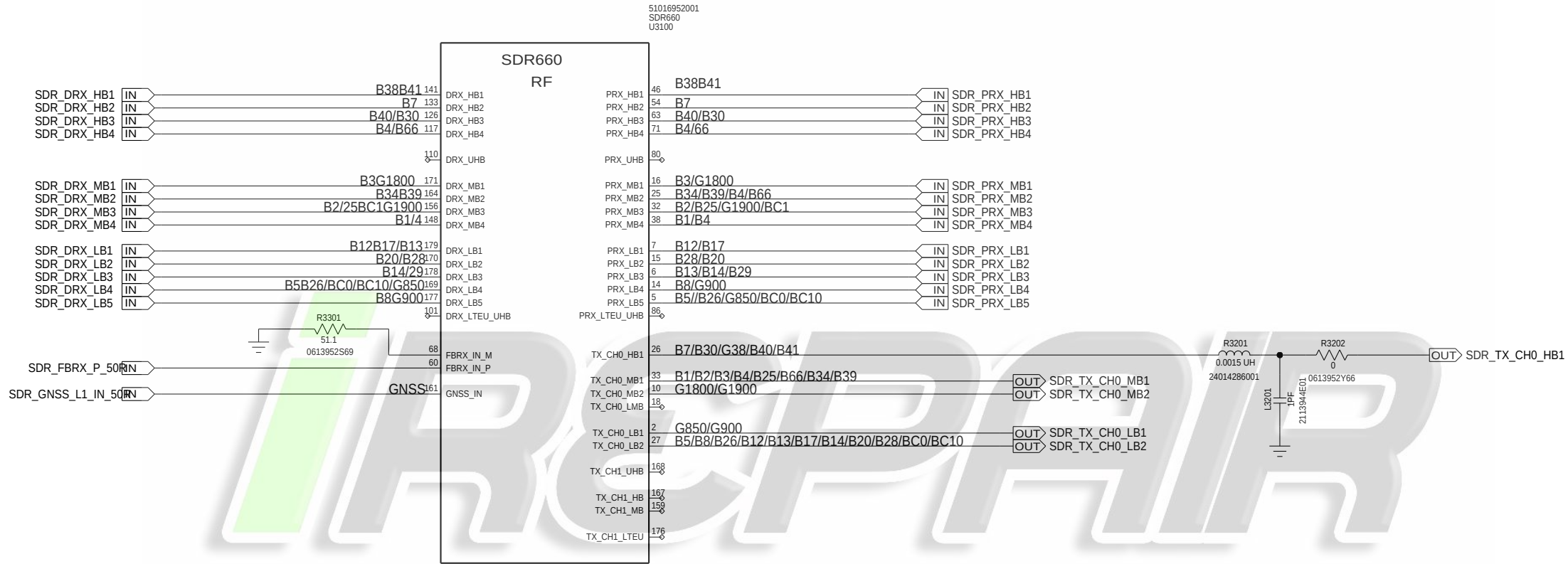
SDR660: TX/PWR/GROUND
REF 3100-3199



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Fri Mar 09 11:08:23 2018		blk_sdr660	
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		CAD FILE NAME: PAGE 30 OF 70	

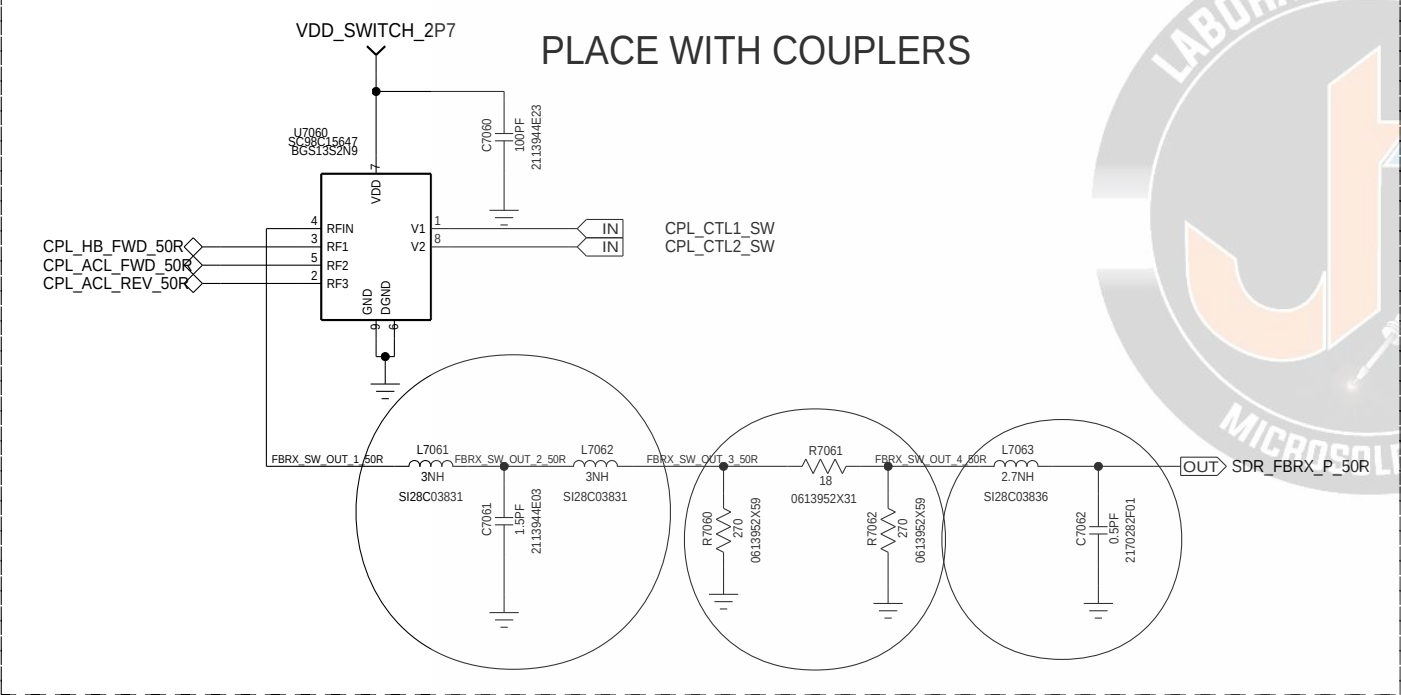
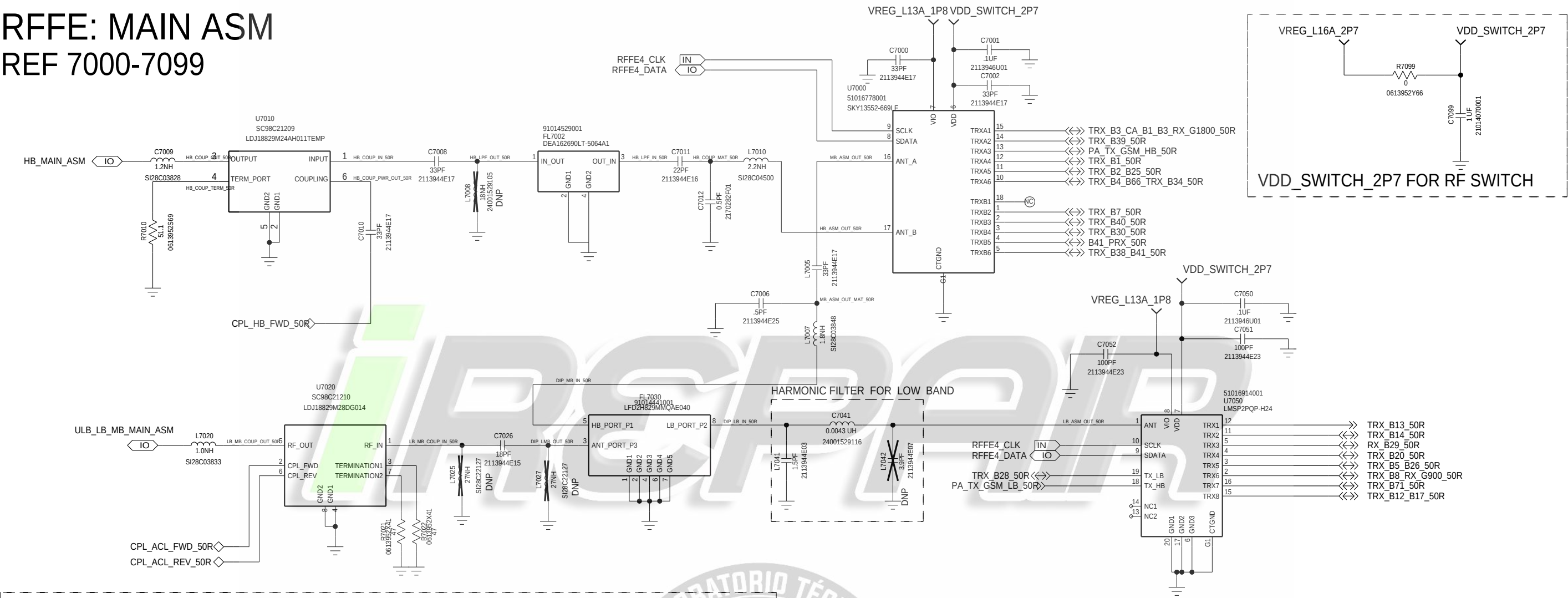
SDR660: PRX/DRX/GPS

REF 3200-3299



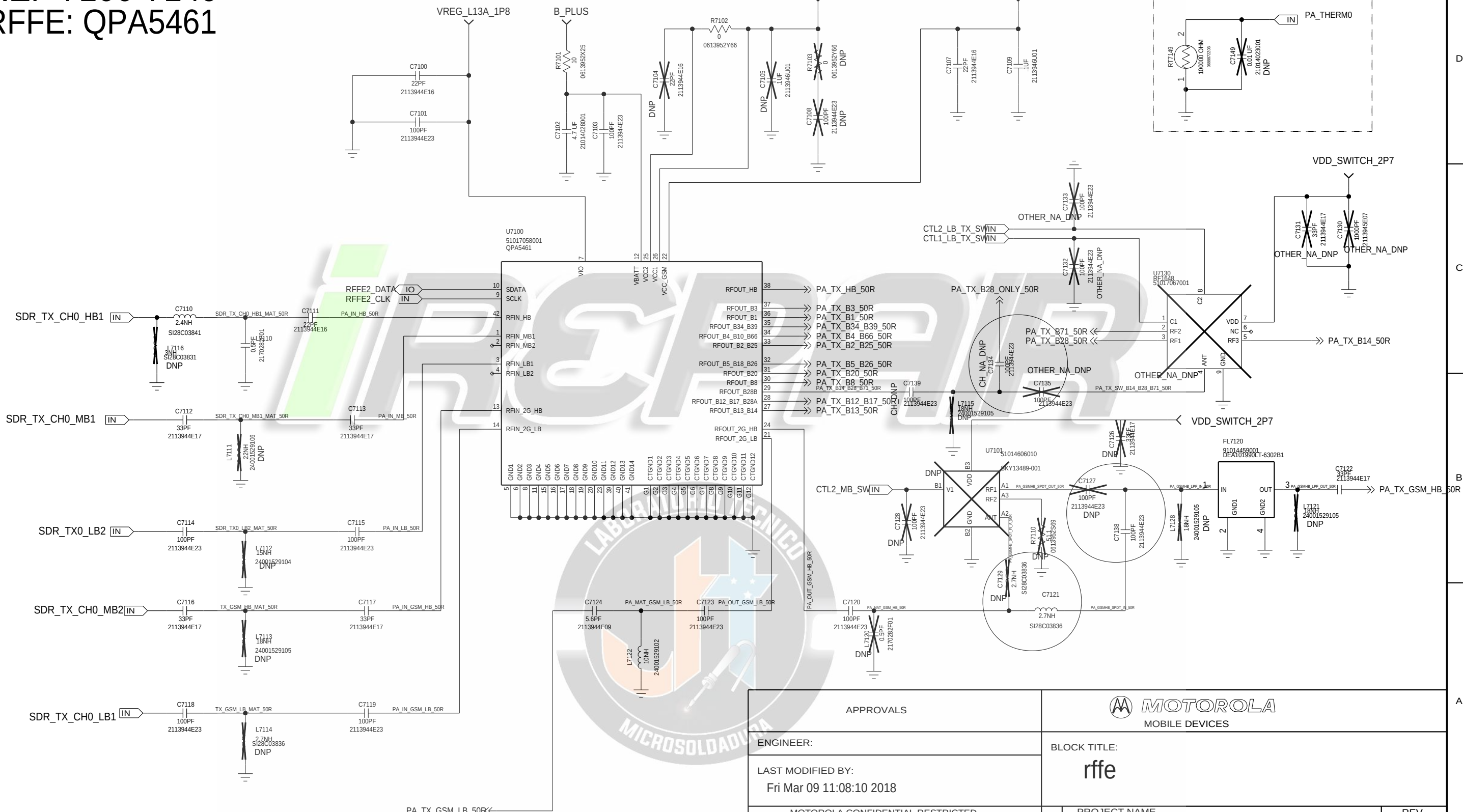
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LAST MODIFIED BY: Thu Mar 01 10:01:01 2018		PROJECT NAME B 8001672201_a_180402.cpm	
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RFFE: MAIN ASM
REF 7000-7099



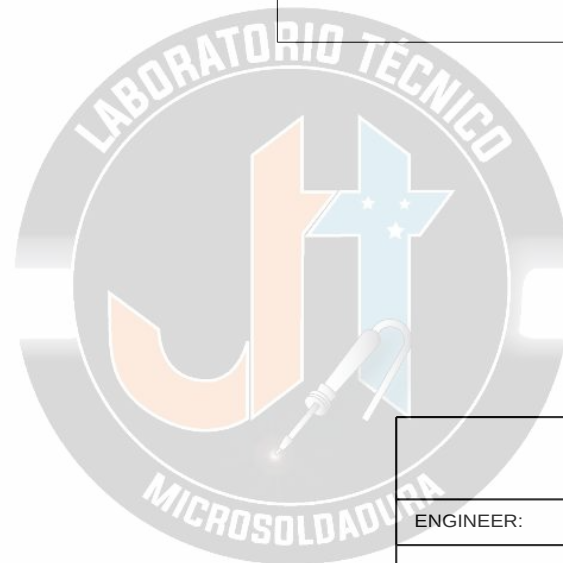
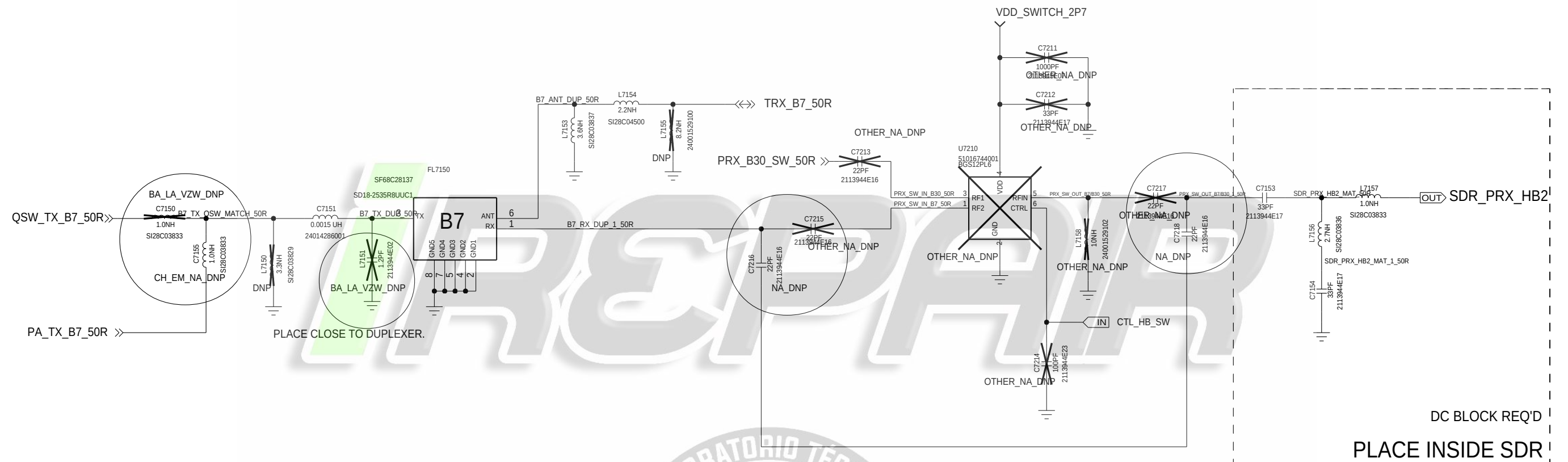
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LAST MODIFIED BY: Thu Mar 01 10:00:52 2018		PROJECT NAME B 8401672201_a_180402.cpm	
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REF 7100-7149
RFFE: QPA5461



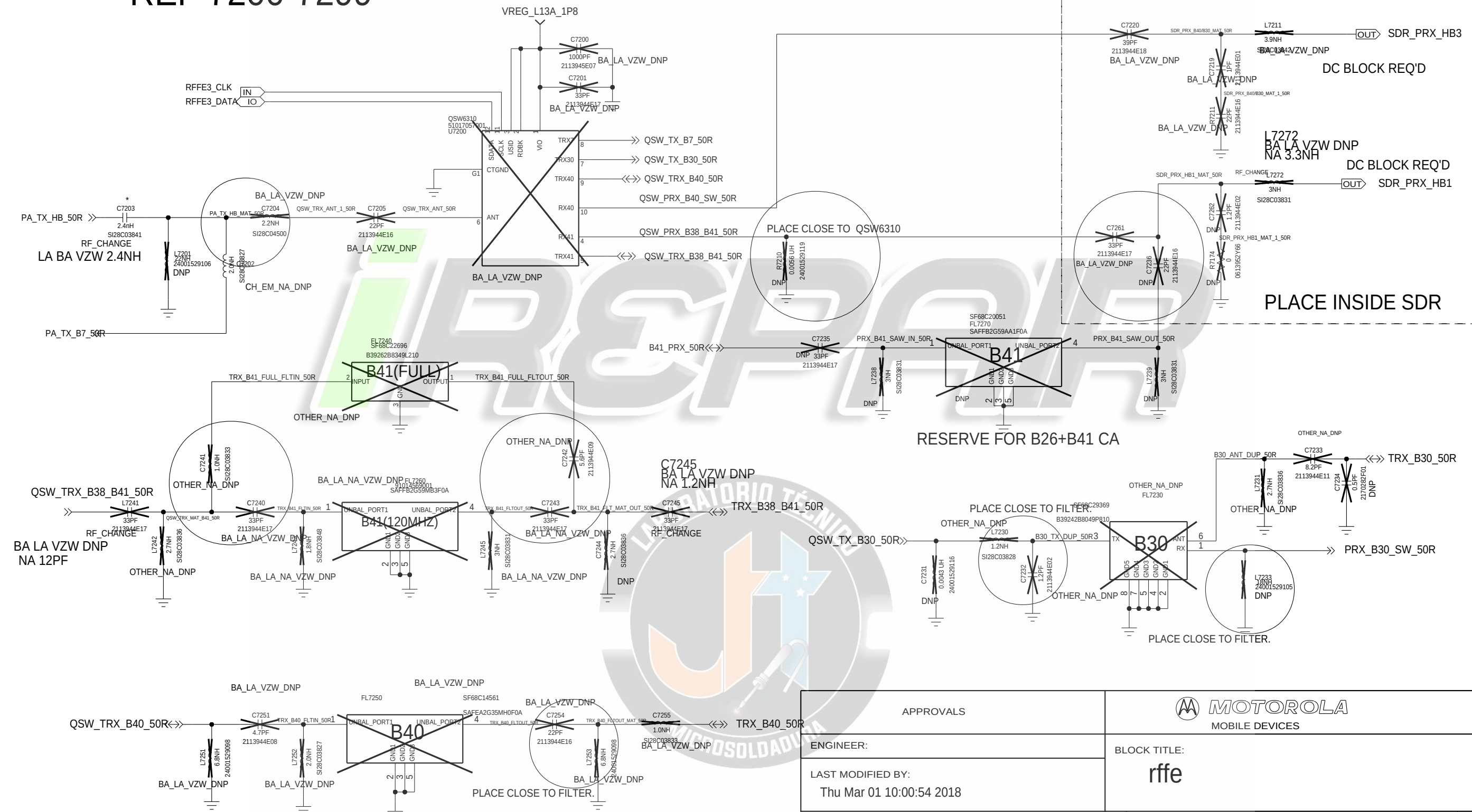
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ENGINEER:		rffe	
LAST MODIFIED BY:		PROJECT NAME	
Fri Mar 09 11:08:10 2018		B 8401672201_a_180402.cpm	
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CAD FILE NAME:		PAGE 33 OF 70	

RFFE: HB TRX B7
REF 7150-7199



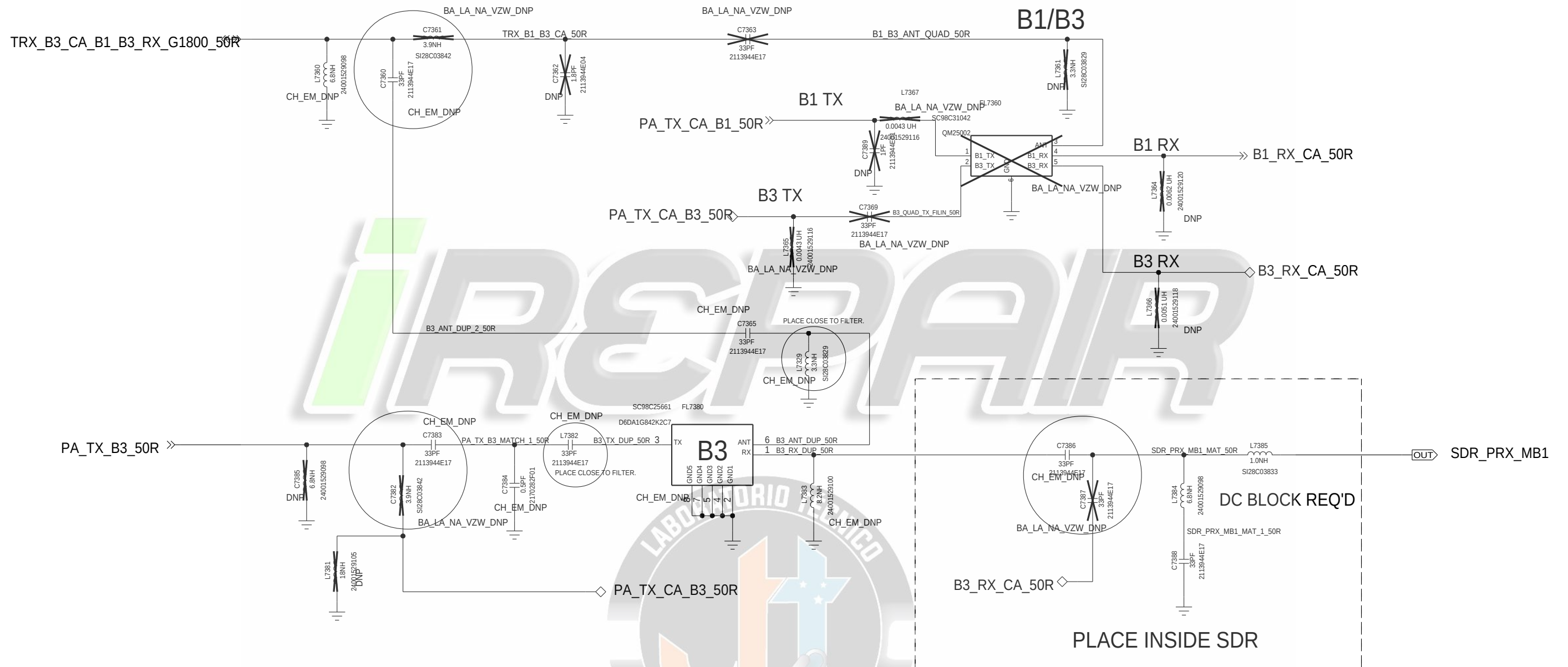
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Mar 01 10:00:54 2018		rffe	
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CAD FILE NAME:			PAGE 34 OF 70

RFFE: HB_TRX_B30,38,40,41
REF 7200-7299



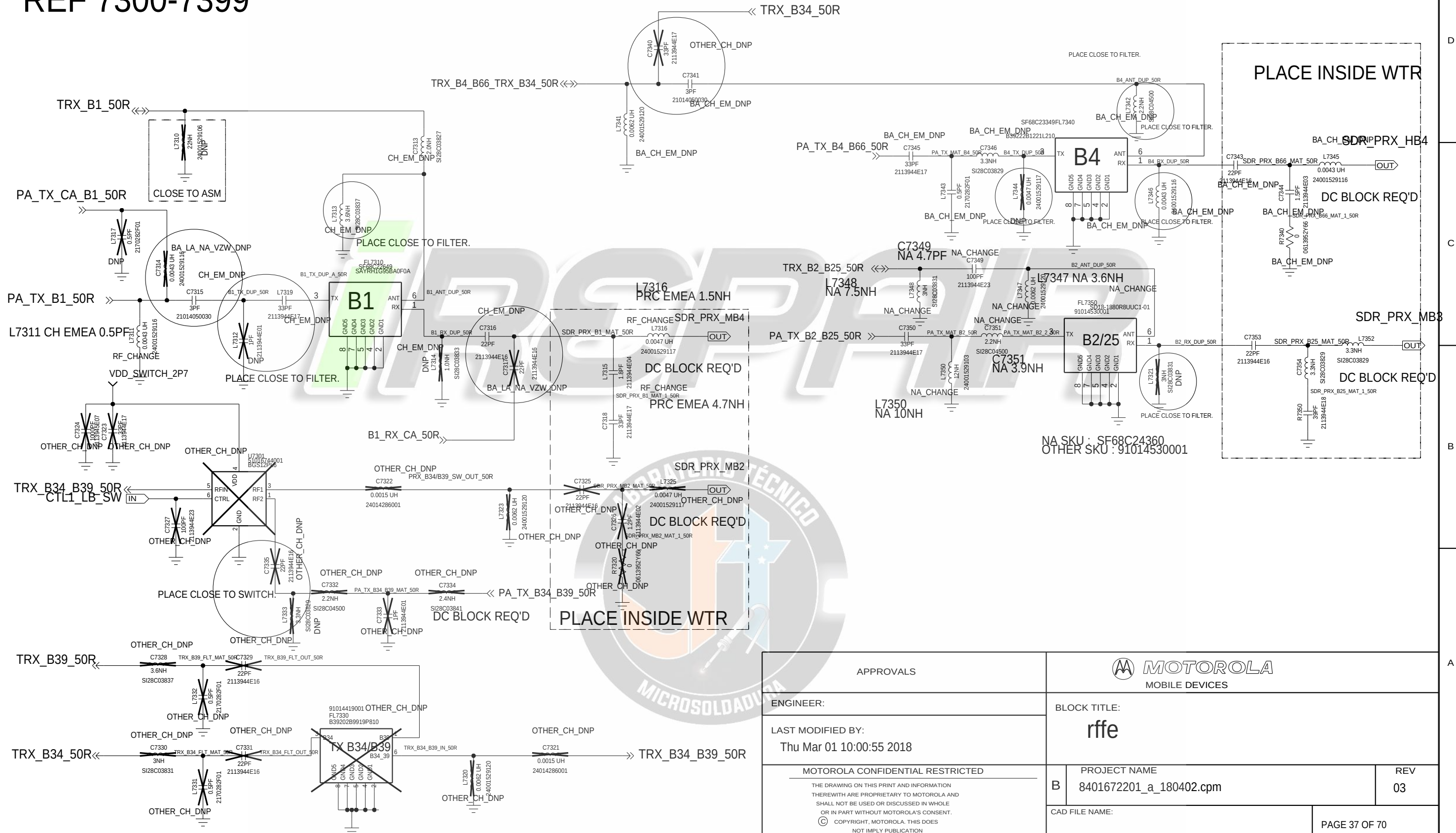
APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:	BLOCK TITLE: rffe			
LAST MODIFIED BY: Thu Mar 01 10:00:54 2018				
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME		REV
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		CAD FILE NAME:		PAGE 35 OF 70

RFFE: MB_TRX_B3_B1/3 Quad
REF 7350-7399

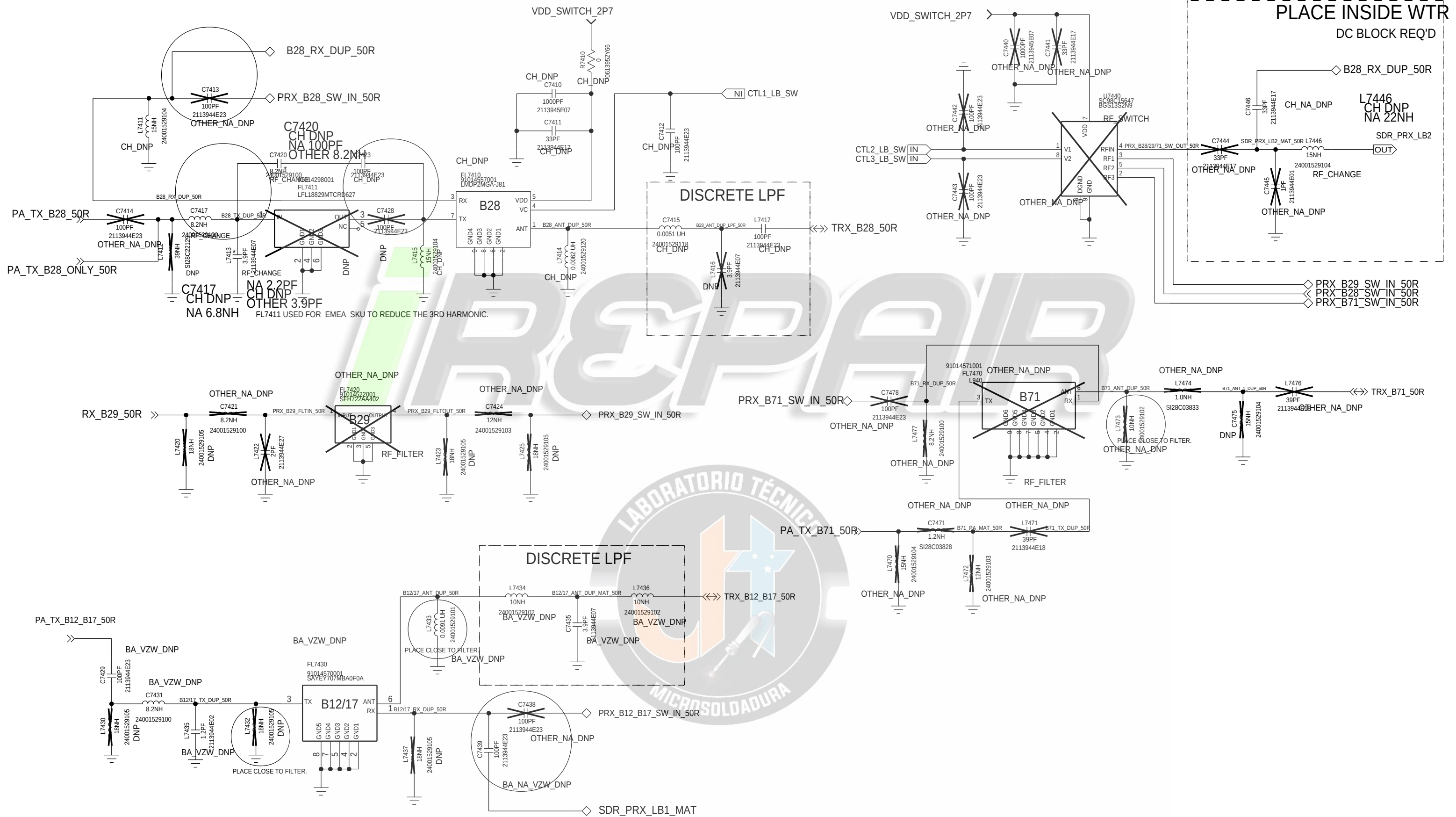


APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:	BLOCK TITLE: rffe			
LAST MODIFIED BY: Thu Mar 01 14:18:14 2018				
MOTOROLA CONFIDENTIAL RESTRICTED		B	PROJECT NAME 8401672201_a_180402.cpm	REV 03
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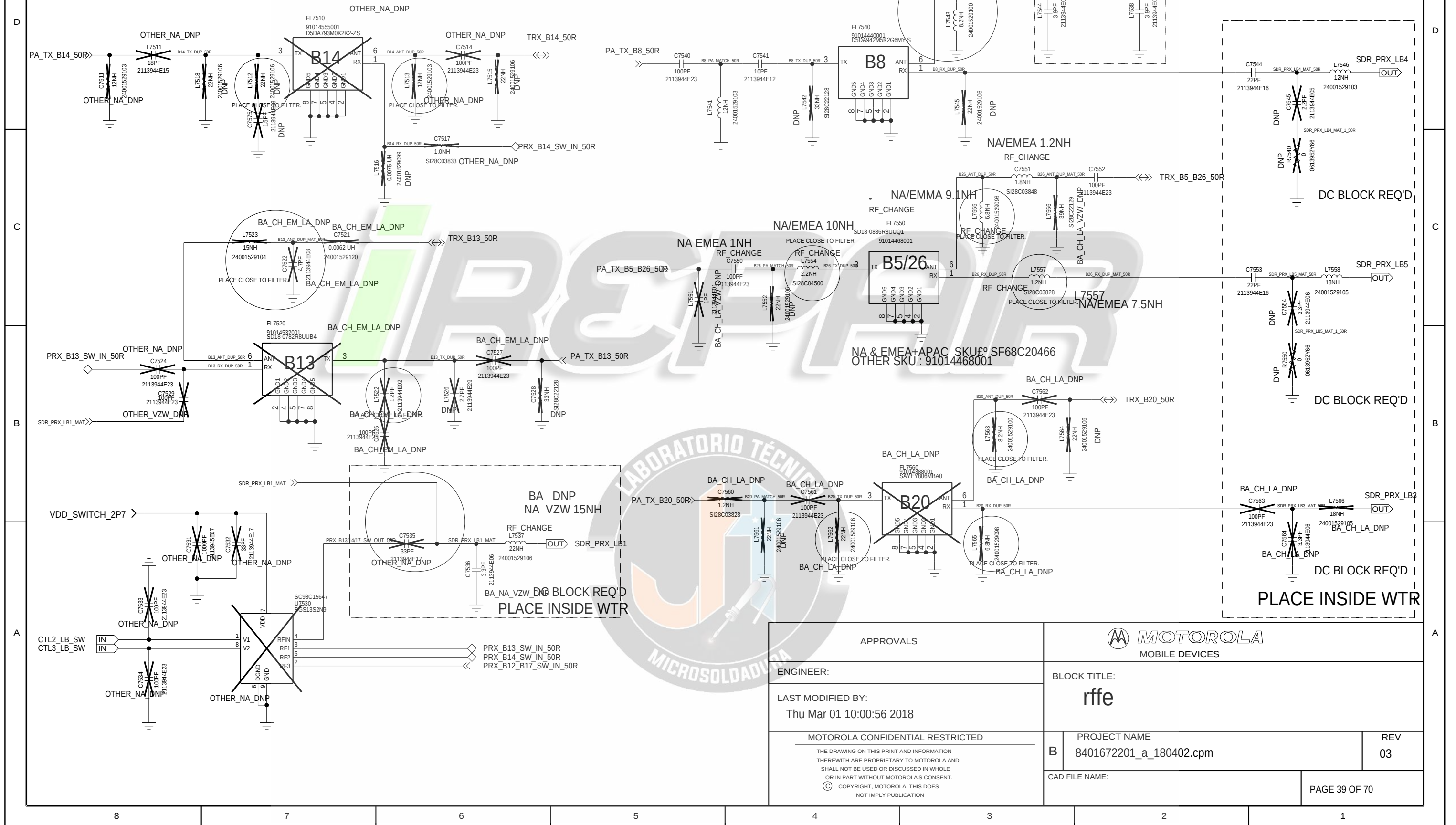
RFFE : MB_TRX_B1,2,3,4,34,39,GSM HB
REF 7300-7399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe		
LAST MODIFIED BY: Thu Mar 01 10:00:55 2018			
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		CAD FILE NAME:	PAGE 37 OF 70

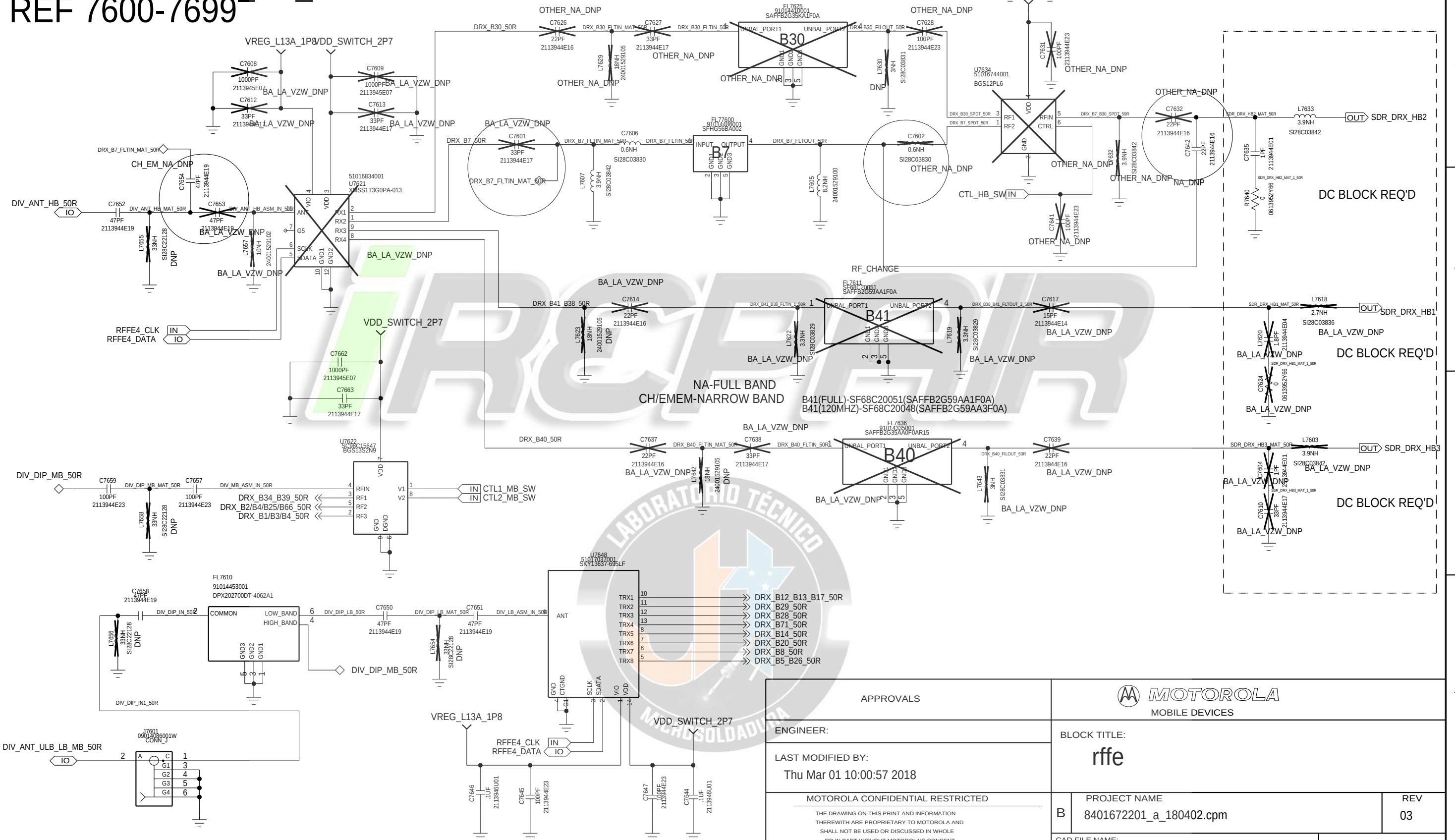


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REF 7500-7599



RFFE: DRX ASM_HB_B7,30,38,40,41

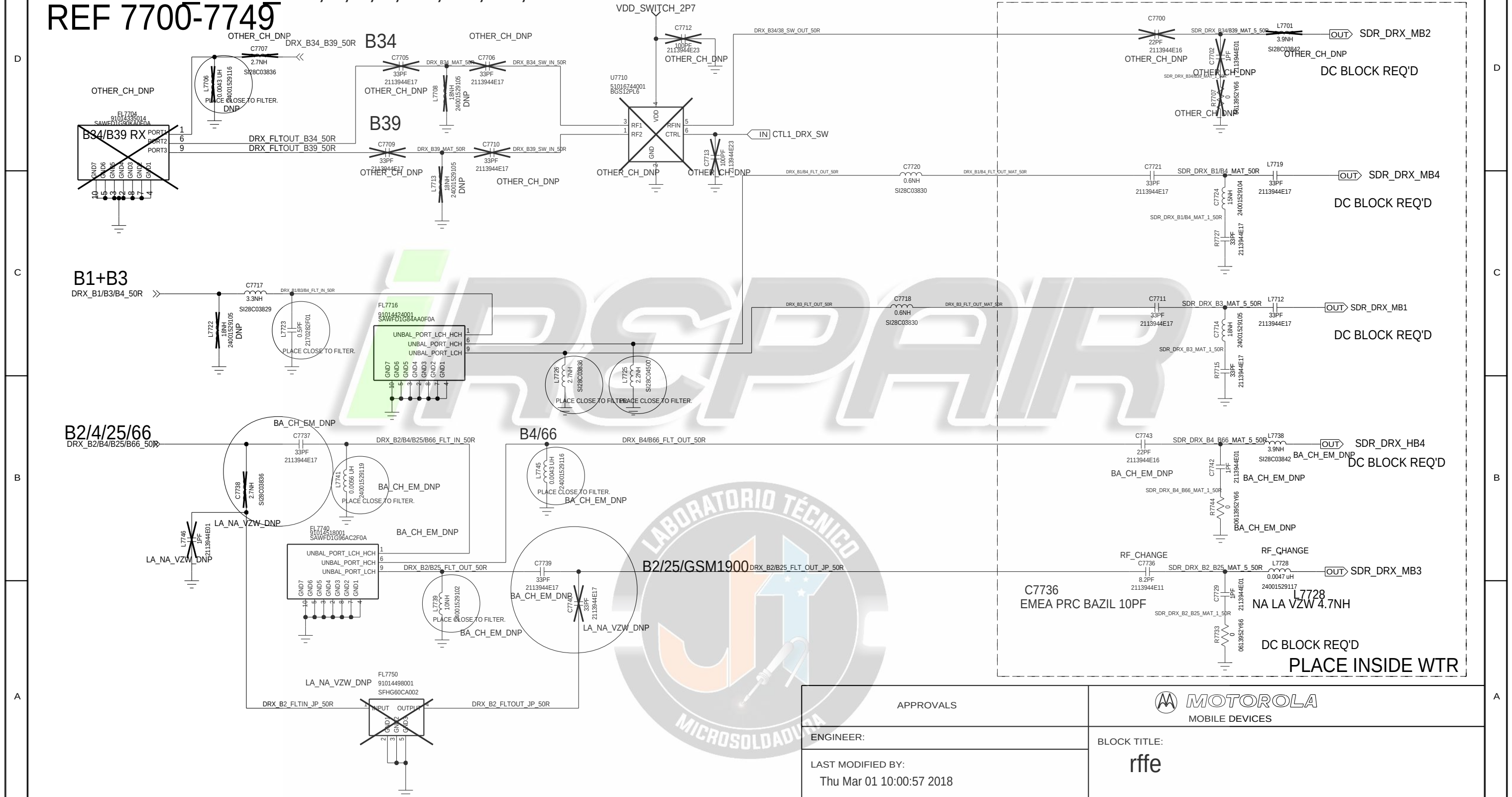
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RFFE: MB DRX B1,2,3,4,25,34,39,66

REF 7700-7749

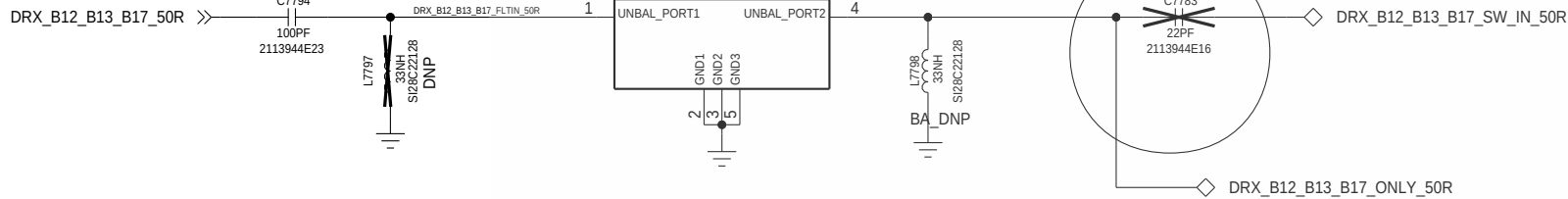


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ENGINEER:		rffe	
LAST MODIFIED BY:		PROJECT NAME	
Thu Mar 01 10:00:57 2018		B 8401672201_a_180402.cpm	
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CAD FILE NAME:		PAGE 41 OF 70	

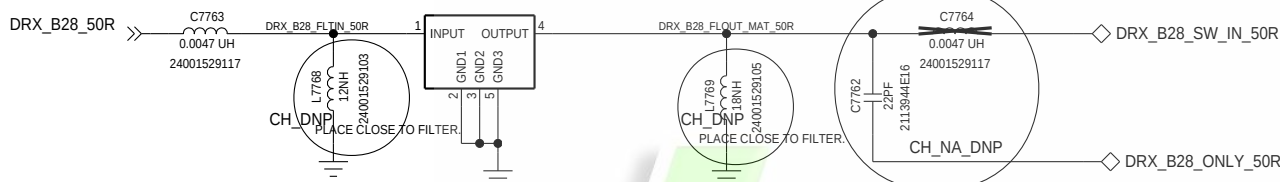
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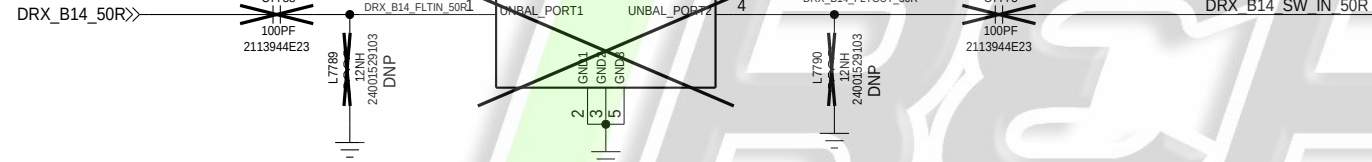
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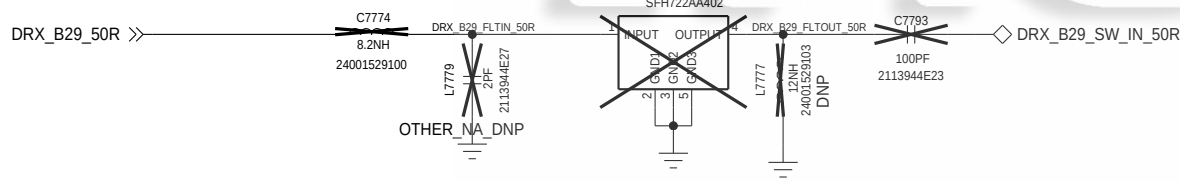
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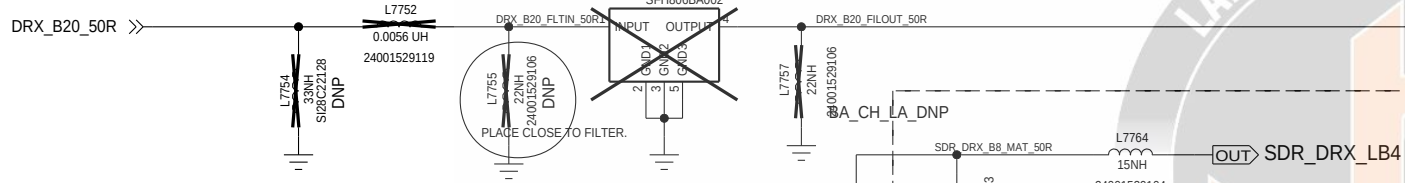
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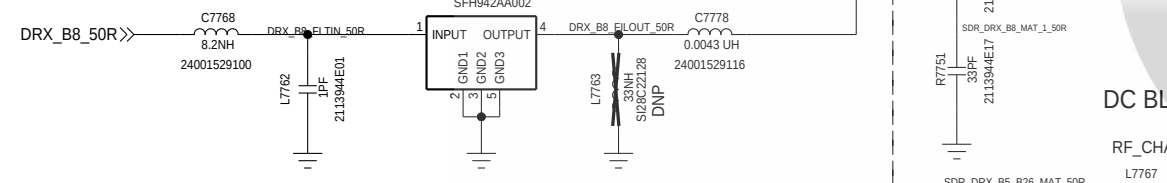
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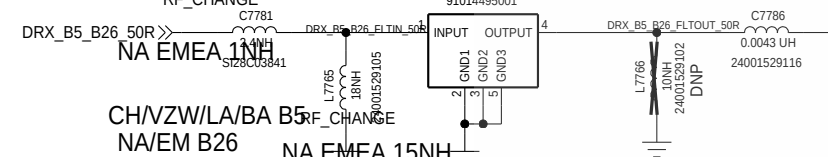
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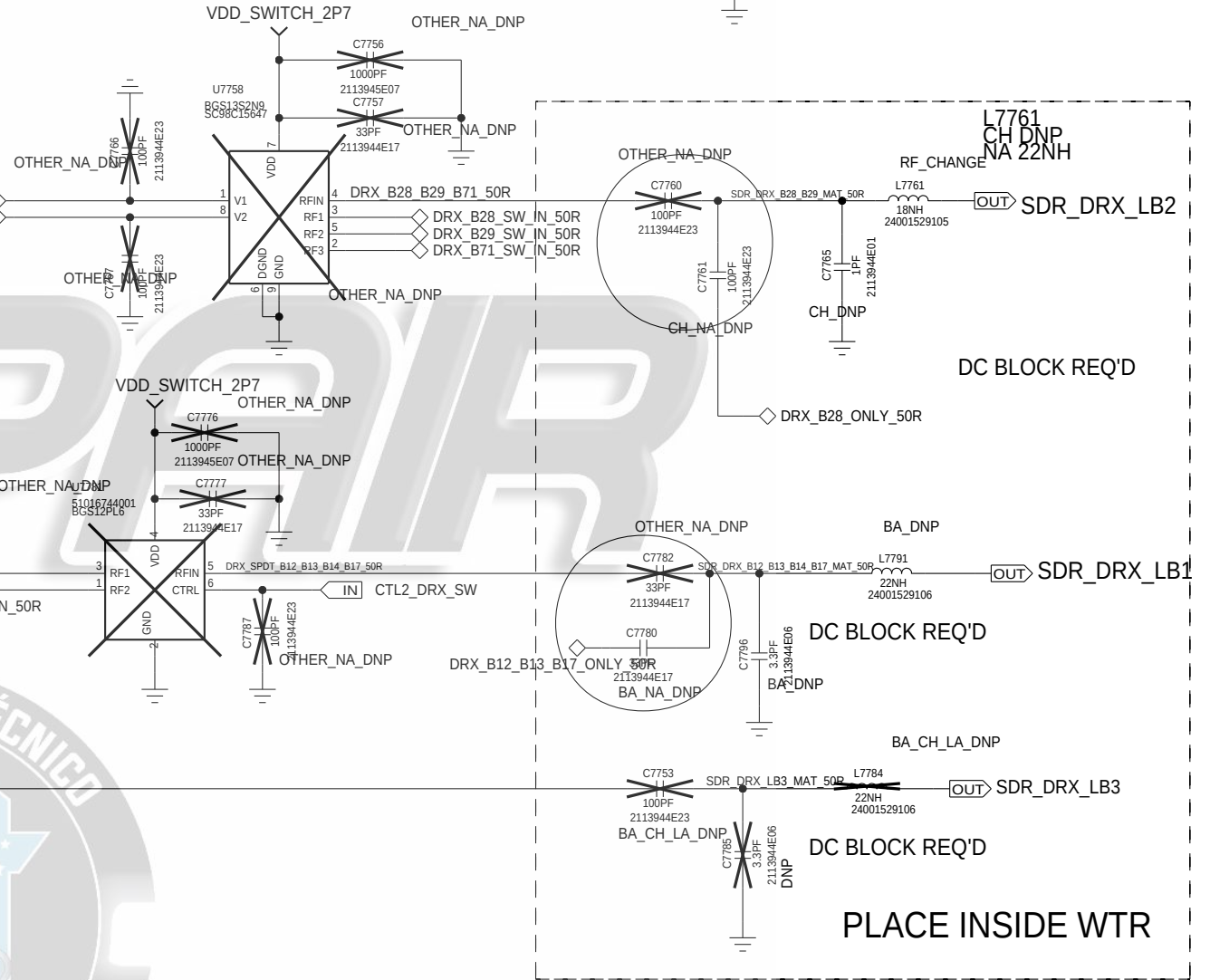
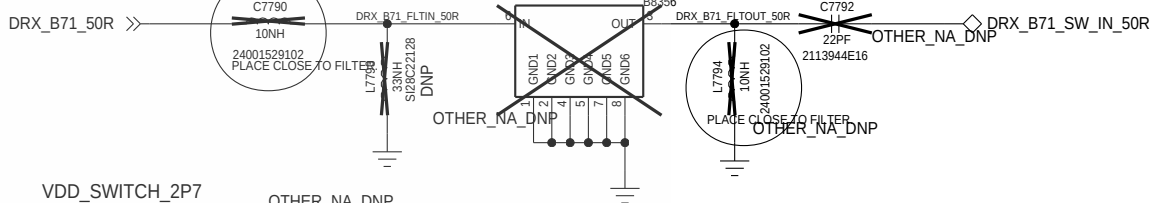
B8/GSM900



B5/B26/GSM850



B71



APPROVALS

MOTOROLA
MOBILE DEVICES

ENGINEER:

BLOCK TITLE:

rffe

LAST MODIFIED BY:

Thu Mar 01 10:00:58 2018

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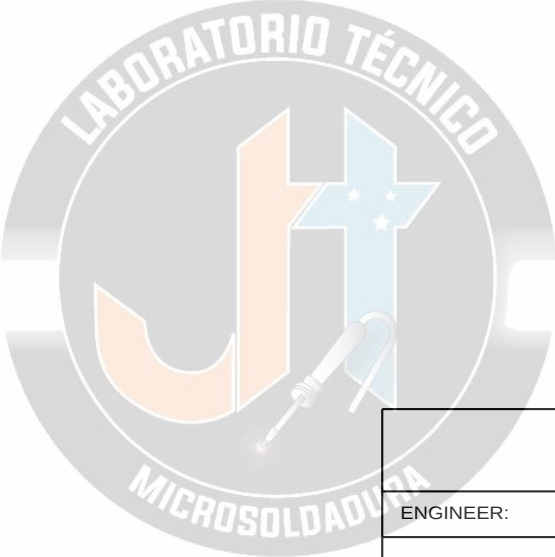
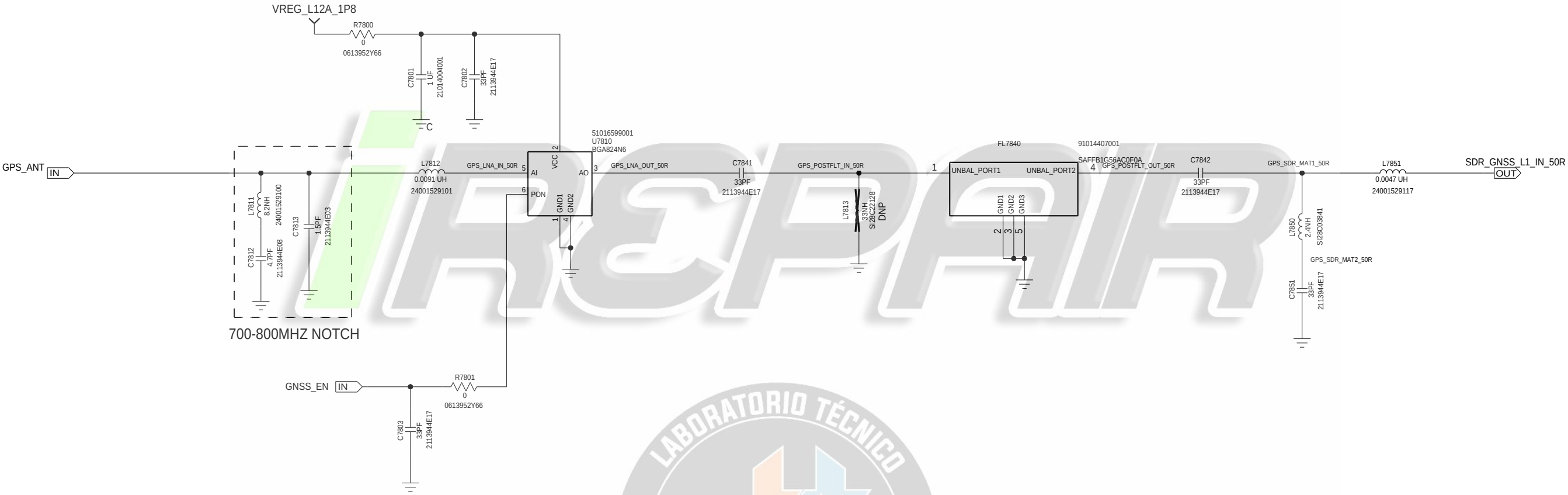
REV

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CAD FILE NAME:

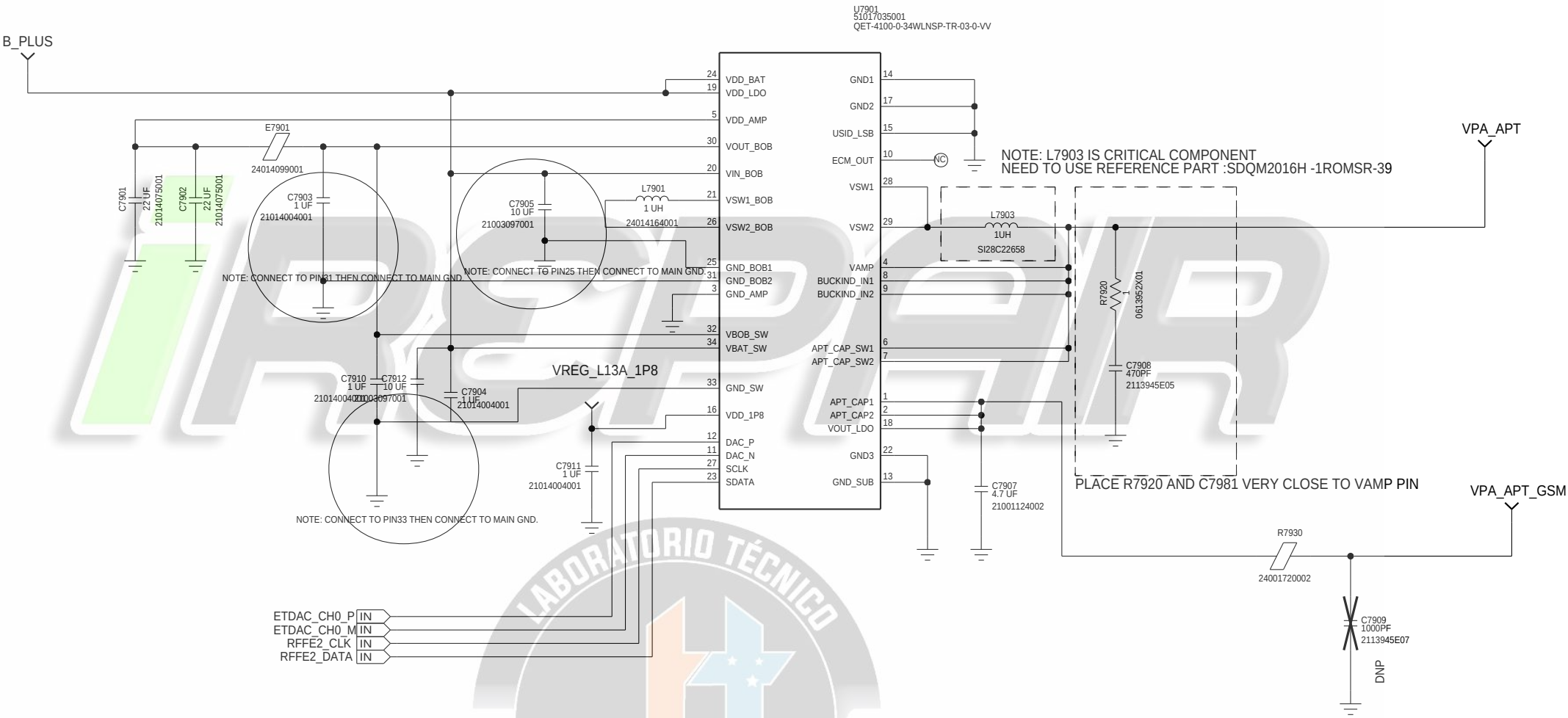
PAGE 42 OF 70

RFFE: GPS
REF7800-7899



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: rffe	
LAST MODIFIED BY: Thu Mar 01 10:00:58 2018			
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		CAD FILE NAME:	
		PAGE 43 OF 70	

RFFE: QET4100
REF7900-7949



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Fri Mar 09 11:08:16 2018			
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		REV	
		03	
		PAGE 44 OF 70	

REF: 5700-5799

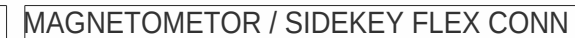


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: sensor	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:00:48 2018			
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		CAD FILE NAME:	PAGE 45 OF 70

REF: 6100-6199



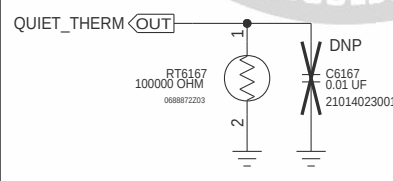
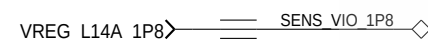
REF:6120-6129
I2C ADDR:0X39



SENS_VIO_1P8



REF:6150-6159



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Thu Mar 01 10:00:49 2018

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BLOCK TITLE:

sensor

	PROJECT NAME
B	8401672201_a_180402.cpm

CAD FILE NAME:

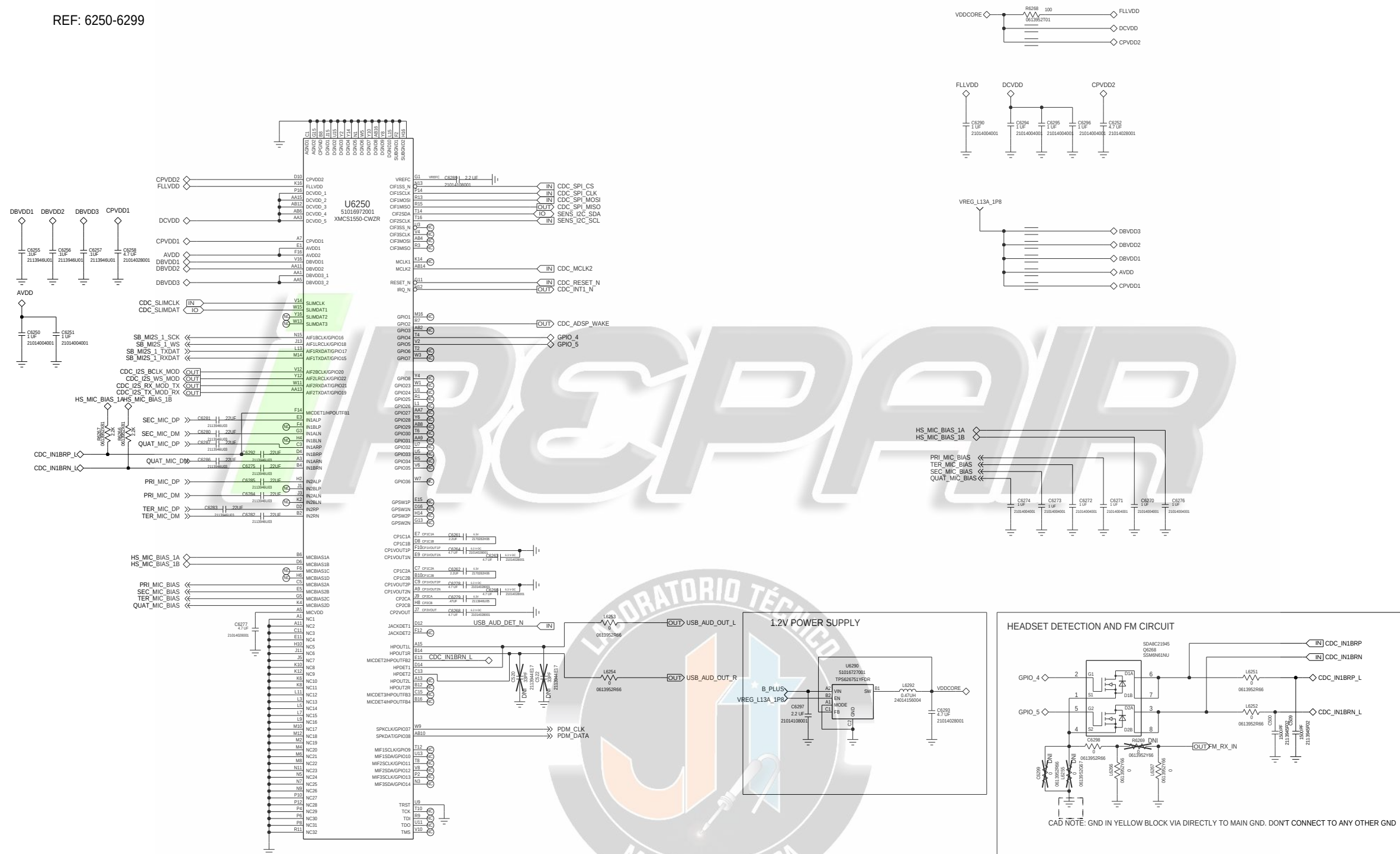
REV
03

PAGE 46 OF 70

XMCS: CIRRUS CODEC

REF: 6250-6299

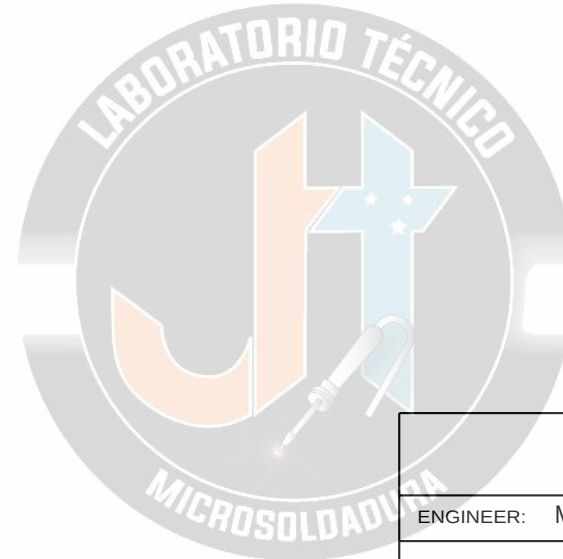
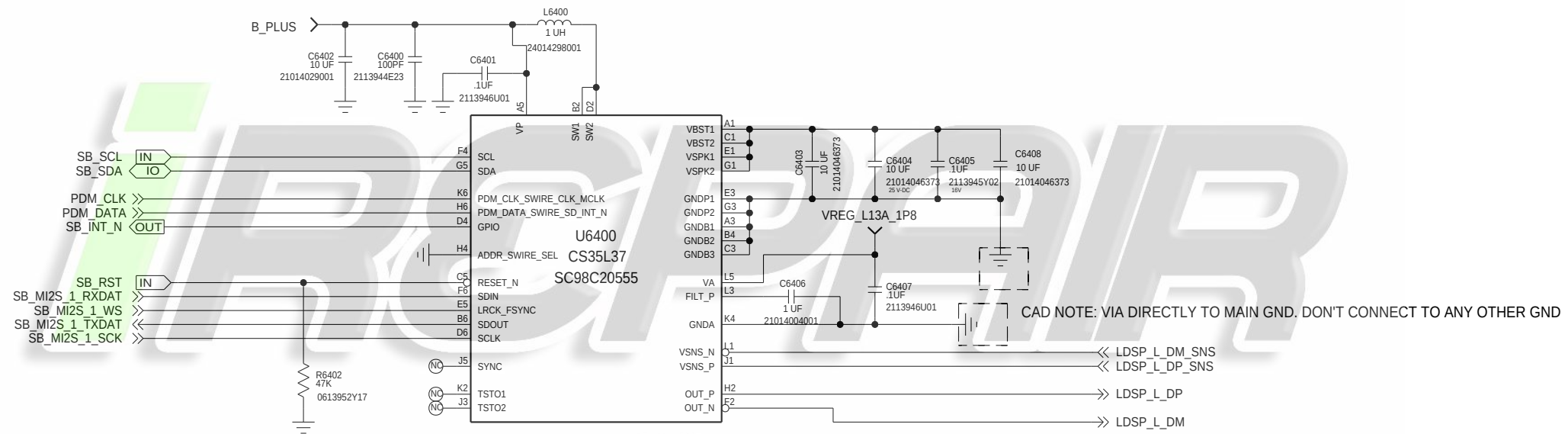
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REVISIONS					
ZONE	LTR	DESCRIPTION		DATE	APPR.



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE:		
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:00:50 2018	audio		
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		PAGE 47 of 70	

XMCS: SMART BOOST

```
*** BOOST IC I2C ADD. 0X80
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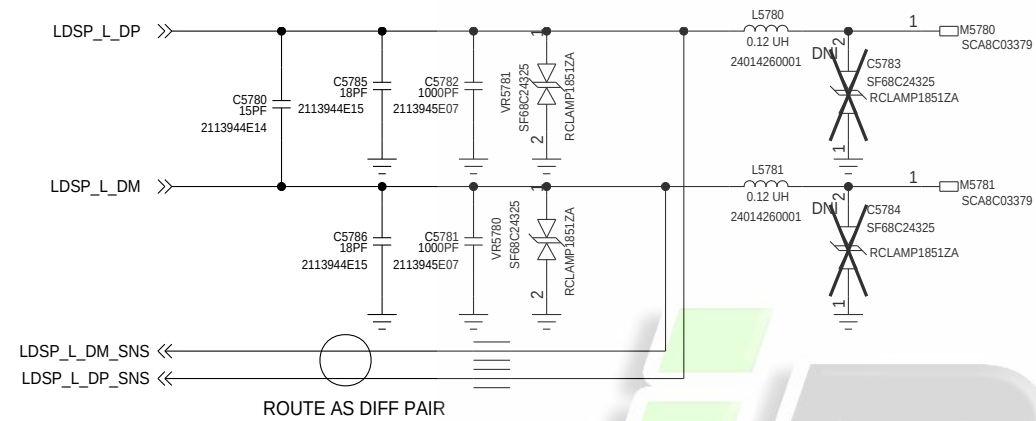


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: audio		
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:00:51 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME B 8401672201_a_180402.cpm	REV 03
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		PAGE 48 OF 70	

AFE: SPEAKERS

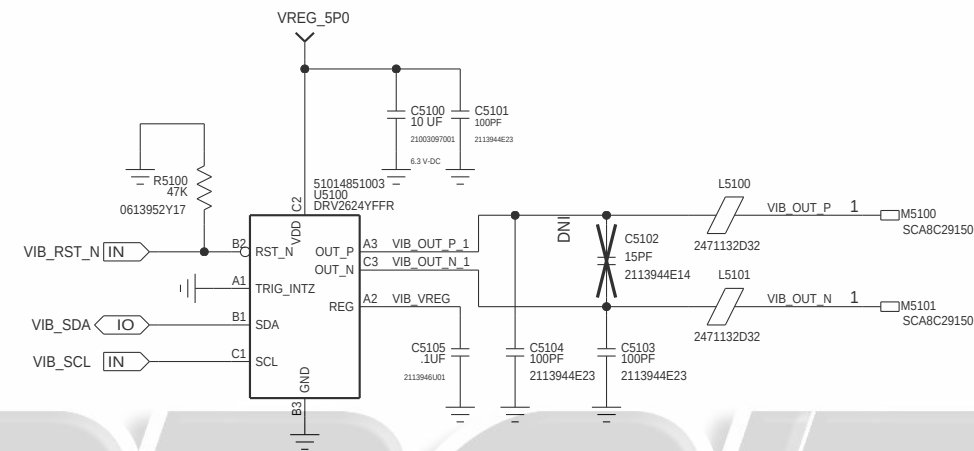
LOUDSPEAKER/EAR PIECE

REF:5780-5789

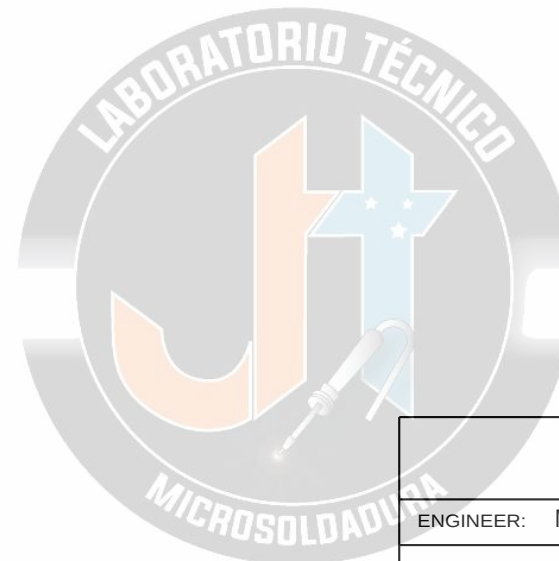


LINEAR VIBRATOR

REF 5100-5119



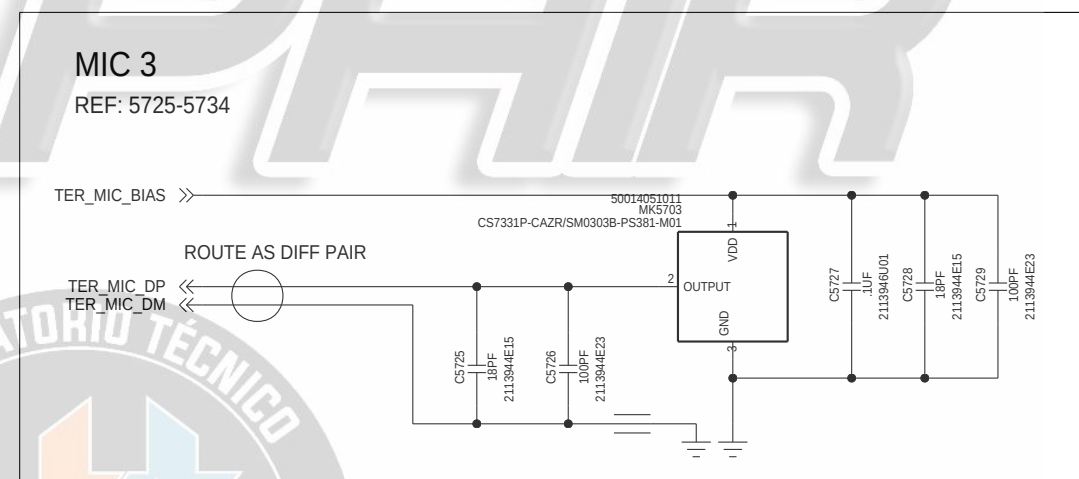
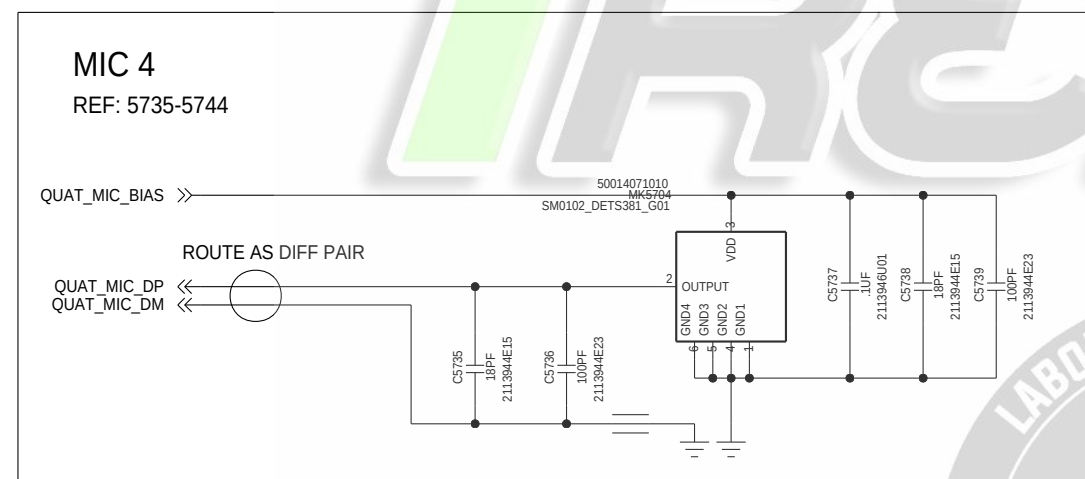
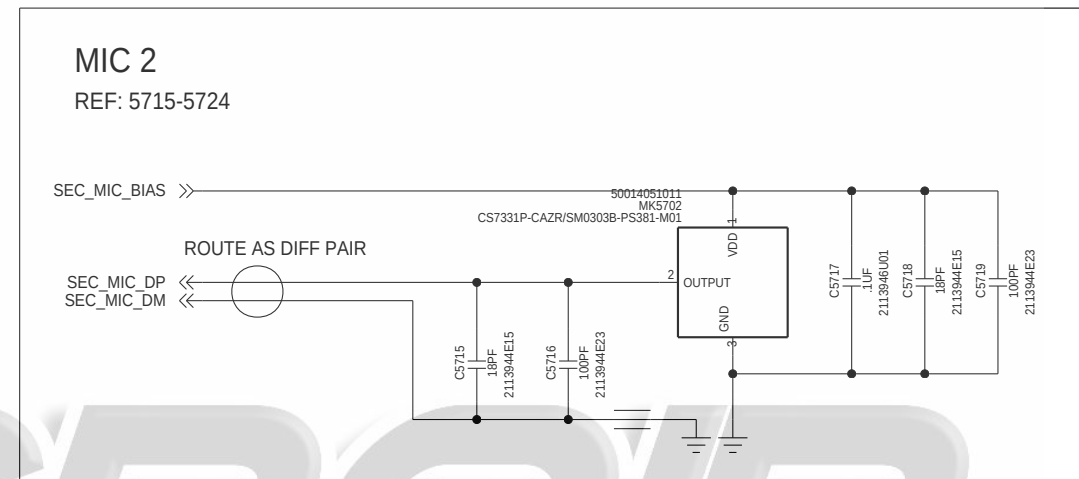
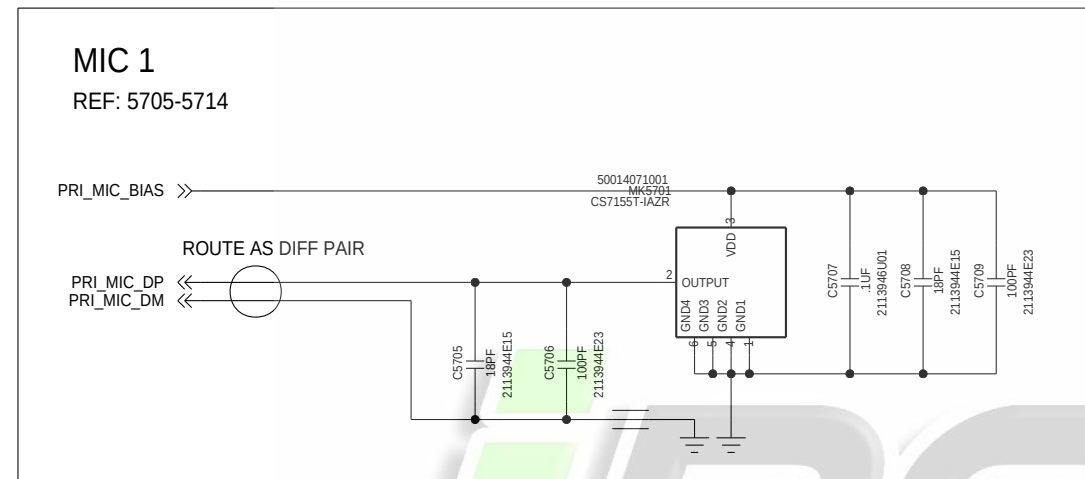
BASE I2C ADDR: 0X5A, WRITE 0XB4, READ 0XB5



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio	
LAST MODIFIED BY: MOTOROLA Mon Apr 02 15:46:33 2018			
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		REV 03	
		CAD FILE NAME:	
		PAGE 49 OF 70	

AFE: MICS

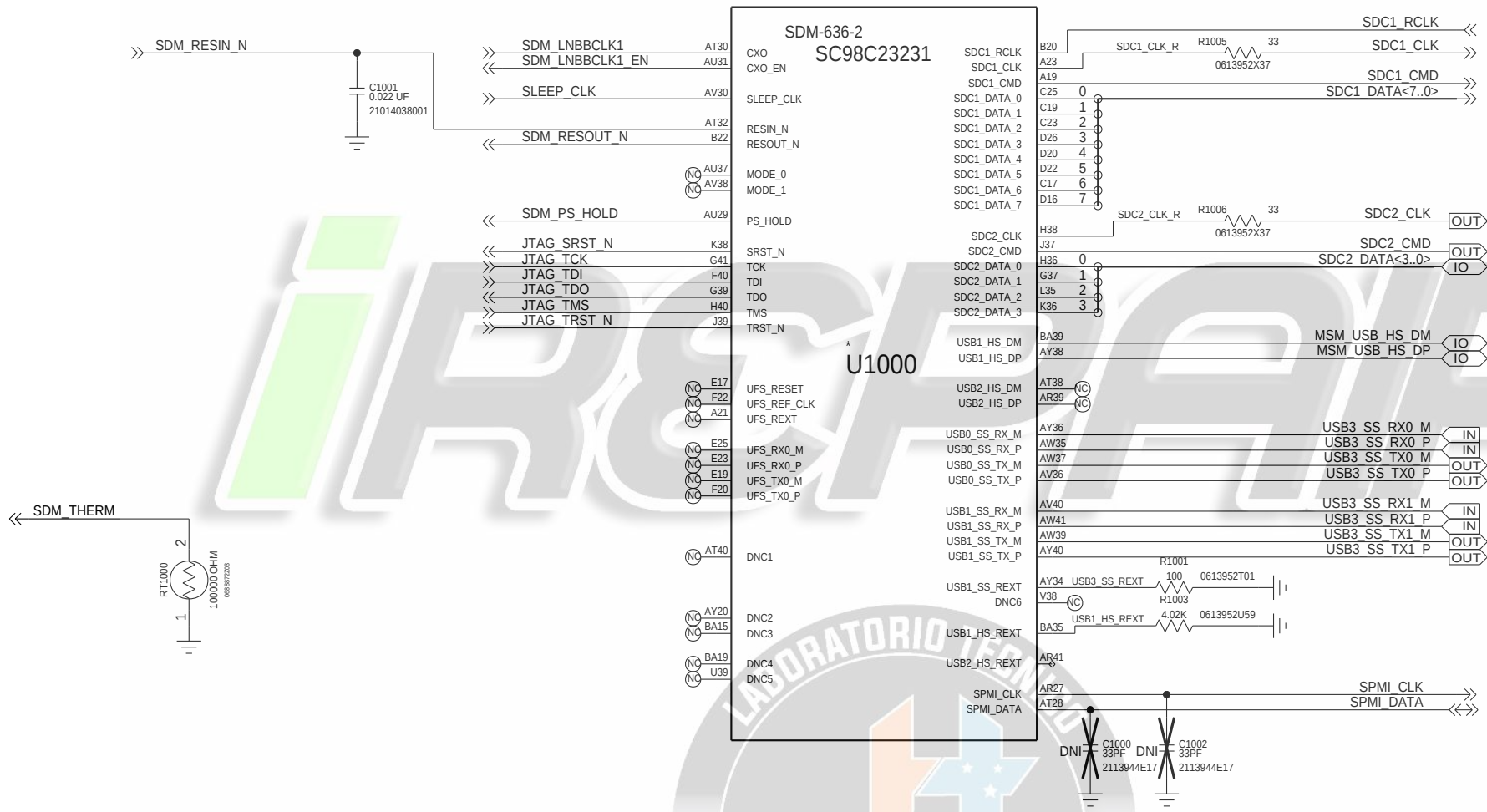
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APPROVALS		 MOTOROLA MOBILE DEVICES		
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LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:00:52 2018				
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		B 8401672201_a_180402.cpm		03
		CAD FILE NAME:		PAGE 50 OF 70

SDM660: CONTROL

REF 1000-1099



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:05 2018		ap_pmic	
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		CAD FILE NAME: PAGE 51 OF 70	

ENSURE THAT THERE ARE NO EXTERNAL PULLS ON THE GPIOs (92, 93, 64, 63, 113, AND 53) IF SECURE BOOT IS NOT REQUIRED.
THE X000D EXTERNAL PULLS CAN FORCE THE SDM DEVICE TO ENTER SECURE BOOT OR TO SELECT A DIFFERENT BOOT DEVICE.

I/O EXPANSION
I2C ADDR: 0X20

VREG_L13A_1P8

U5040 SC98C20324

PCAL618AEX1

I/O PINS:

- OUT CAM R MAIN RST N B1
- OUT CAM R DEP RST N C0
- OUT CAM F MAIN RST N C0
- OUT CAM CS0 MPH SET C0
- OUT CAM R MAIN DVDVD EN G0
- OUT CAM F MAIN DVDVD EN G0
- OUT CAM DEP DVDVD EN G0
- OUT CAM REG EN G0

I/O PINS:

- OUT CAM AFVDD EN D3
- OUT SB RST N E4
- OUT VIB RST N E5

I/O PINS:

- INT INT
- TP_IO_INT 1

I/O PINS:

- BLSP2 I2C SCL A4
- BLSP2 I2C SDA A4

I/O PINS:

- GPIOX_RST_N A2

I2C ADDR: 0X21

VREG_L13A_1P8

U5041 SC98C20324

PCAL618AEX1

I/O PINS:

- OUT FP VDDIO EN B1
- OUT MOD BPLUS EN C1
- OUT MOD BPLUS LIM E2

I/O PINS:

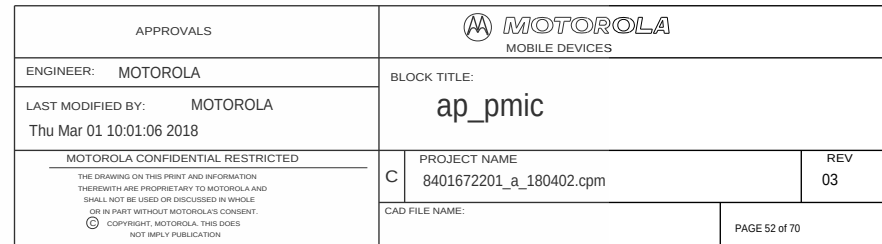
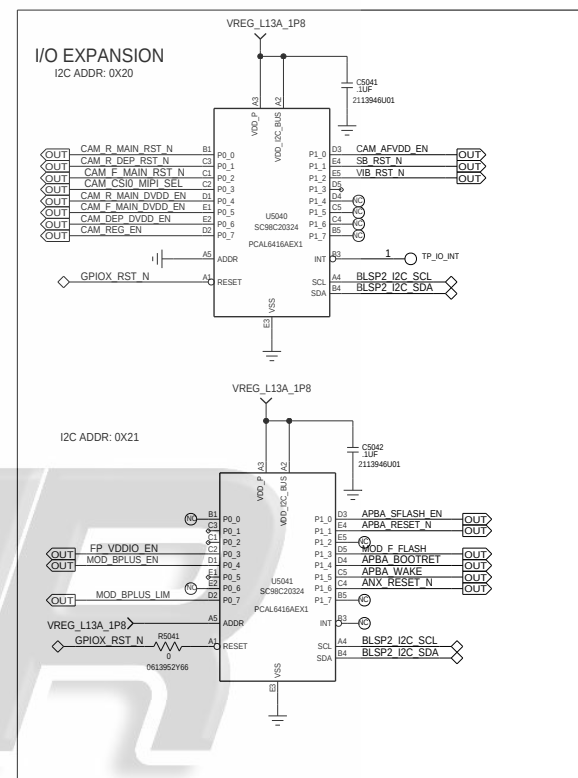
- OUT APBA FLASH EN D3
- OUT APBA RESET N E4
- OUT MOD F FLASH E5
- OUT APBA BOOTRET D4
- OUT APBA WAKE C5
- OUT ANX RESET N E6

I/O PINS:

- BLSP2 I2C SCL A4
- BLSP2 I2C SDA A4

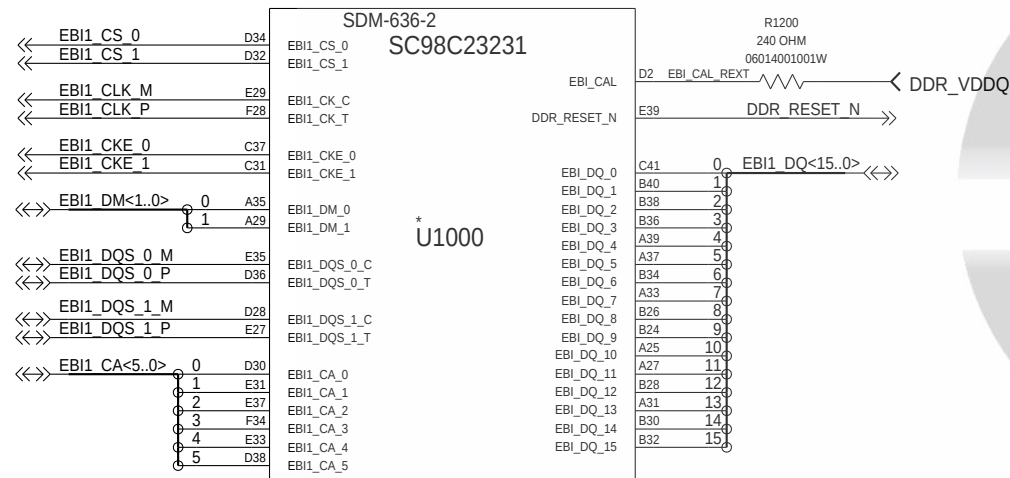
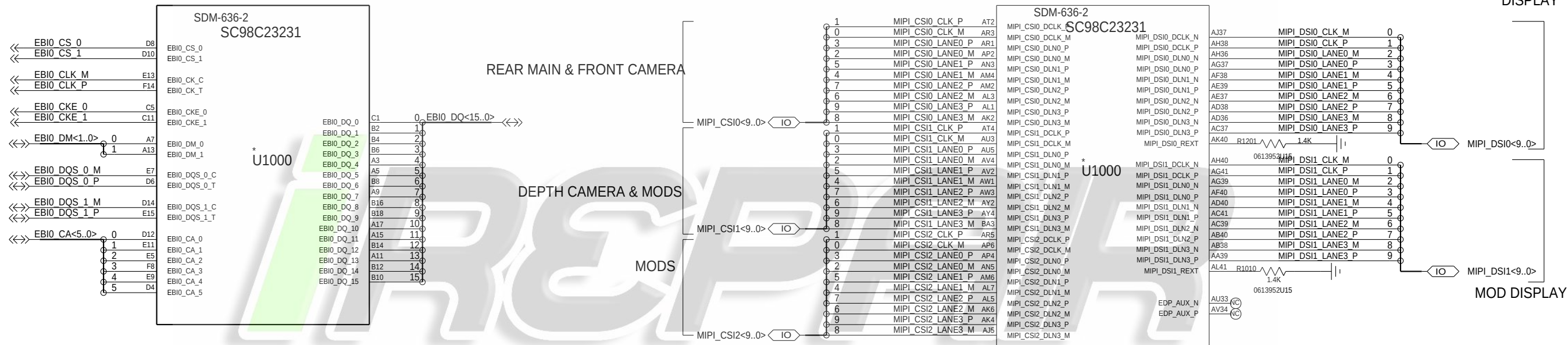
I/O PINS:

- GPIOX_RST_N A2



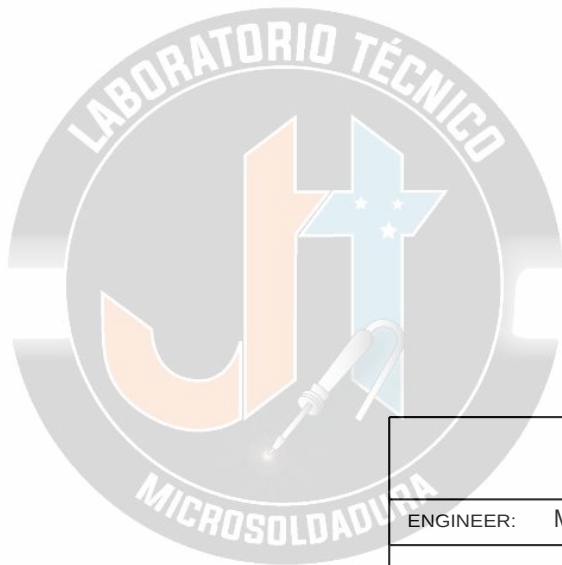
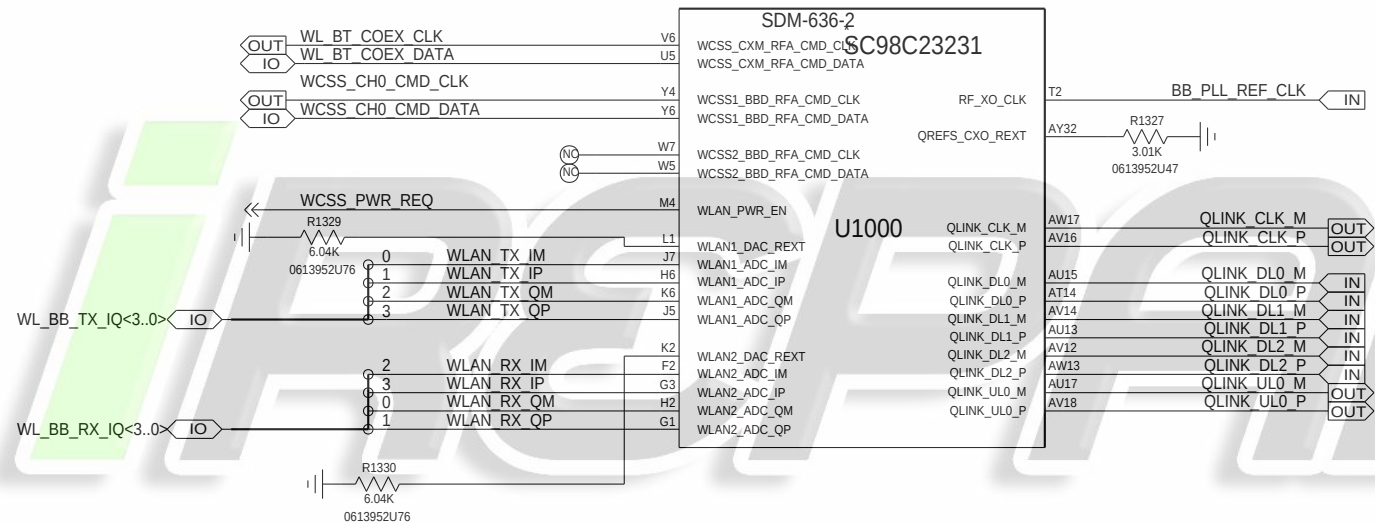
SDM660: MIPI/DSI/CSI/EBIO

REF 1200-1299



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 53 OF 70	

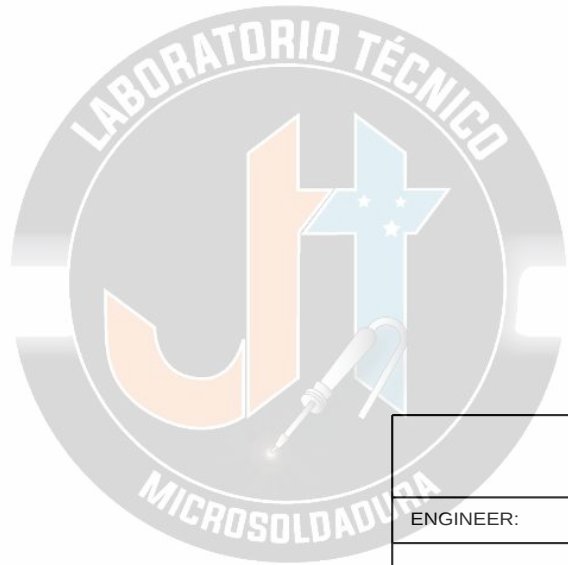
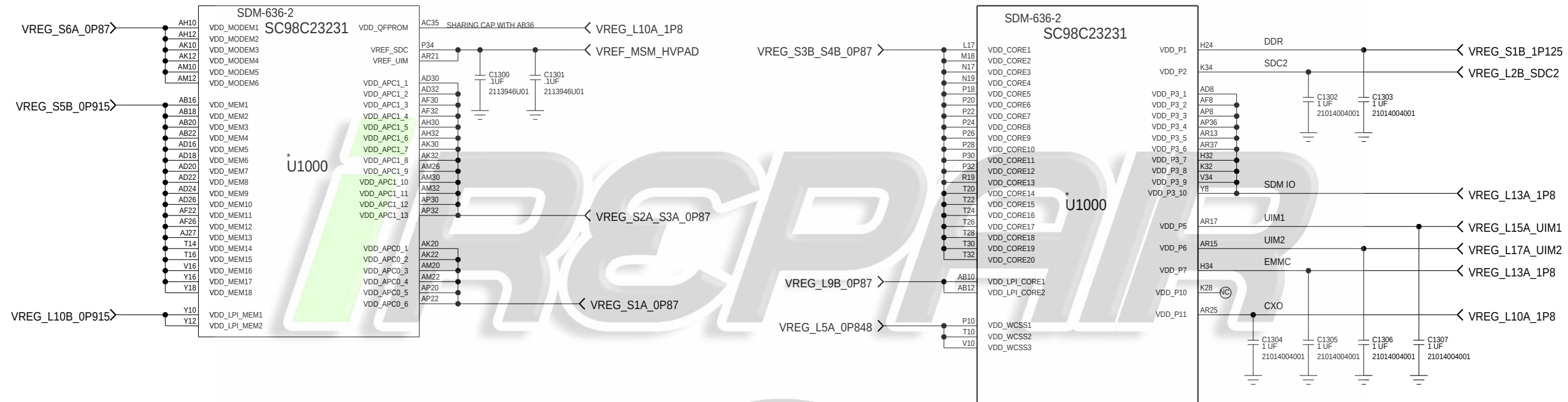
SDM660: WCSS/RF
REF 1200-1299



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:07 2018		ap_pmic	
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		CAD FILE NAME: PAGE 54 OF 70	

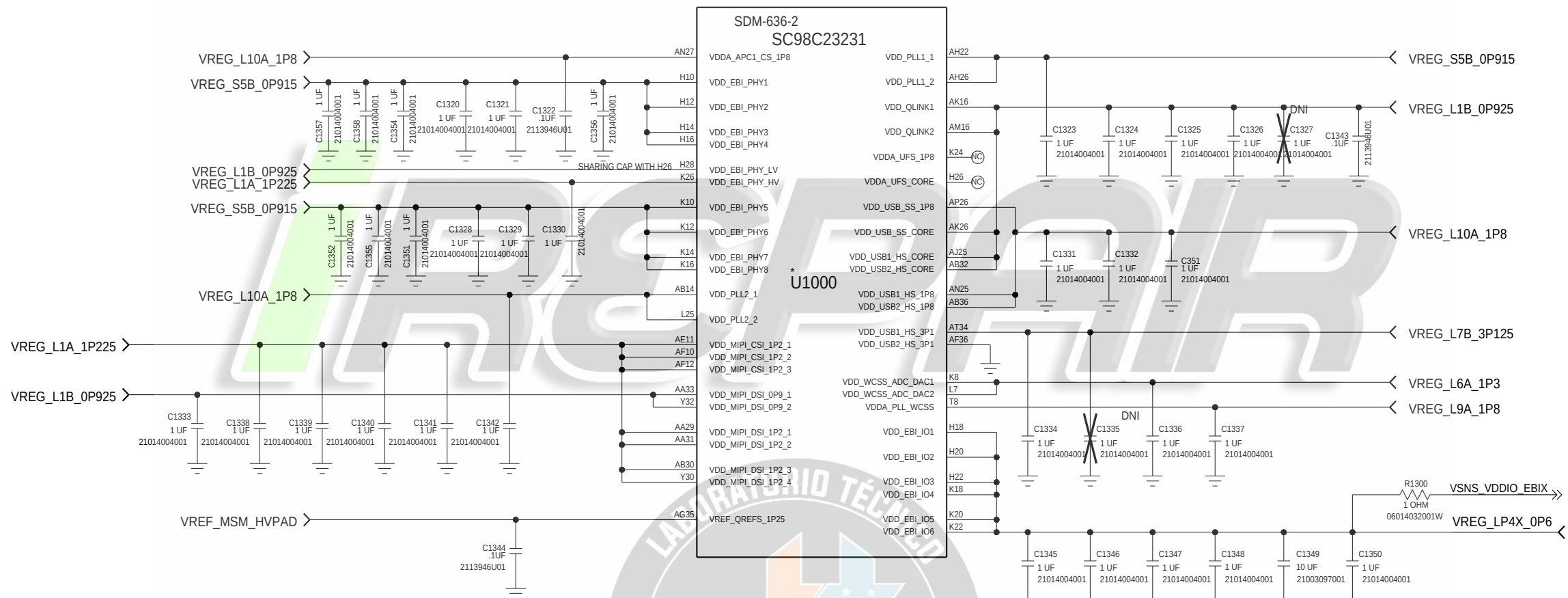
SDM660: PWR1

REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES		
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LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:07 2018				
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				PAGE 55 OF 70

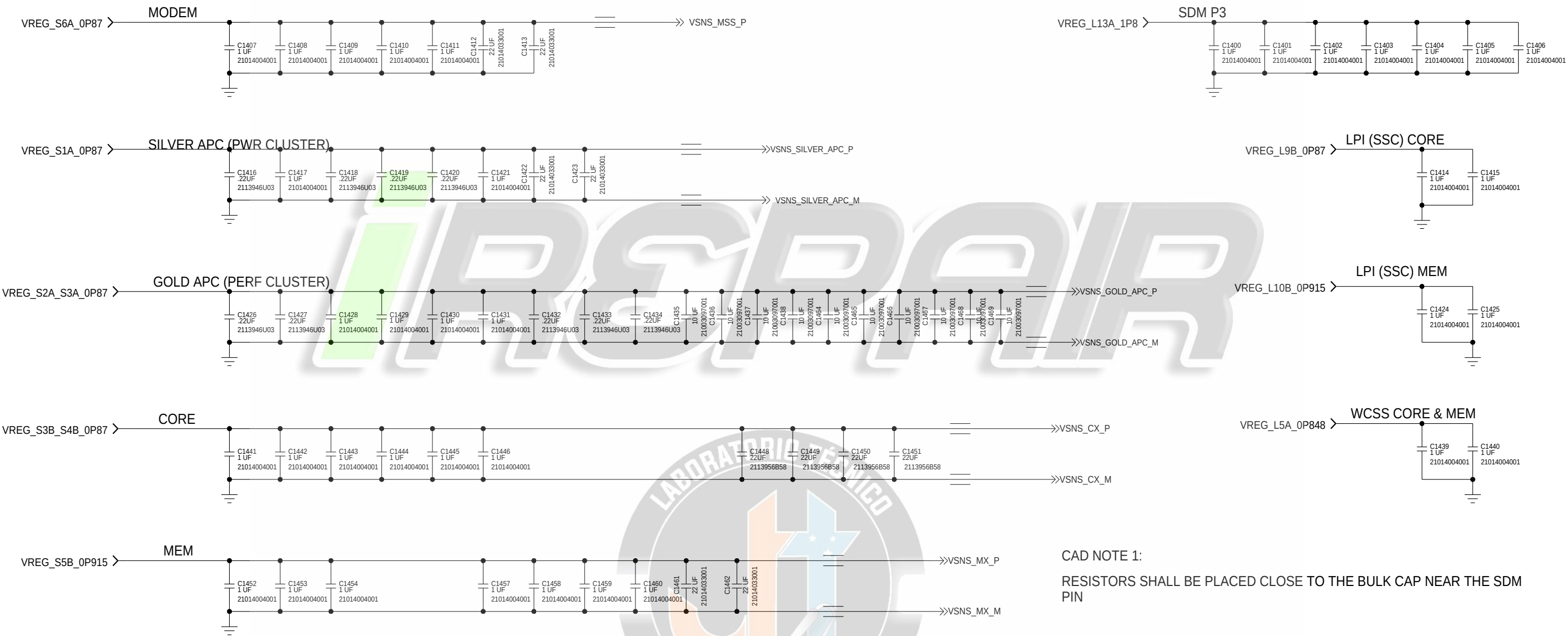
SDM660: PWR2
REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:08 2018		ap_pmic	
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		CAD FILE NAME: PAGE 56 OF 70	

SDM660: DECAPS

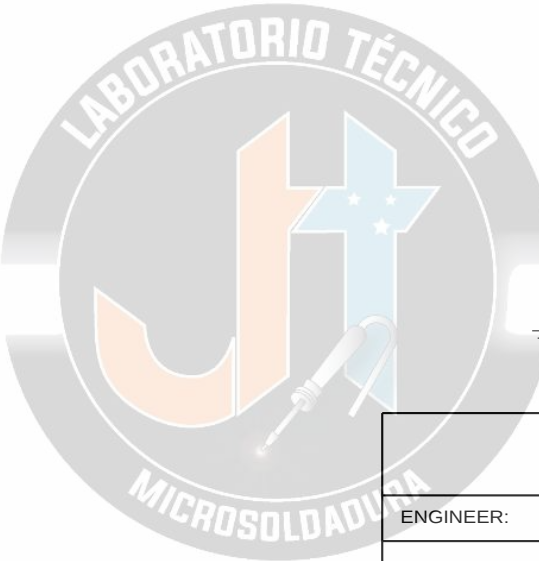
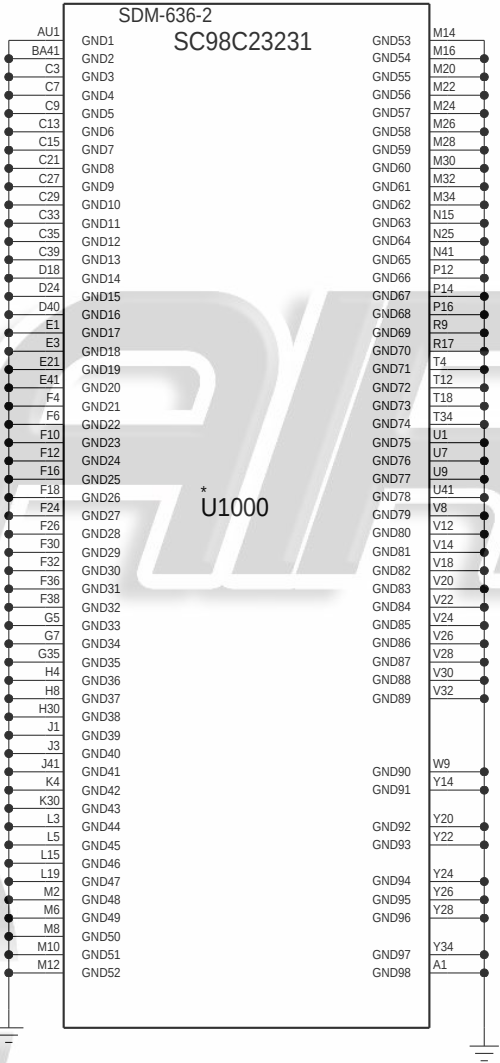
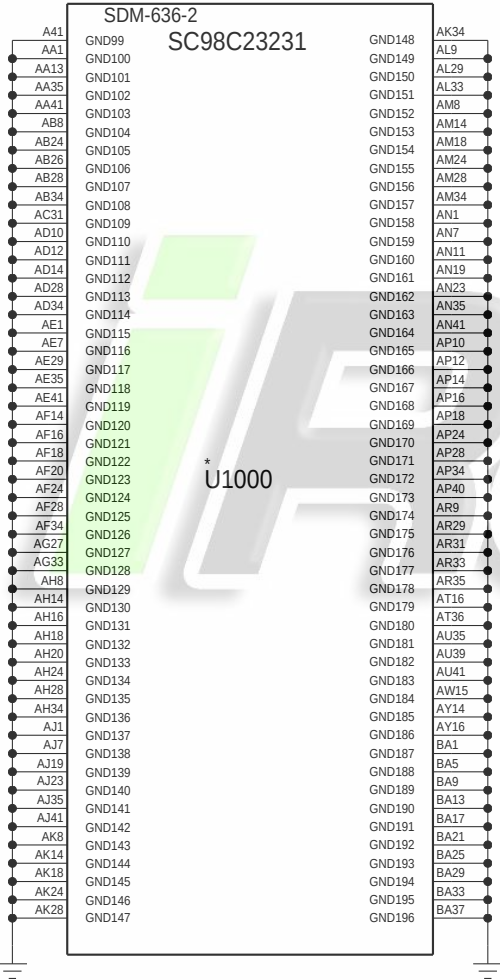
REF 1400-1499



CAD NOTE 1:
RESISTORS SHALL BE PLACED CLOSE TO THE BULK CAP NEAR THE SDM PIN

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:08 2018		ap_pmic	
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		CAD FILE NAME: PAGE 57 OF 70	

SDM660: GND

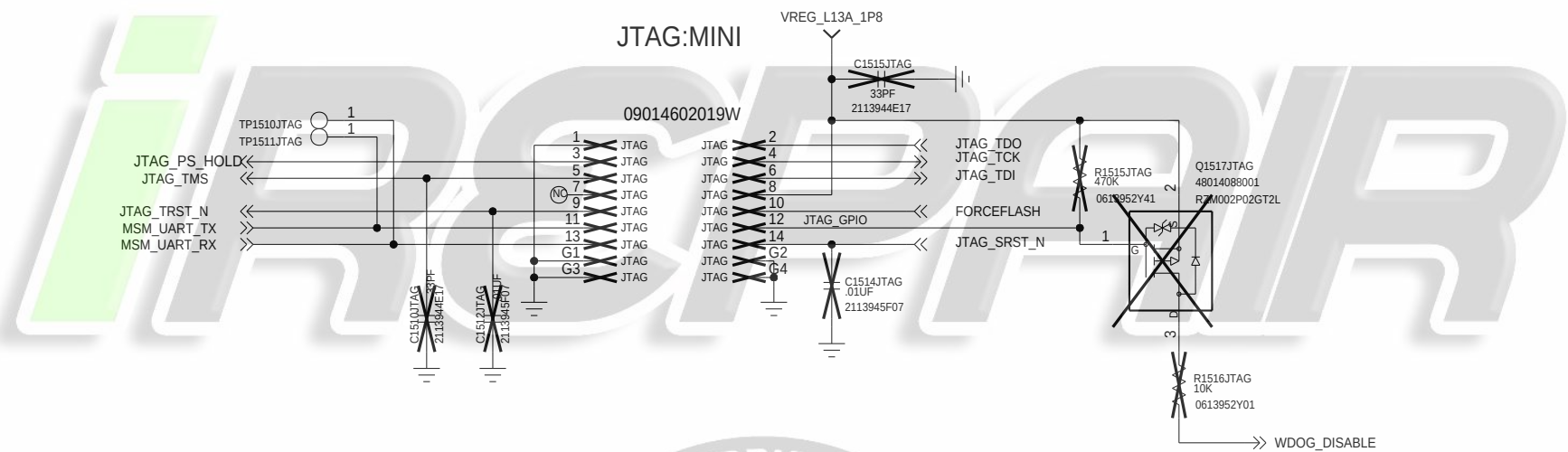


APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:09 2018				
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		B 8401672201_a_180402.cpm		03
		CAD FILE NAME:		PAGE 58 OF 70

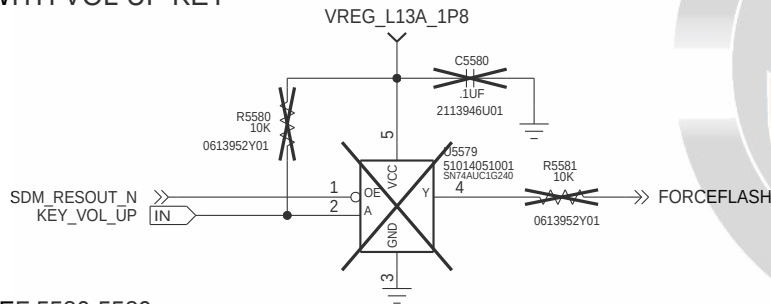
SDM660: DEBUG

REF 1000-1099

(DNI FOR PRODUCTION)



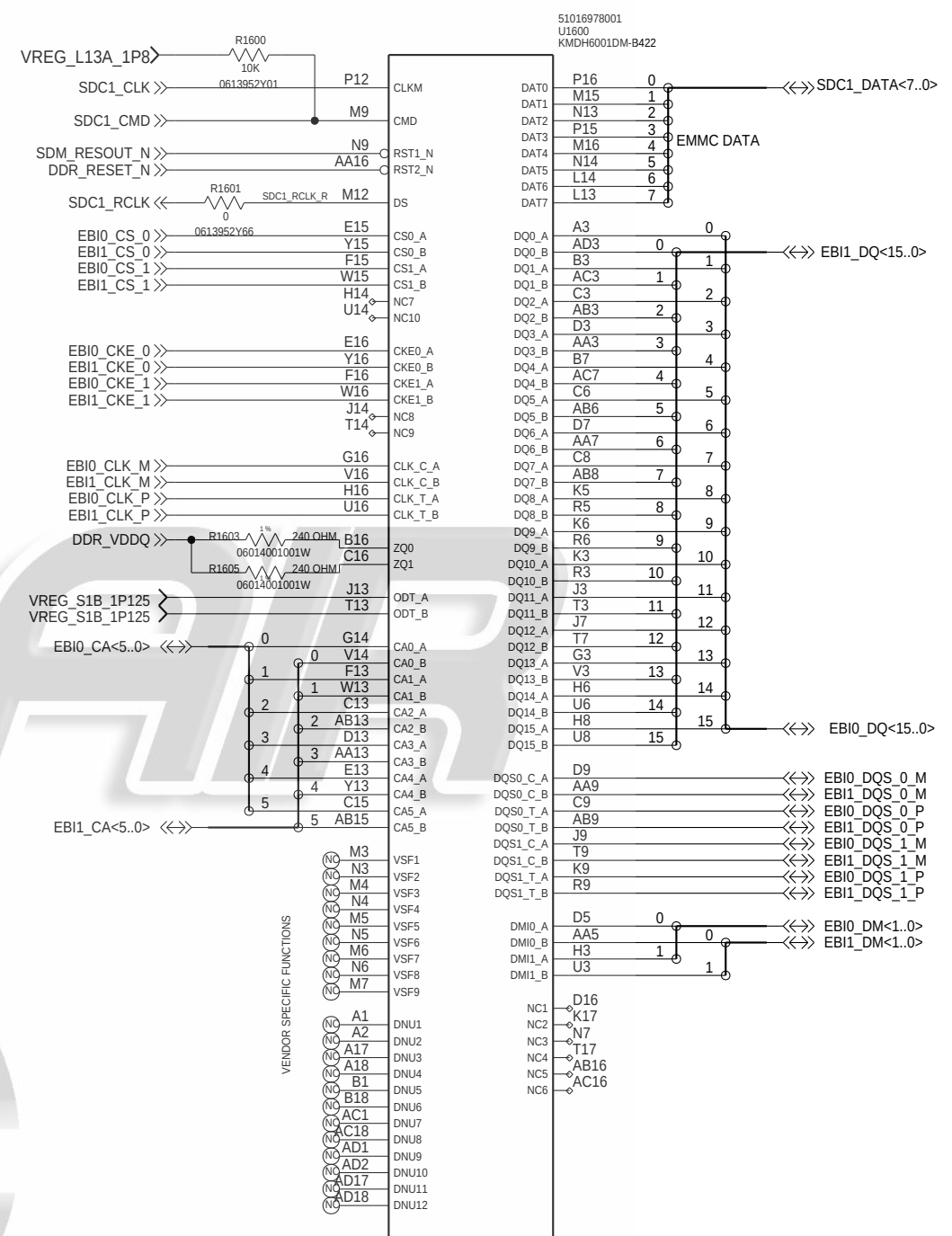
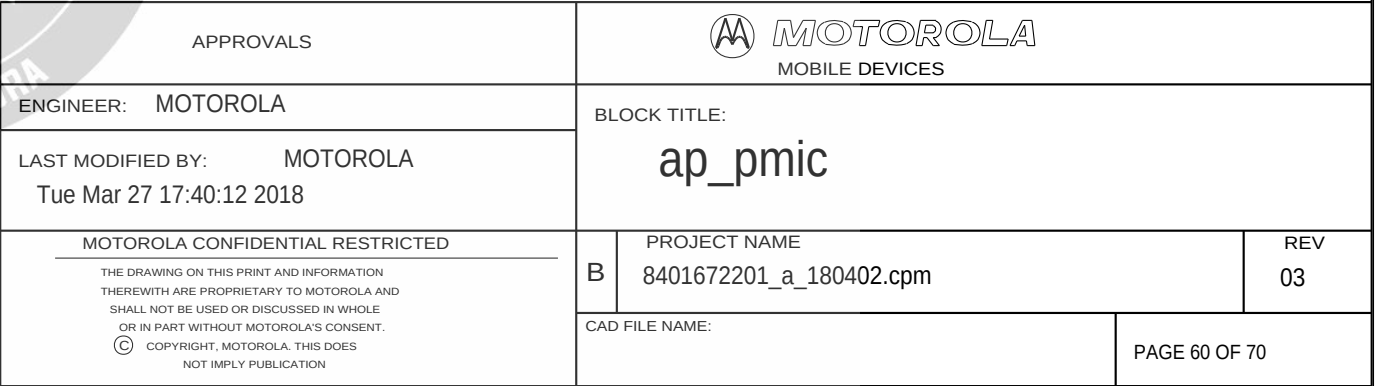
FORCE FLASH SUPPORT WITH VOL UP KEY



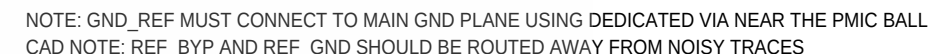
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:09 2018		ap_pmic	
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		REV 03	
		PAGE 59 OF 70	

REF: 1600-1699

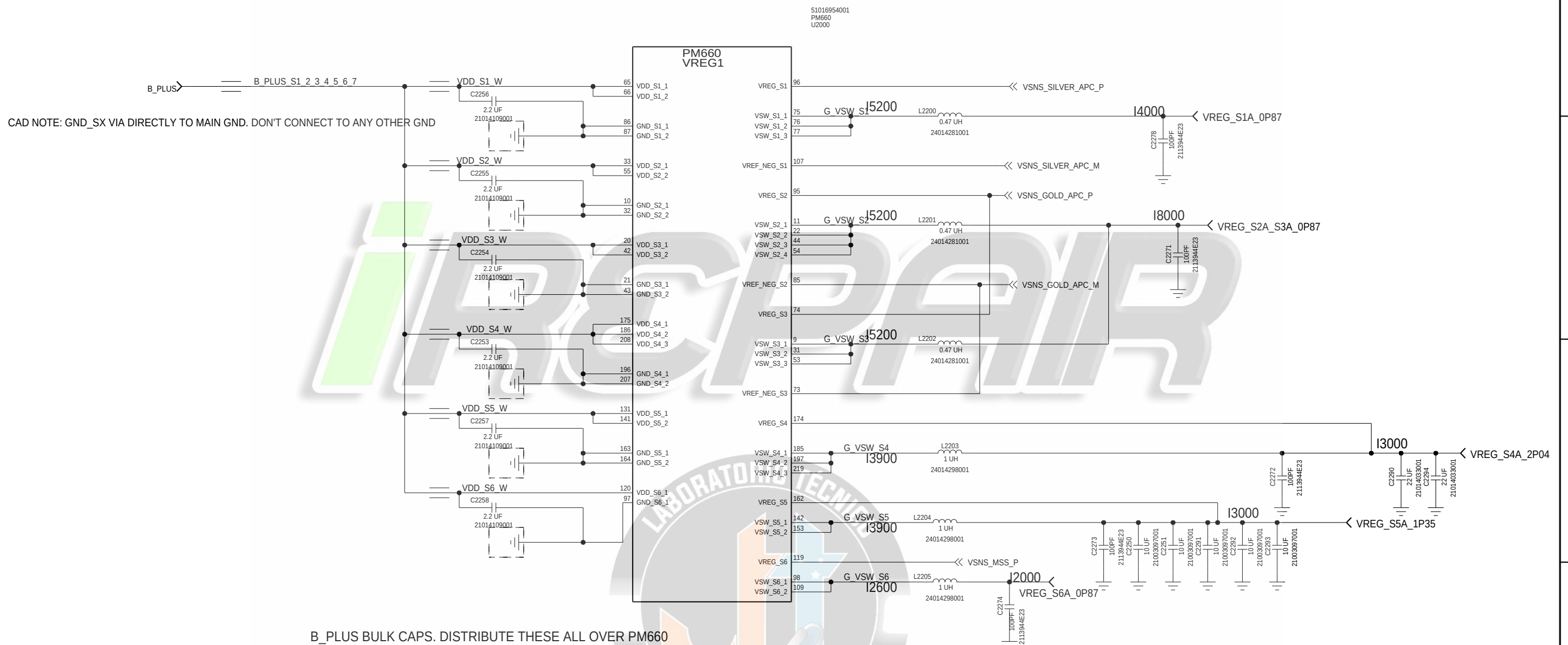


REF: 2000-2099

PAGE 61 OF 70

PM660: SMPS

REF: 2200-2299

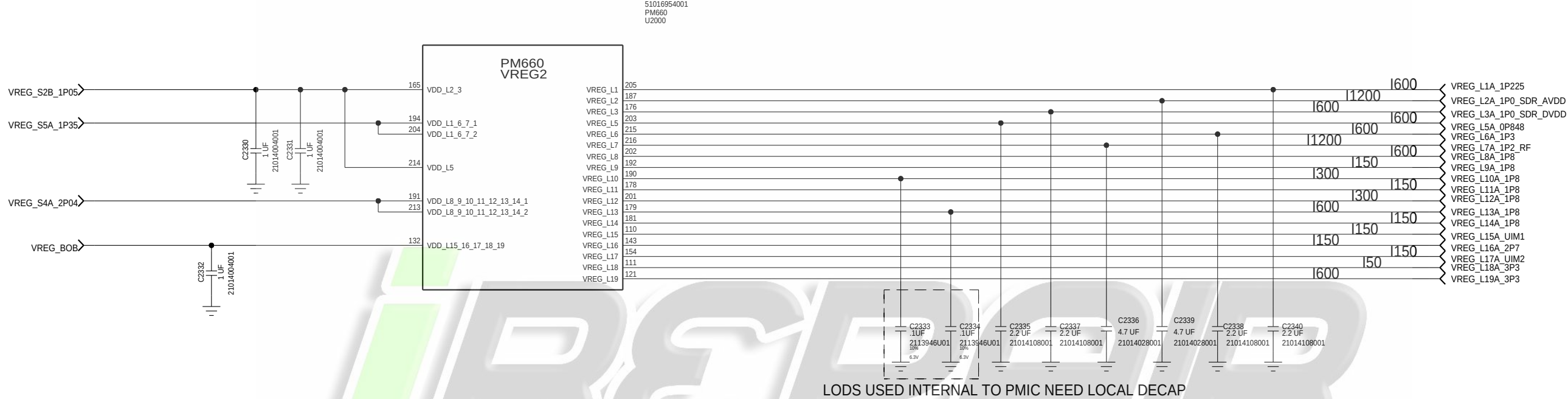


NOTE:LOCAL CAP PLACE HOLDER PROVIDED FOR LOAD CURRENT MEASUREMENTS

APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:11 2018				
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		B 8401672201_a_180402.cpm		03
		CAD FILE NAME:		PAGE 62 OF 70

PM660: VREGS

REF: 2300-2399



LOADS USED INTERNAL TO PMIC NEED LOCAL DECAP

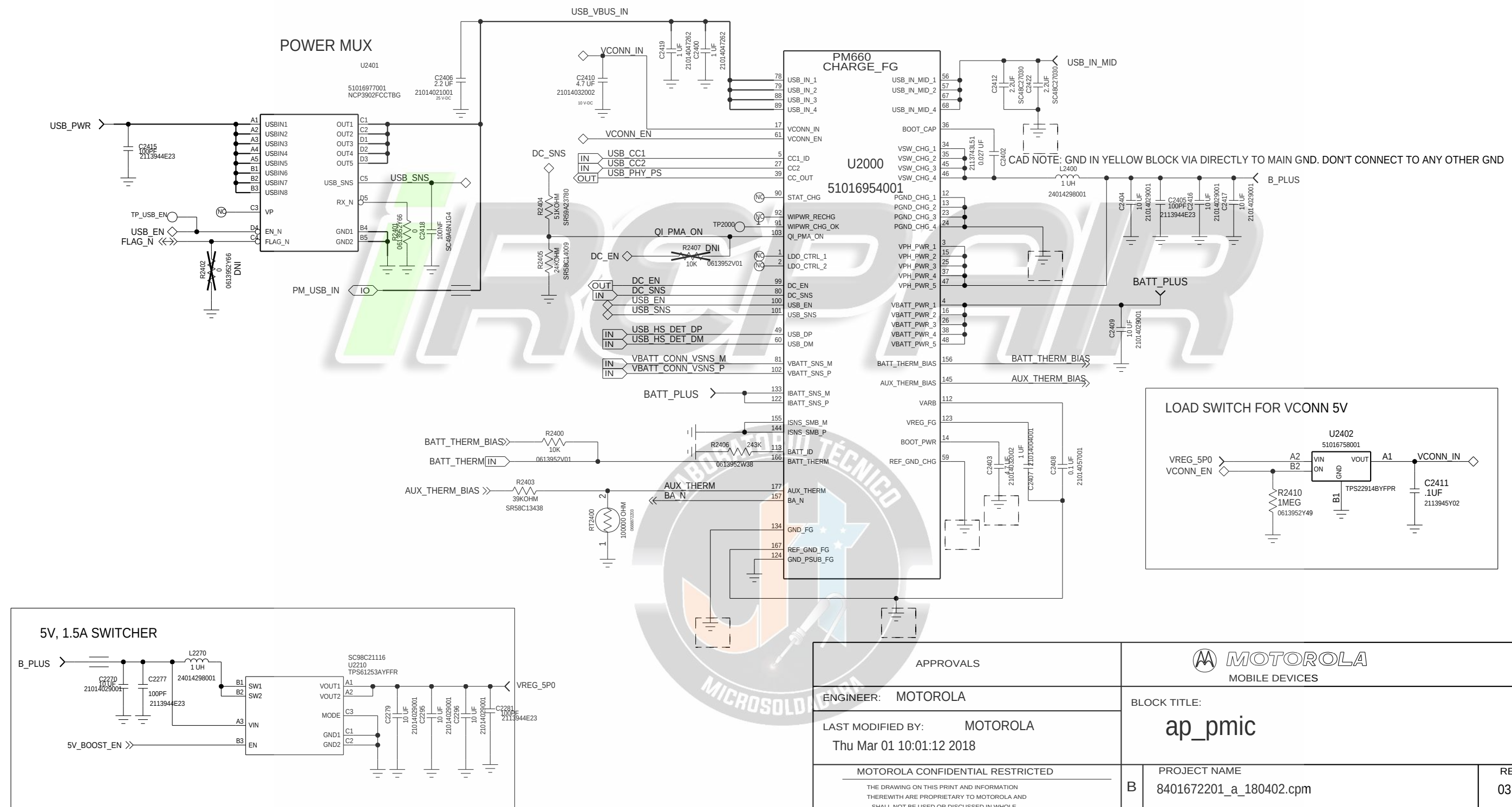


PSEUDO CAPLESS LDOs
FOR CAPLESS LDOs: IF DECAPS ON THE LOAD SIDE DO NOT
ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:12 2018		ap_pmic	
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		CAD FILE NAME: PAGE 63 OF 70	

PM660: CHARGER

REF: 2400-2419



APPROVALS

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Thu Mar 01 10:01:12 2018

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BLOCK TITLE:

ap_pmic

PROJECT NAME
8401672201_a_180402.cpm

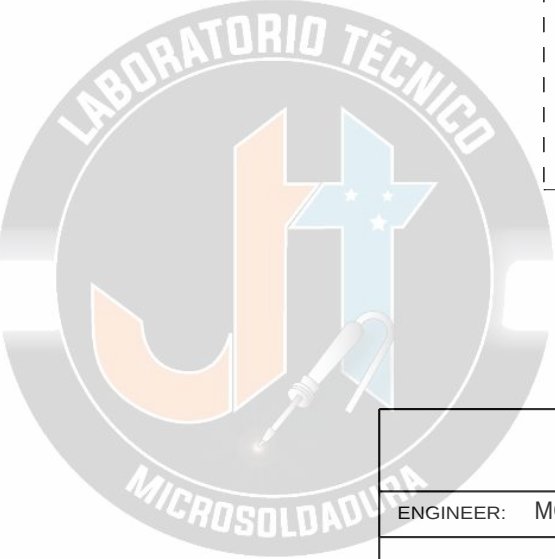
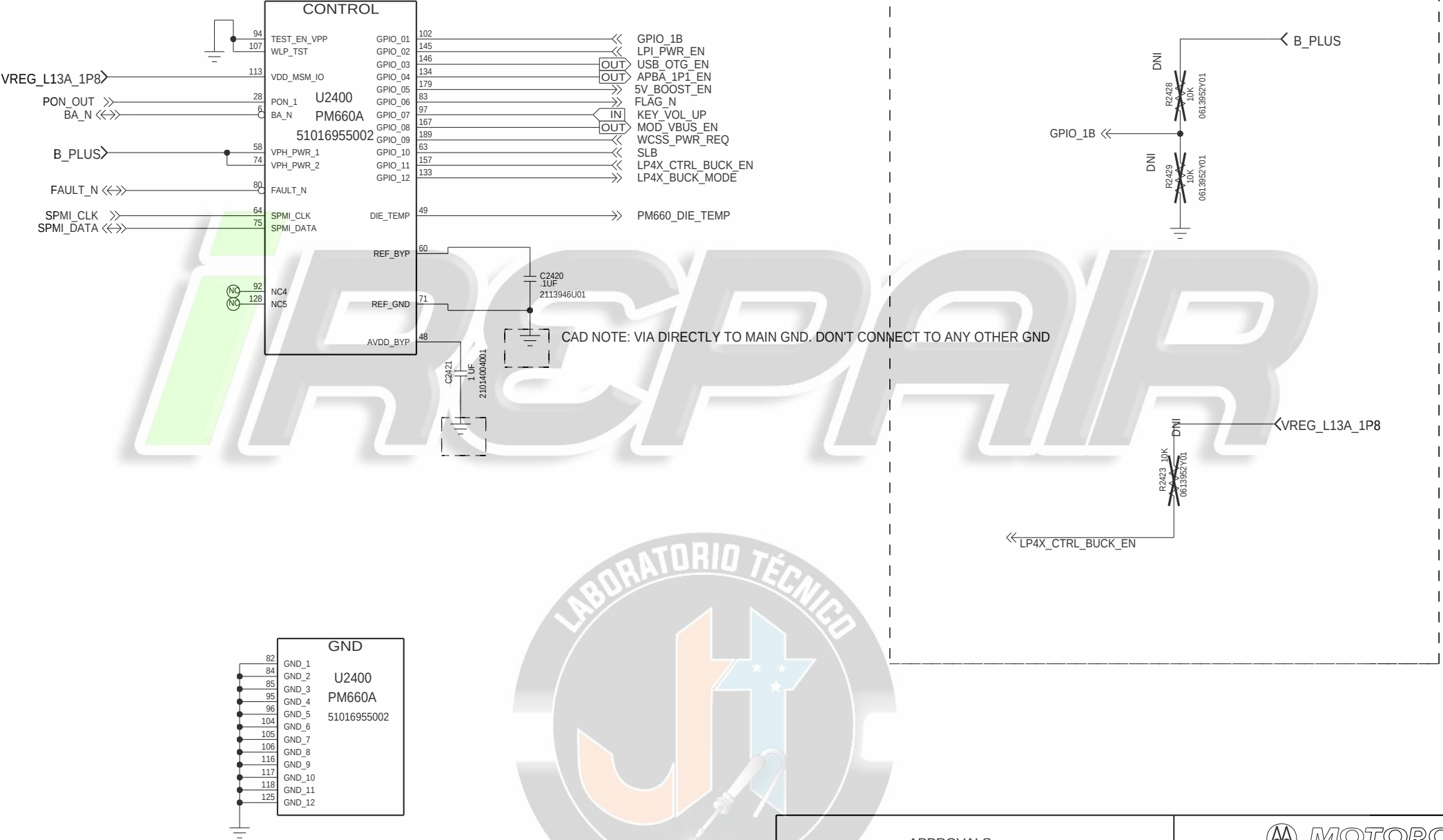
CAD FILE NAME:

REV
03

PAGE 64 OF 70

PM660A: GPIOS

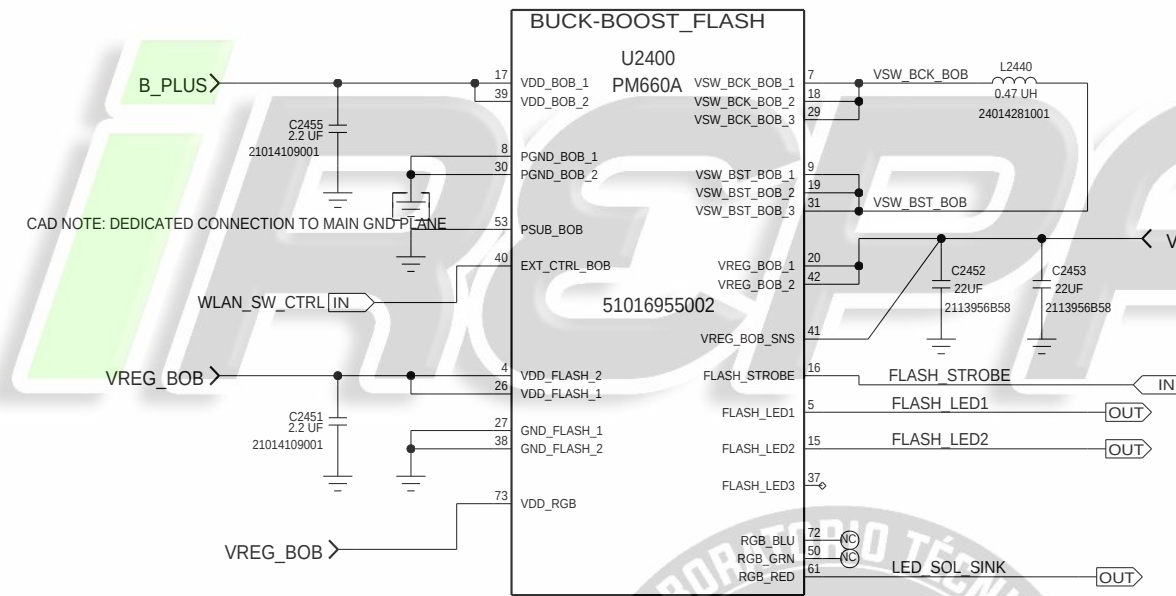
REF: 2420-2439



APPROVALS		MOTOROLA MOBILE DEVICES	
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		REV	
		03	
		PAGE 65 OF 70	

PM660A: BOB_FLASH

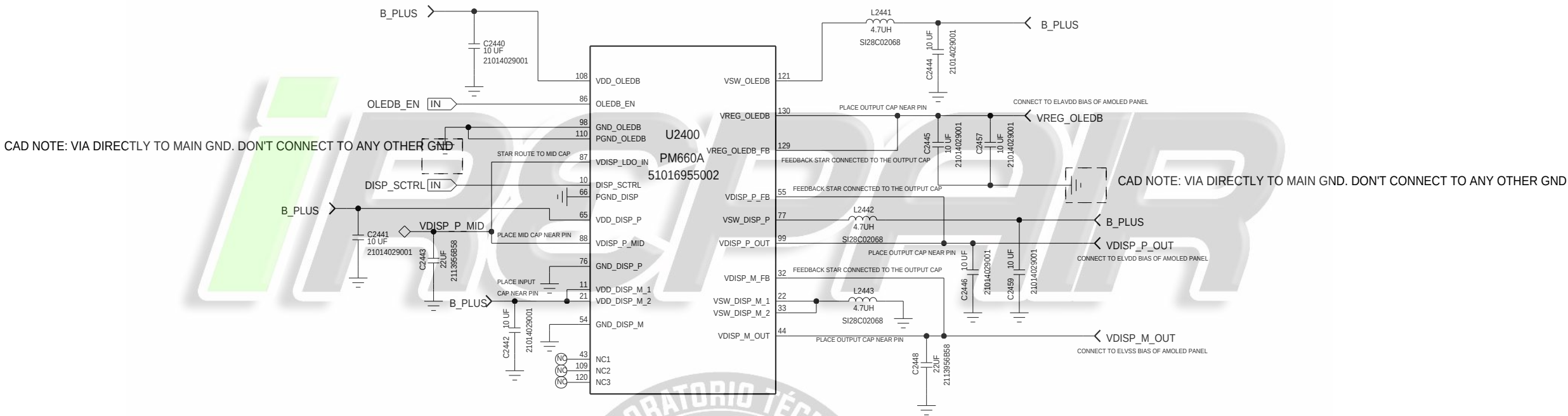
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 66 OF 70	

PM660A: IBB_LAB_OLEDB

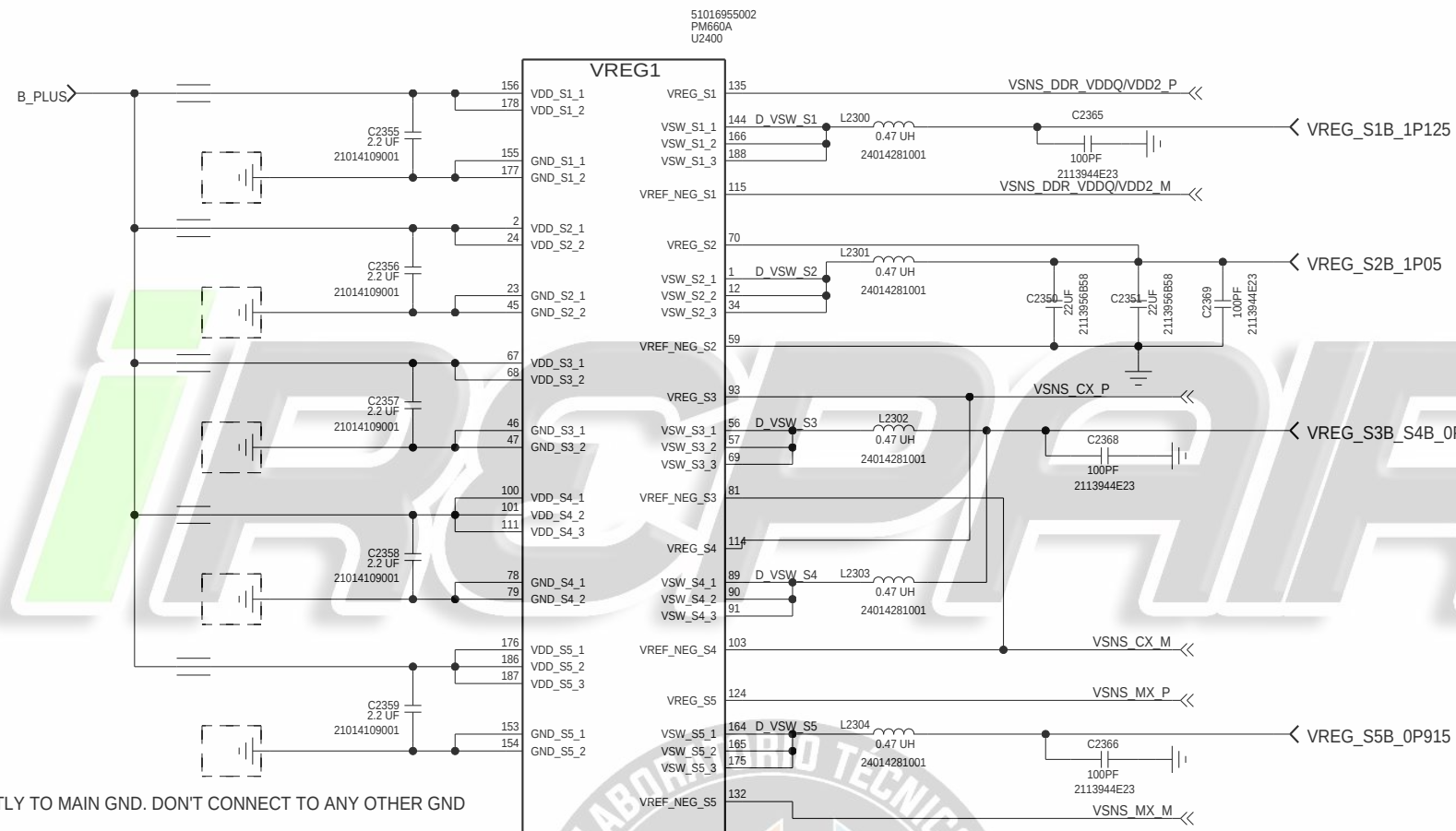
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 67 OF 70	

PM660A: SMPS

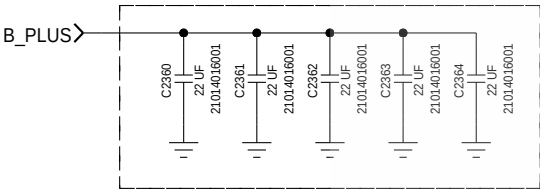
REF: 2300-2399



CAD NOTE: GND_SX VIA DIRECTLY TO MAIN GND. DON'T CONNECT TO ANY OTHER GND

NOTE: LOCAL CAP PLLACE HOLDER PROVIDED FOR LOAD CURRENT MEASUREMENTS

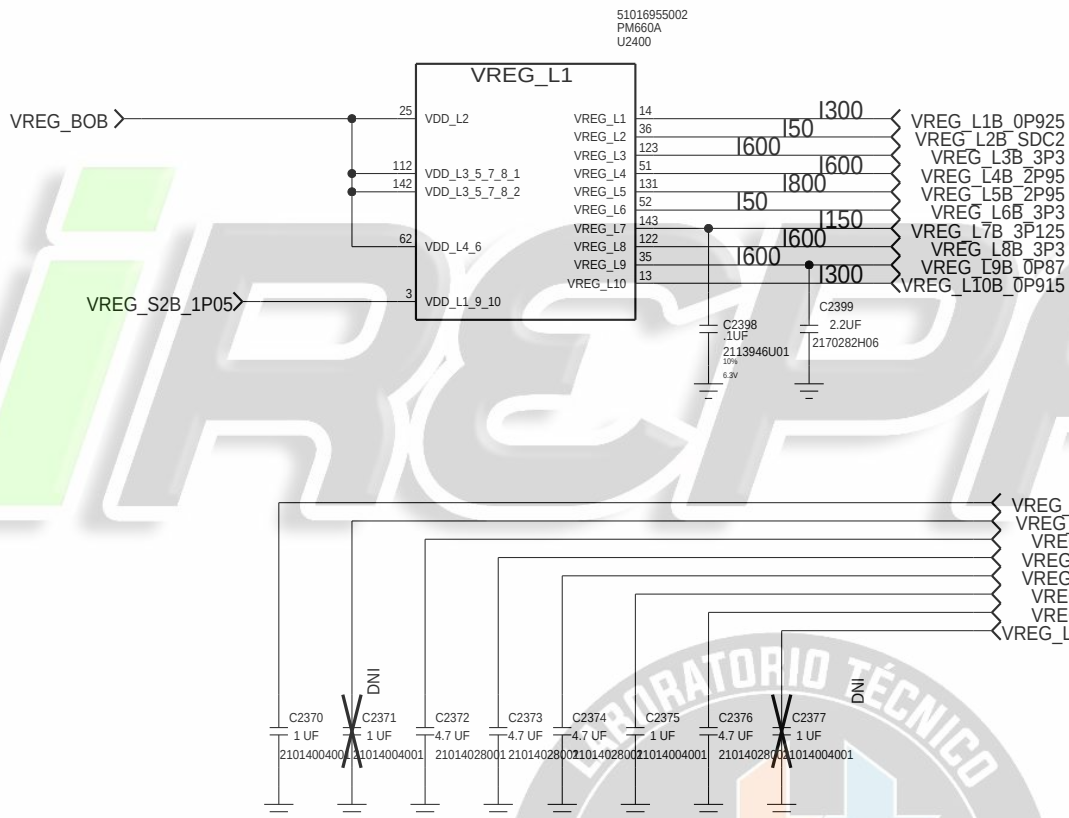
B_PLUS BULK _CAPS.DISTRIBUTE THESE ALL OVER U2400



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Mar 01 10:01:14 2018		ap_pmic	
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		CAD FILE NAME: PAGE 68 OF 70	

PM660A: LDOS

REF: 2300-2399

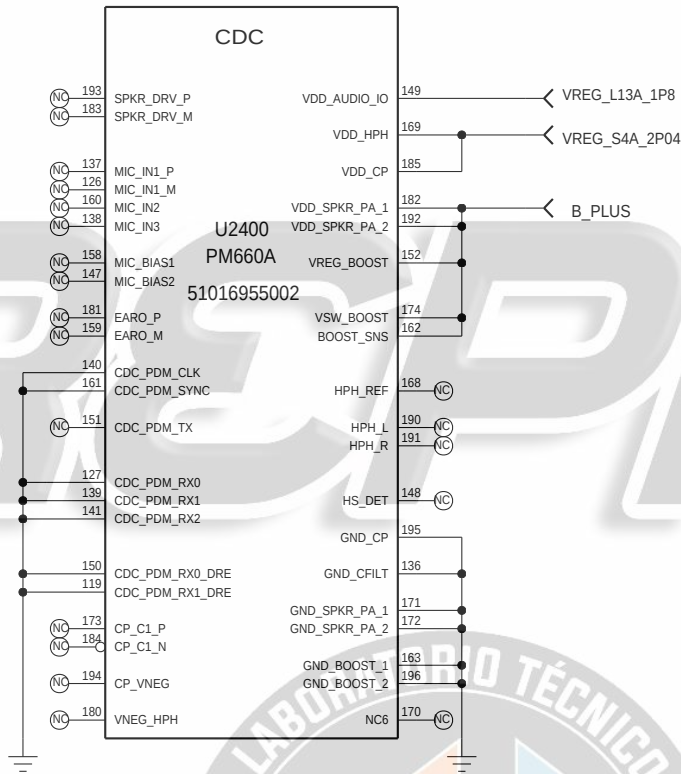


PSEUDO CAPLESS LDOS
FOR CAPLESS LODS:IF DECAPS ON THE LOAD SIDE DO NOT ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 69 OF 70	

PM660A: INTERNAL AUDIO CODEC

REF 2500-2579



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		CAD FILE NAME: PAGE 70 OF 70	